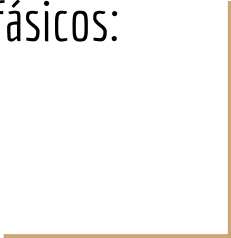




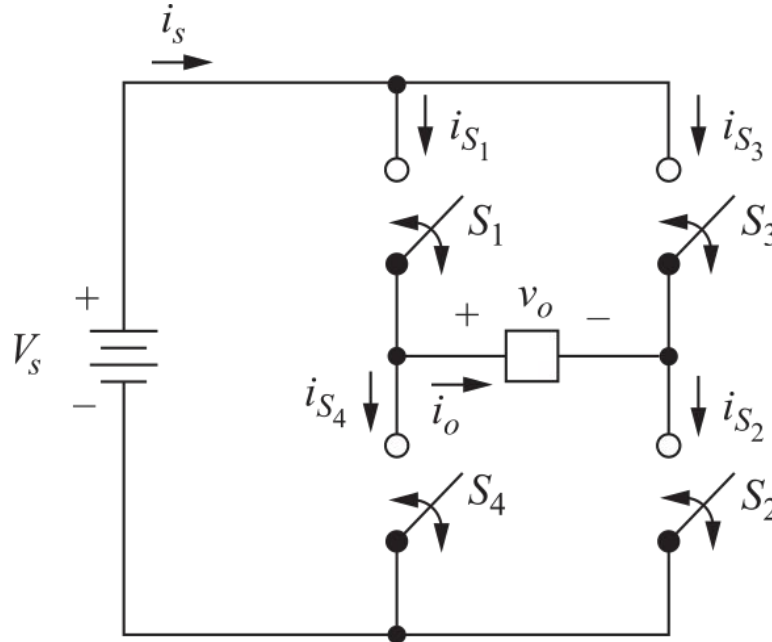
Conversión CC/CA

Inversores Puente Monofásicos:
PWM Monopulso

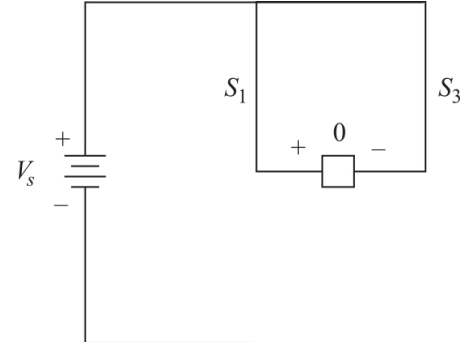
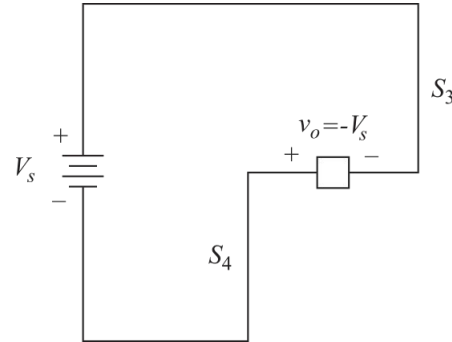
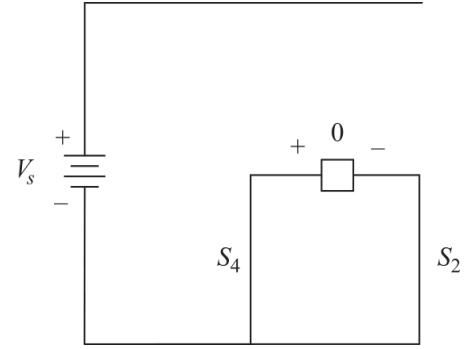
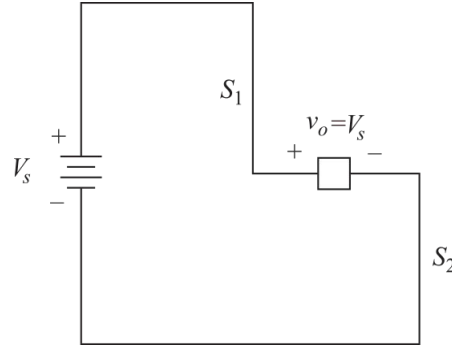
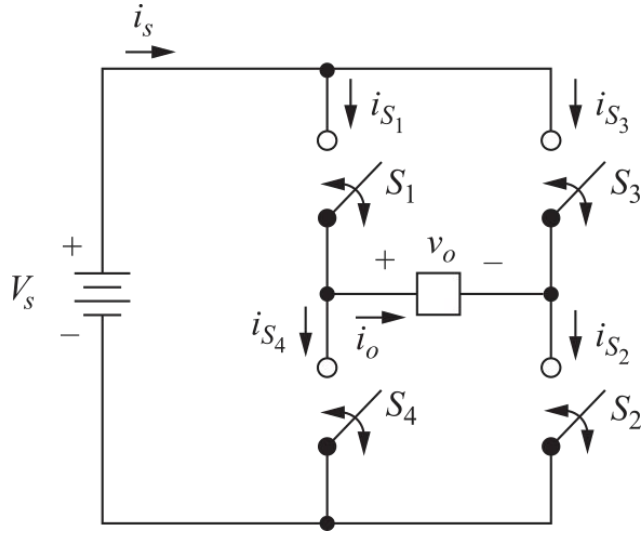


Inversor puente: esquema circuital

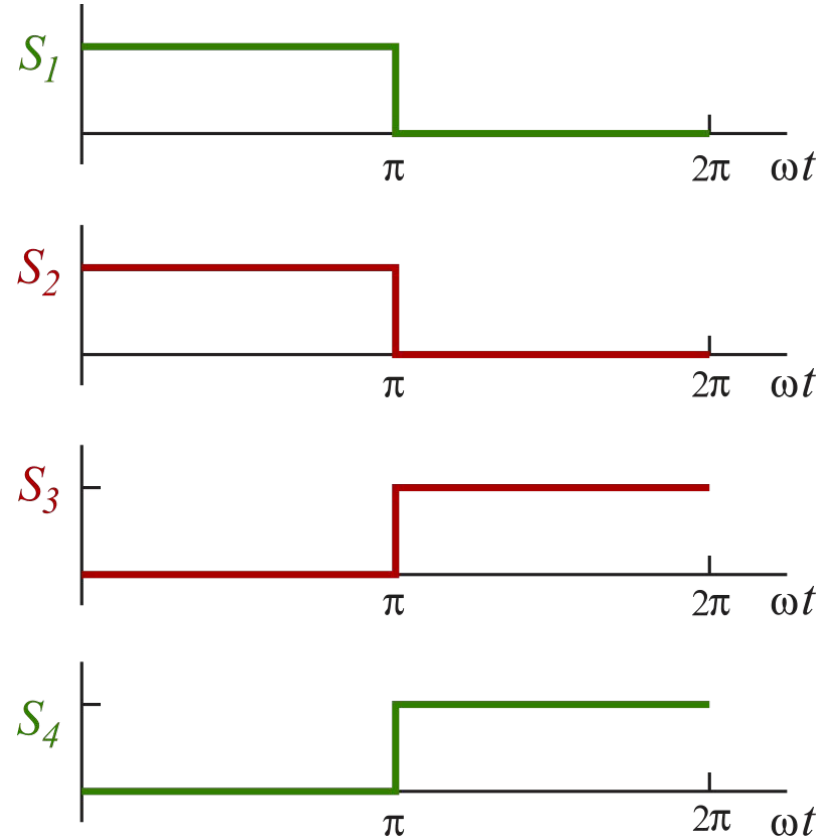
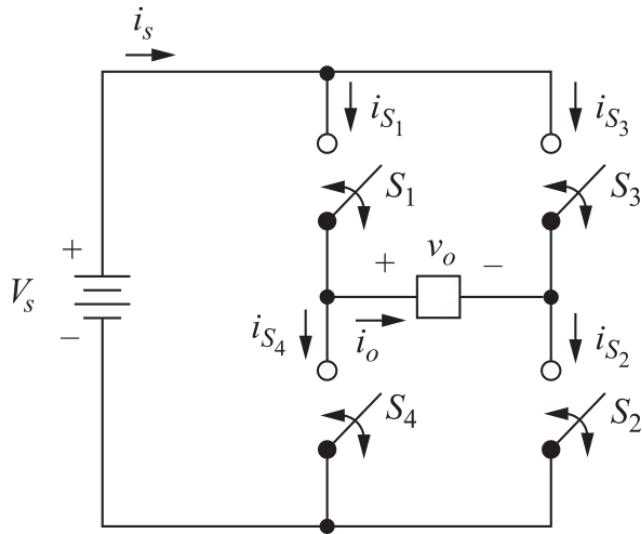
El inversor monofásico tipo puente consta de cuatro llaves dispuestas en dos “columnas”. Puede considerarse la “unión” de dos inversores semi-puente



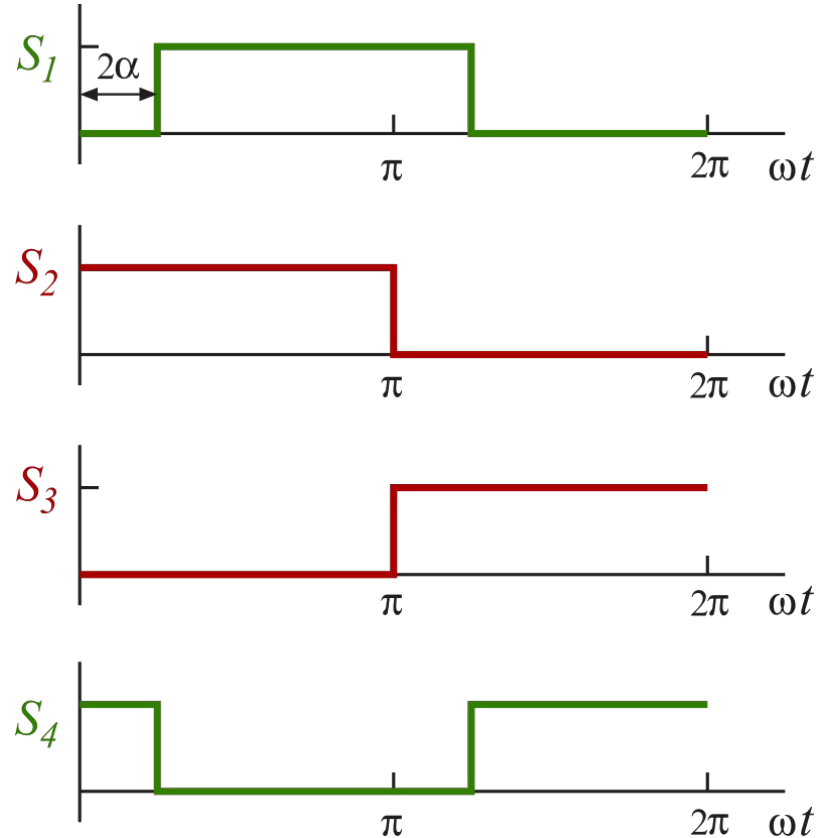
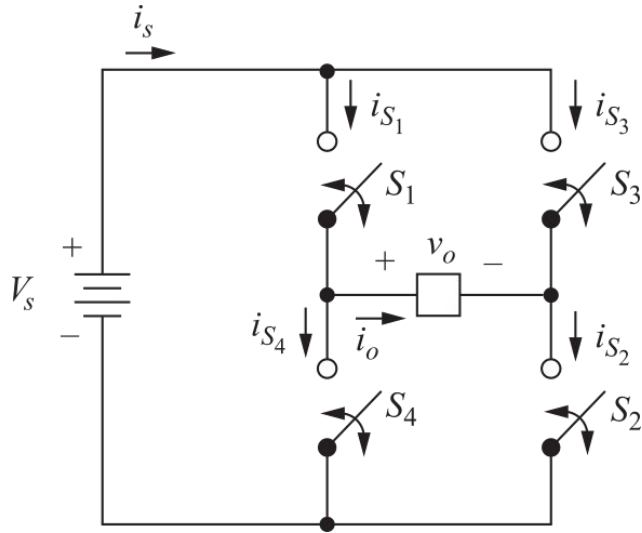
Inversor puente: estados



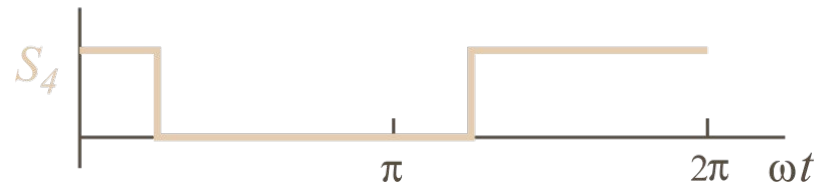
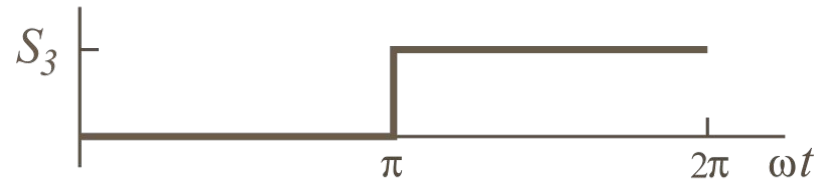
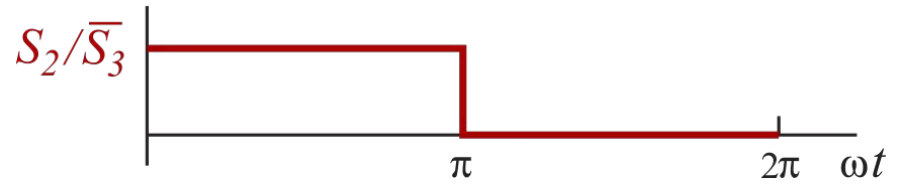
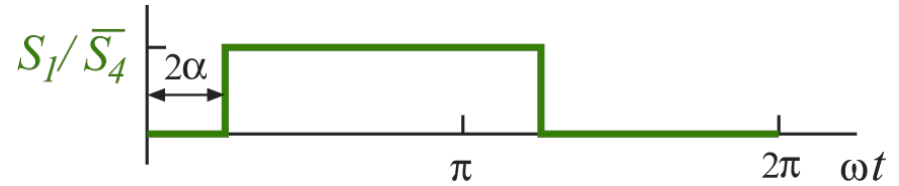
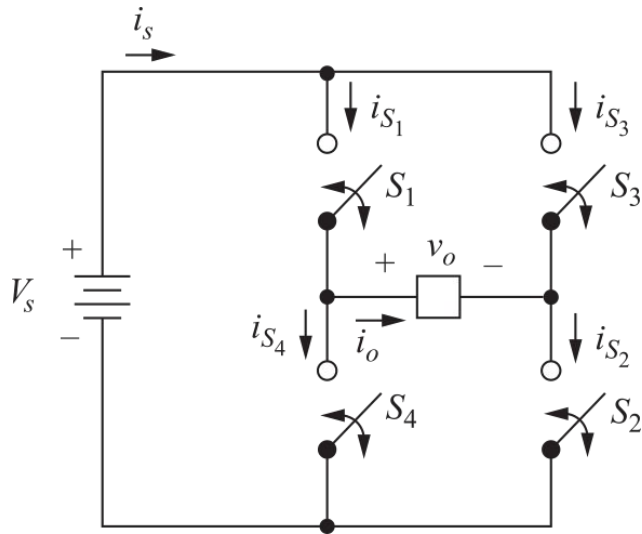
Inversor Onda Cuadrada: señales de disparo



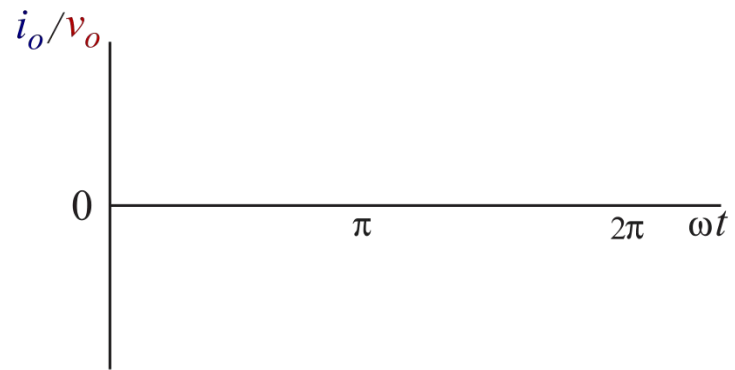
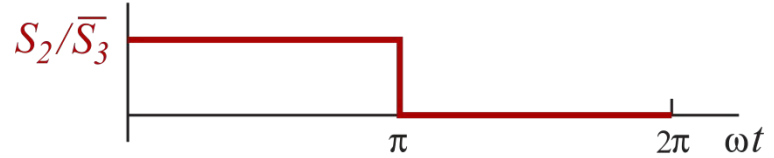
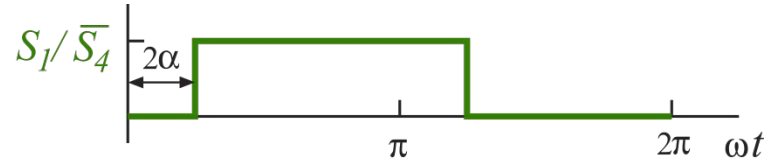
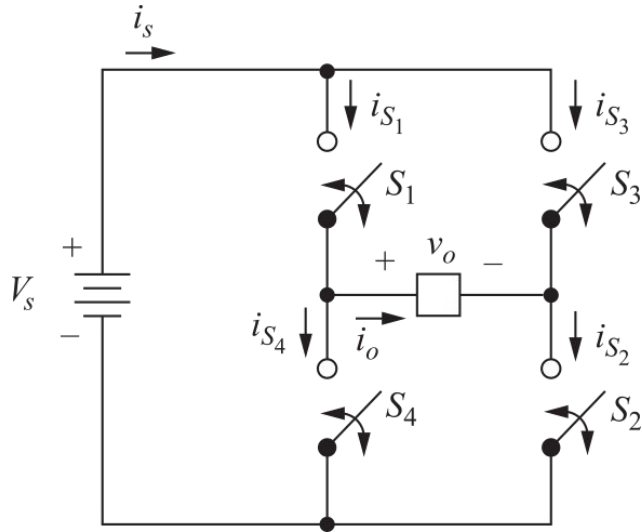
Inversor PWM Monopulso: señales de disparo



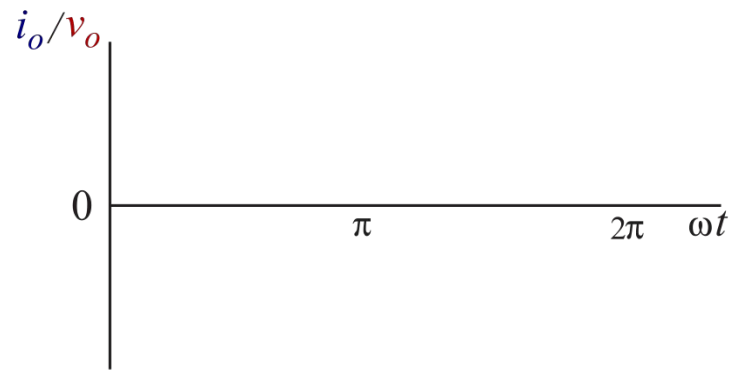
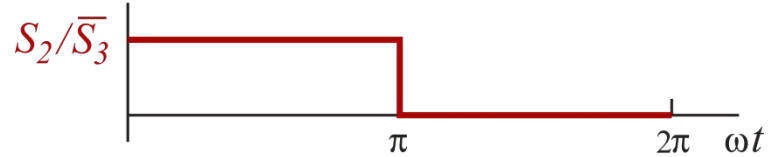
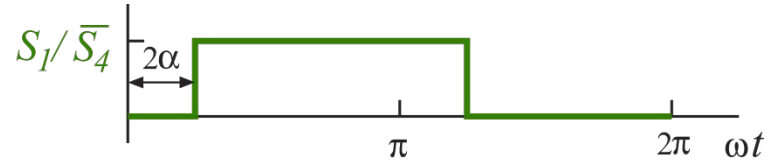
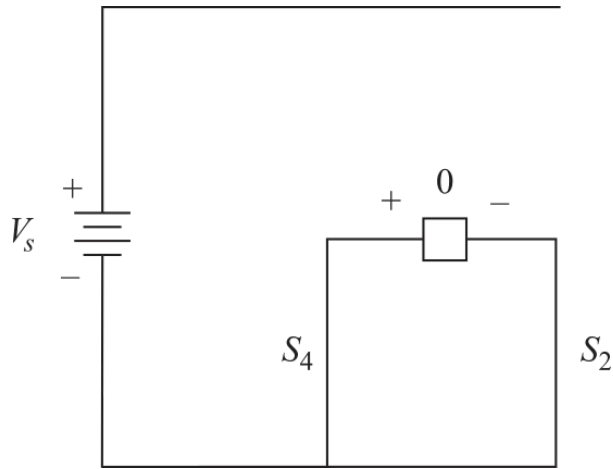
Inversor PWM Monopulso: señales de disparo



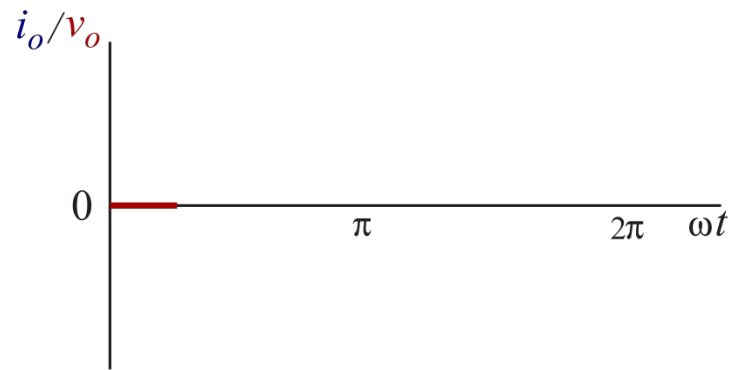
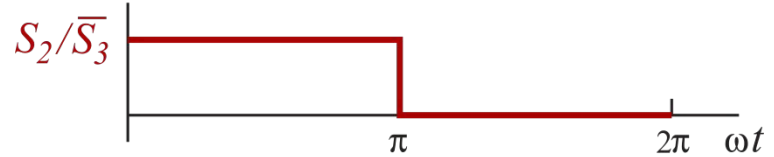
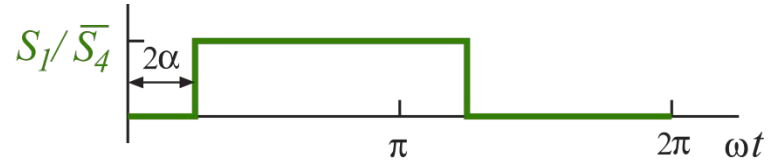
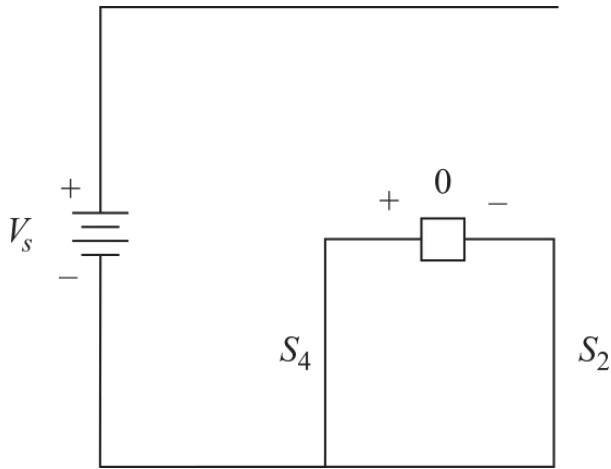
Inversor PWM Monopulso: formas de onda



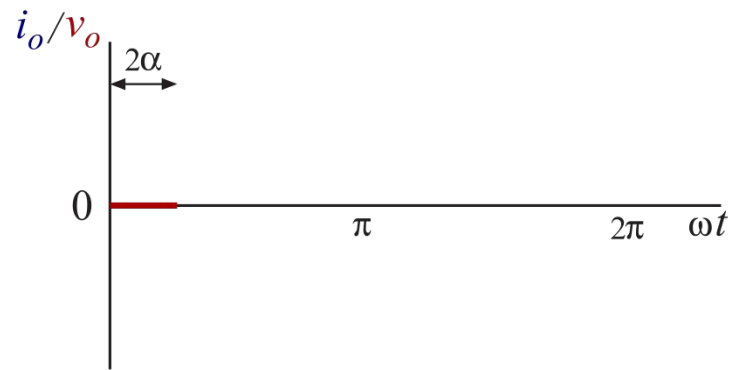
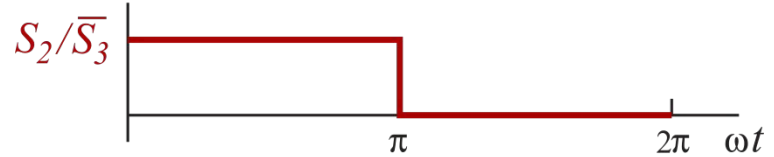
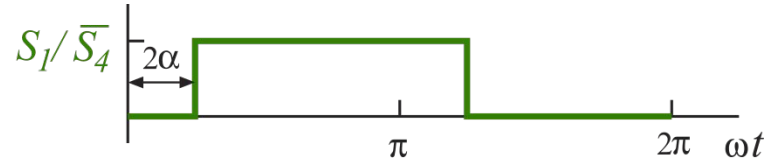
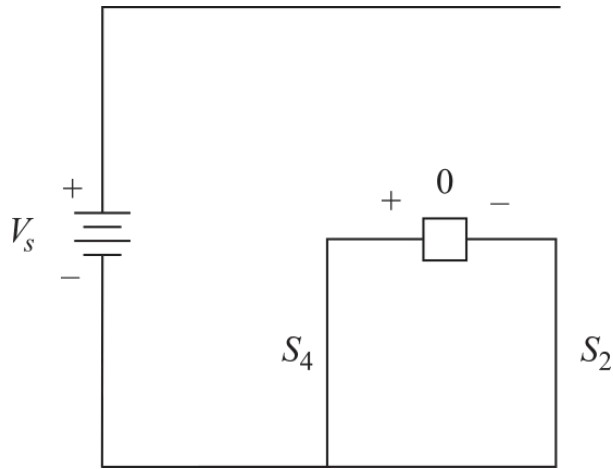
Inversor PWM Monopulso: formas de onda



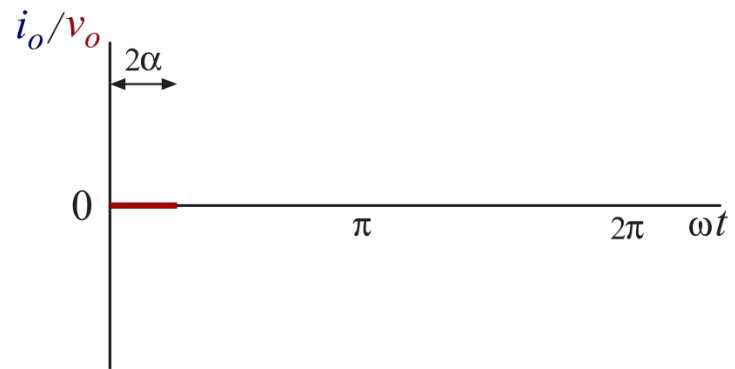
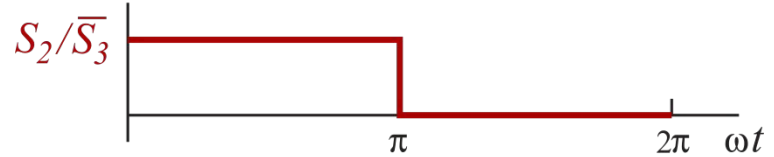
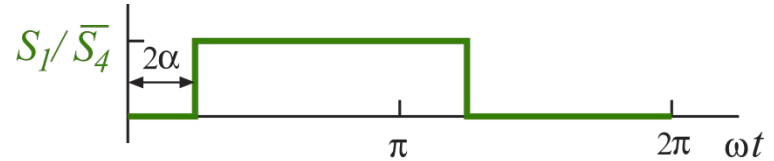
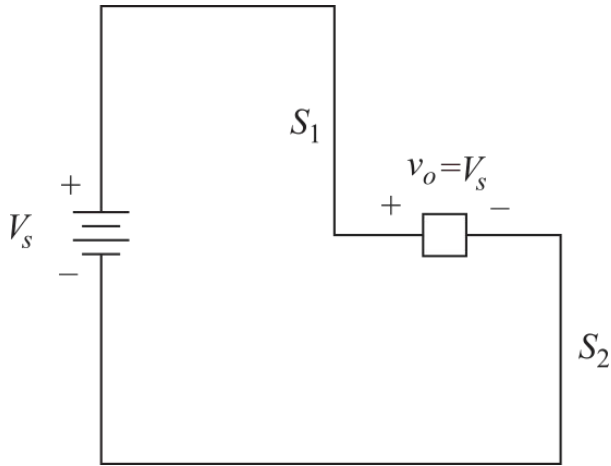
Inversor PWM Monopulso: formas de onda



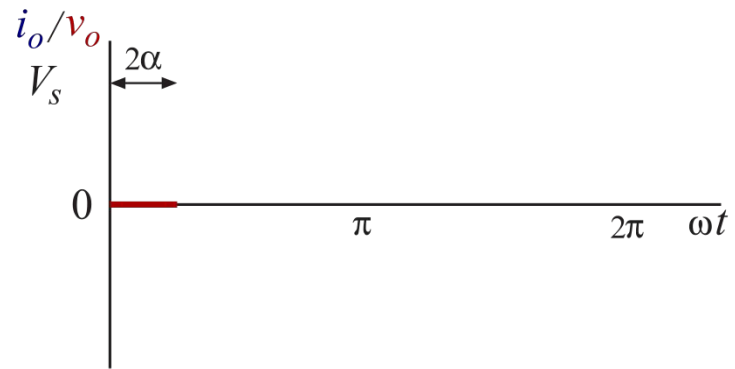
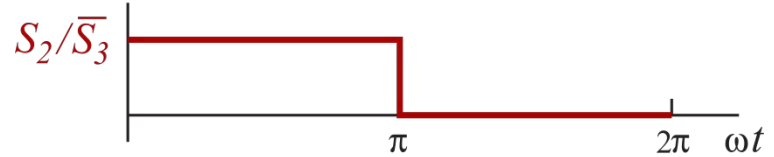
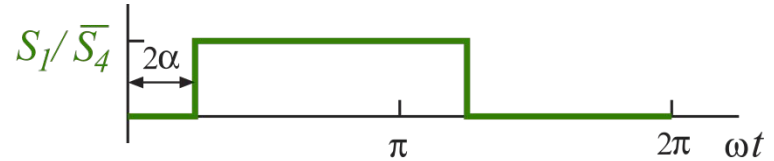
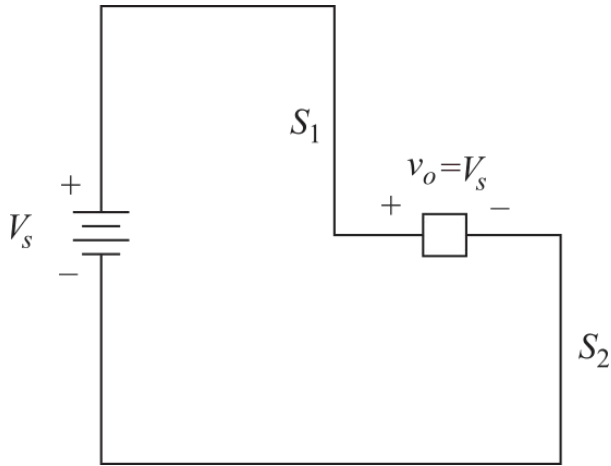
Inversor PWM Monopulso: formas de onda



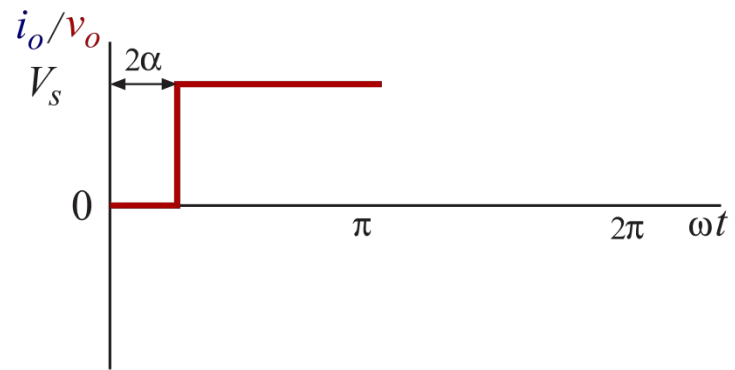
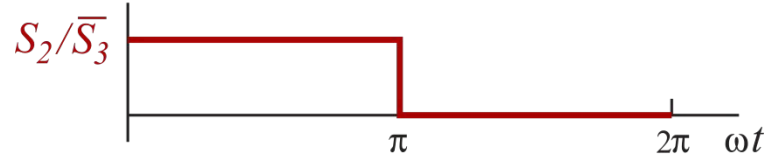
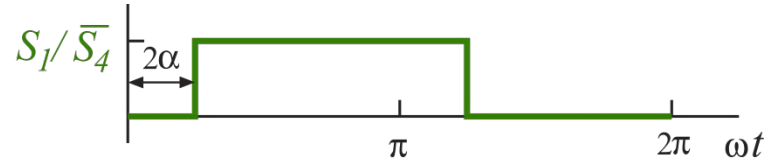
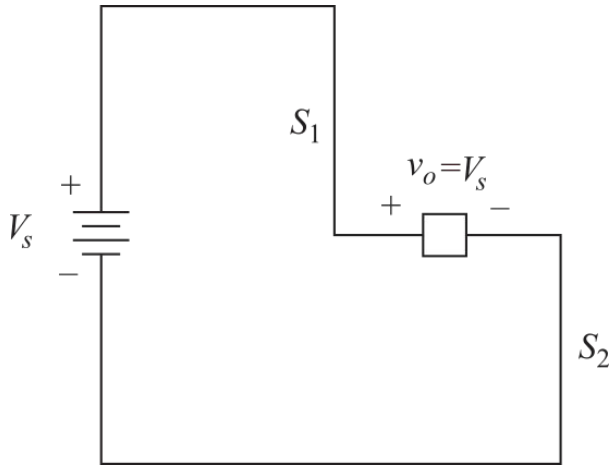
Inversor PWM Monopulso: formas de onda



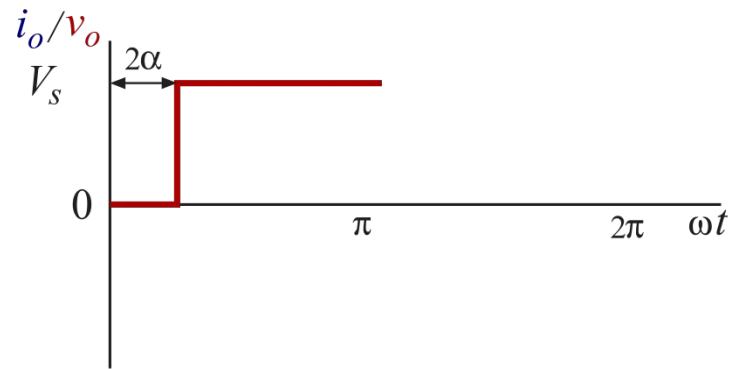
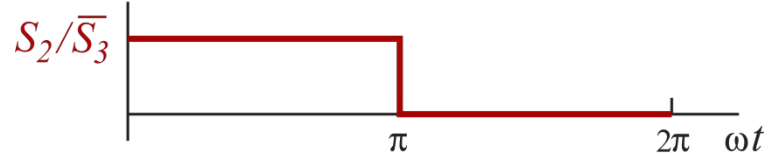
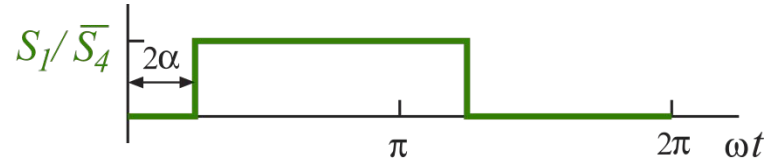
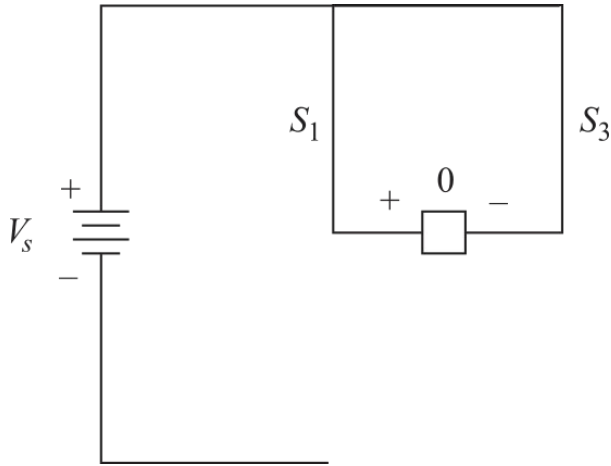
Inversor PWM Monopulso: formas de onda



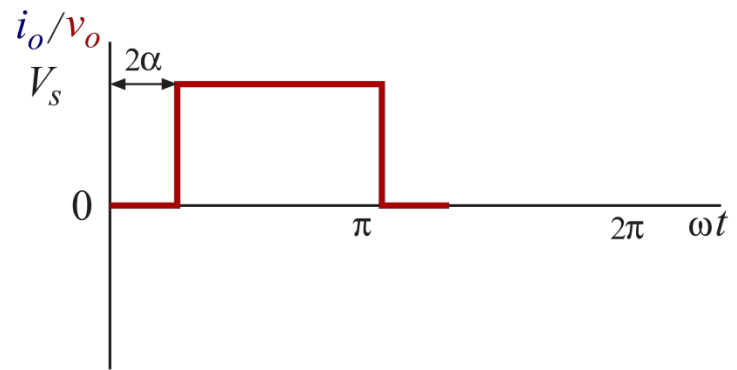
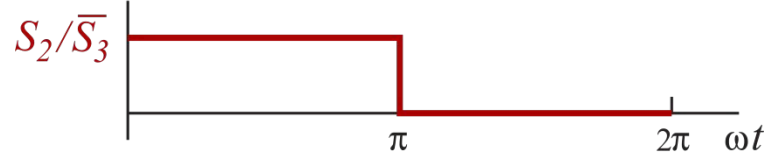
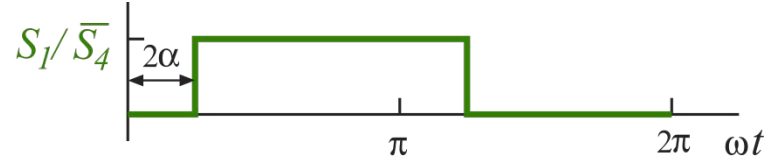
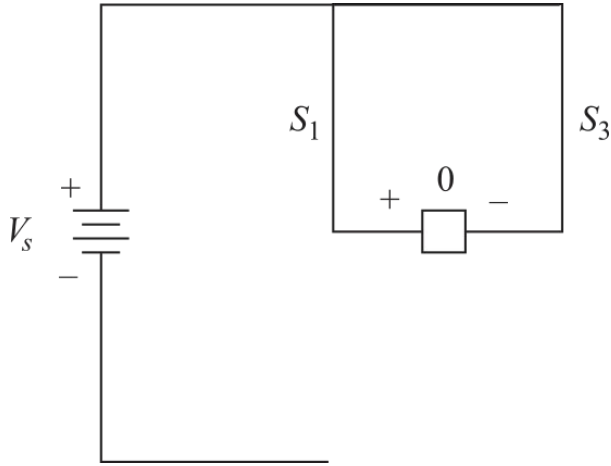
Inversor PWM Monopulso: formas de onda



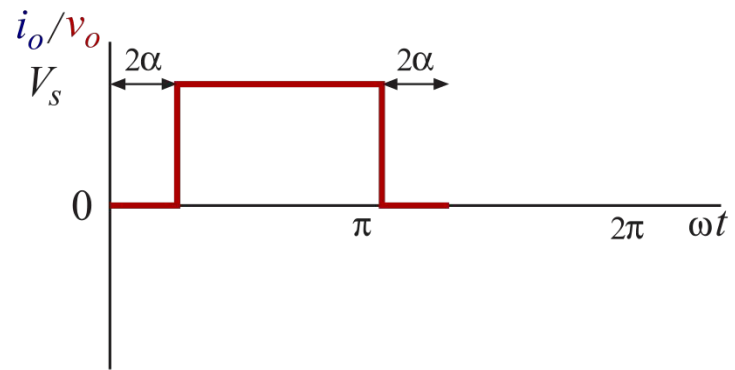
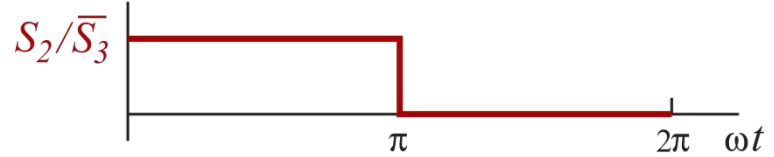
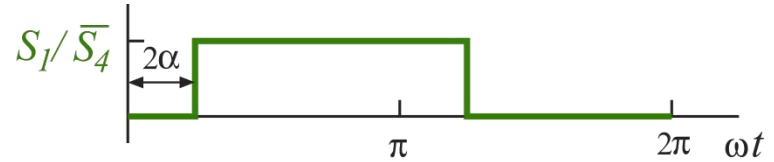
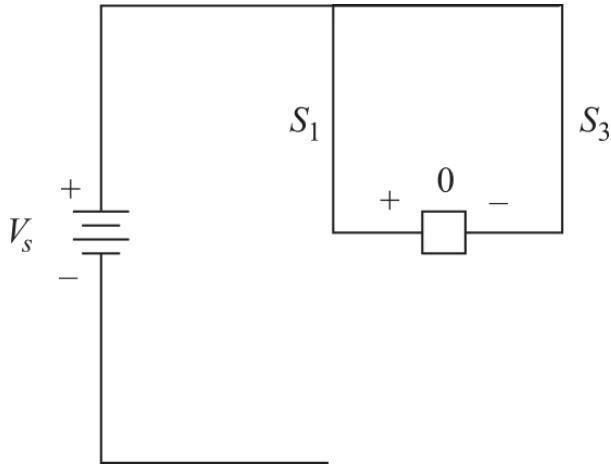
Inversor PWM Monopulso: formas de onda



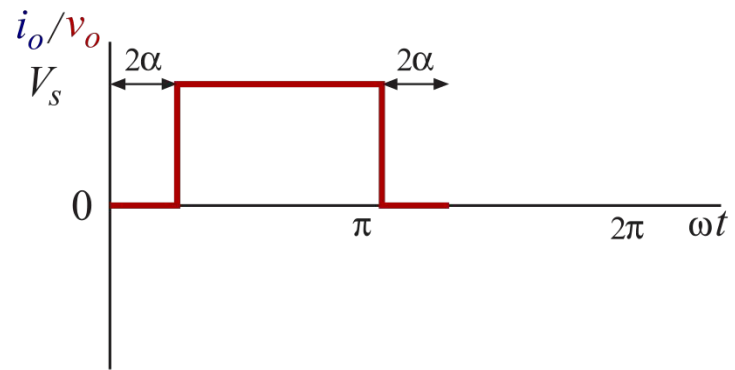
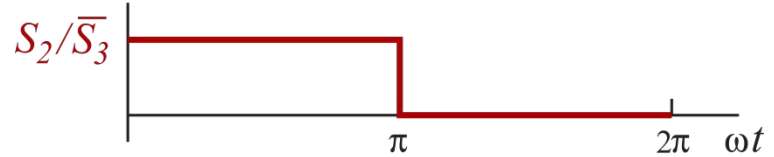
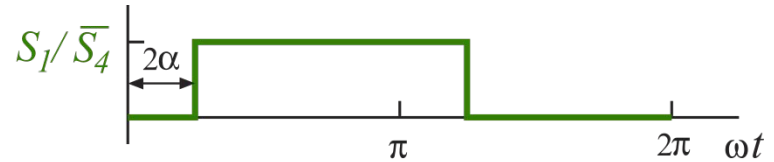
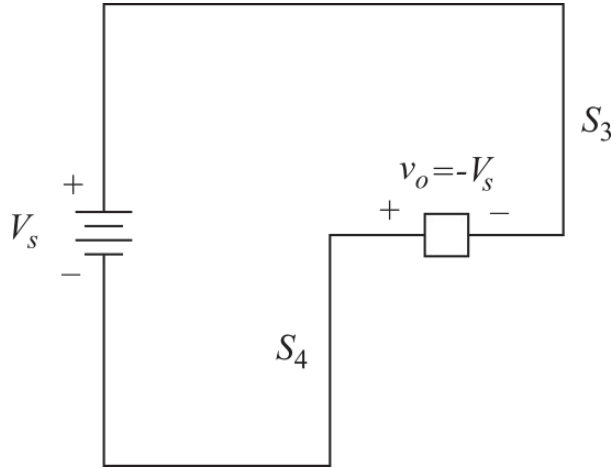
Inversor PWM Monopulso: formas de onda



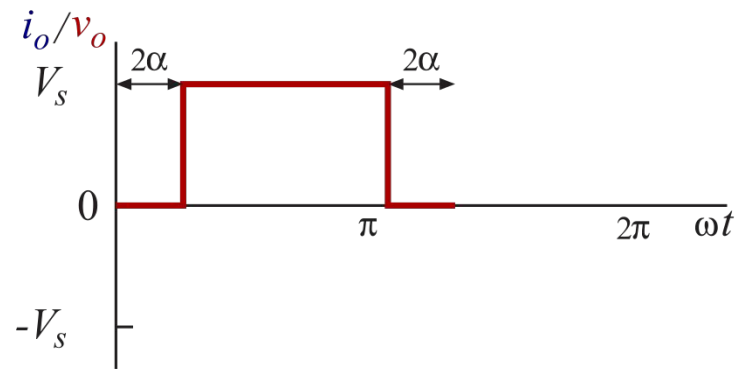
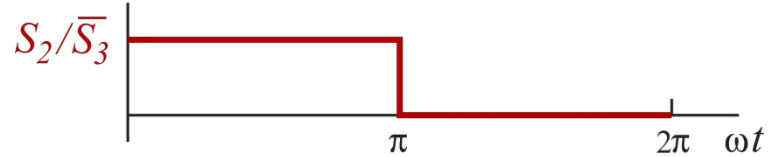
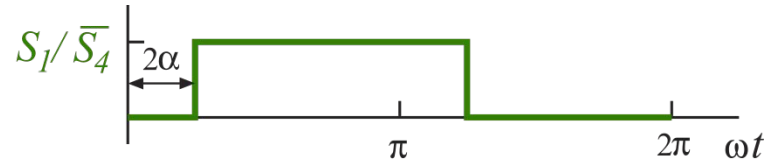
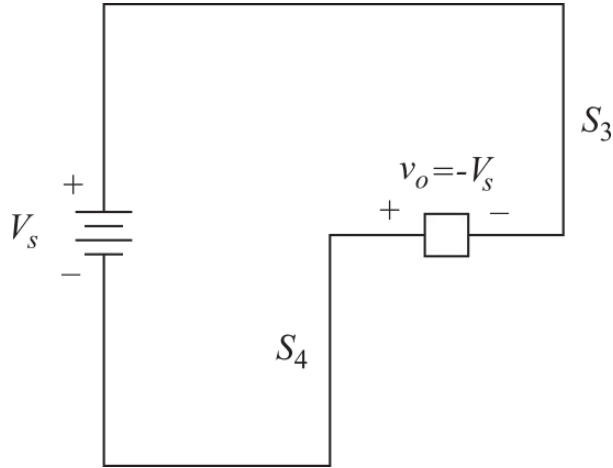
Inversor PWM Monopulso: formas de onda



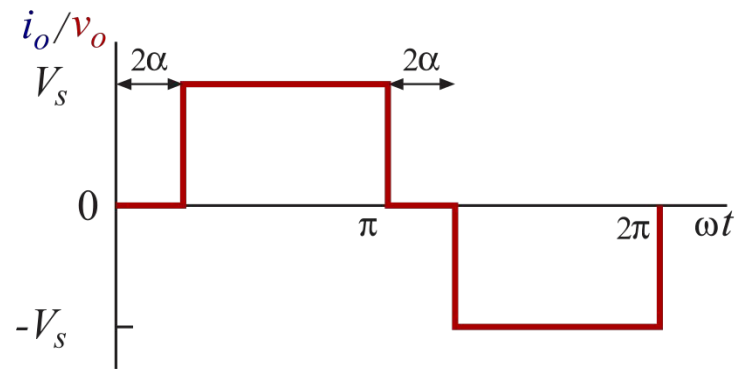
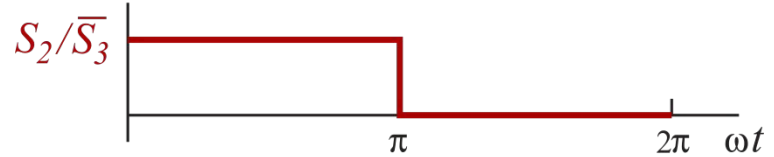
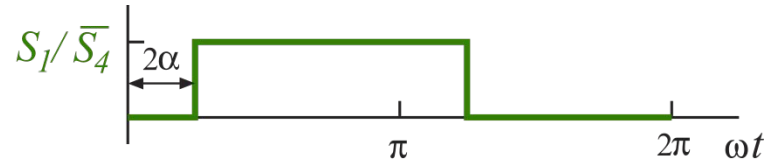
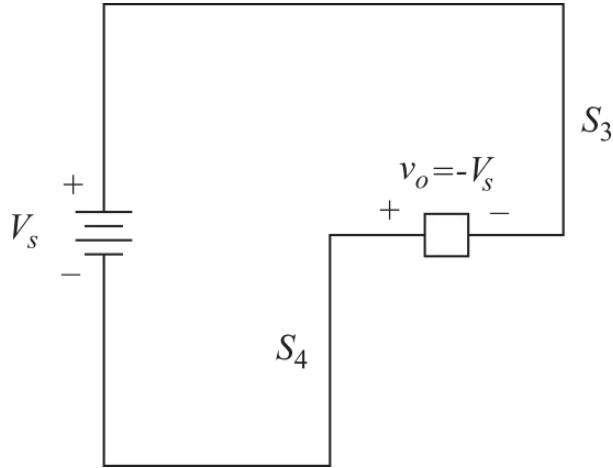
Inversor PWM Monopulso: formas de onda



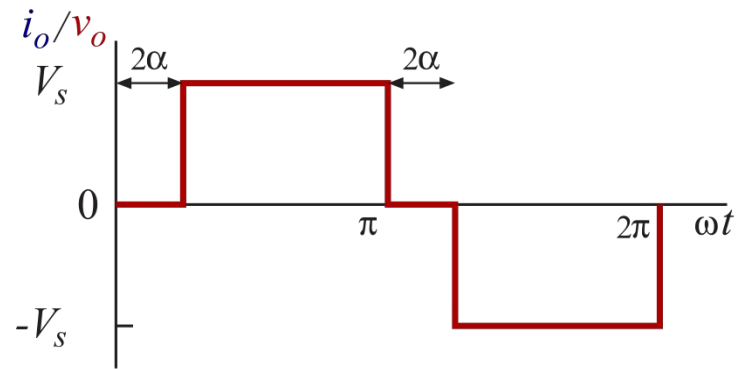
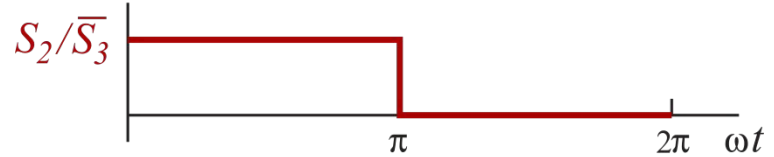
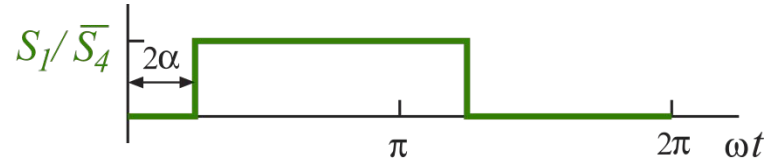
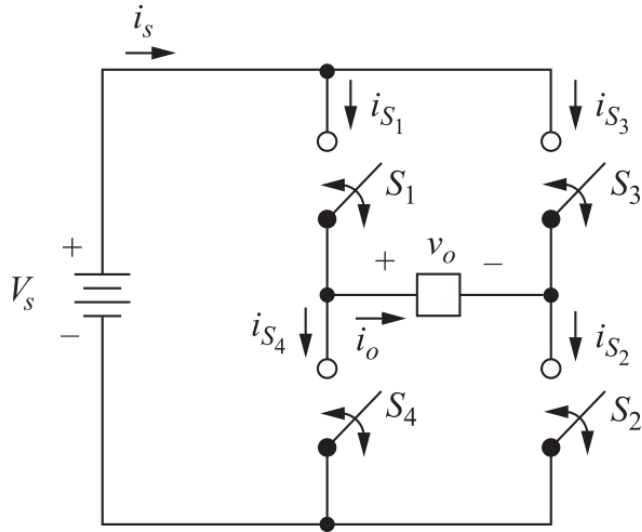
Inversor PWM Monopulso: formas de onda



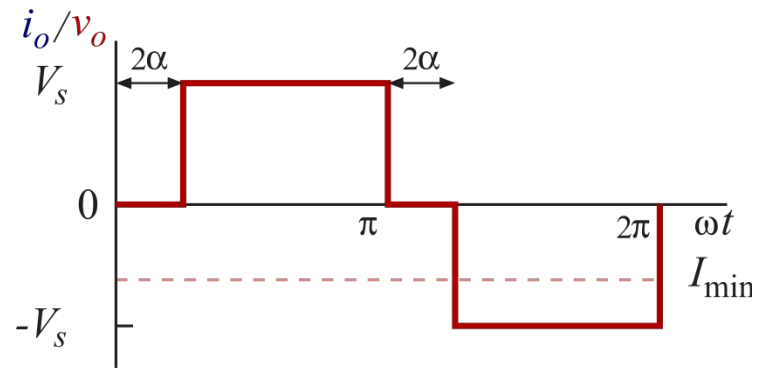
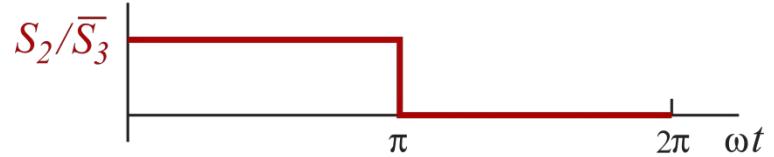
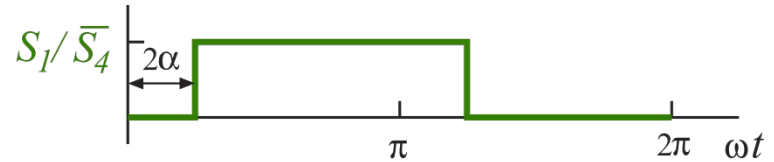
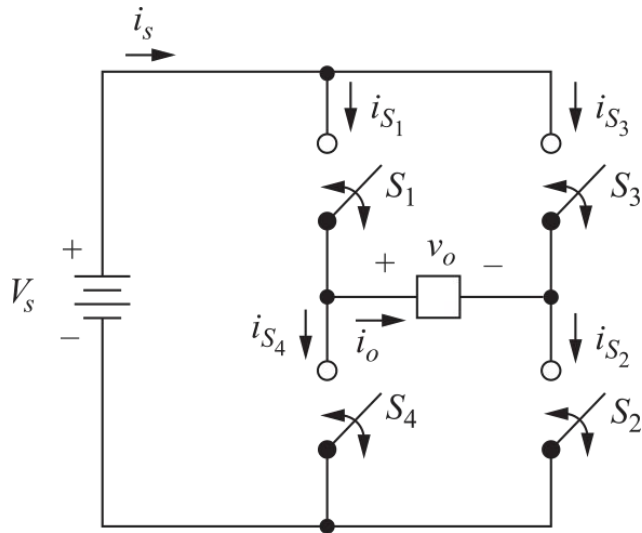
Inversor PWM Monopulso: formas de onda



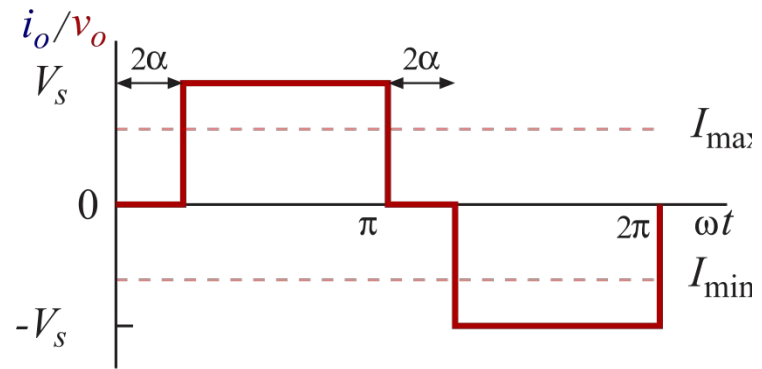
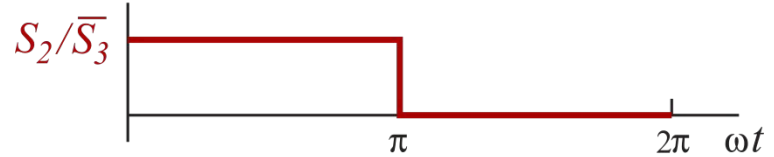
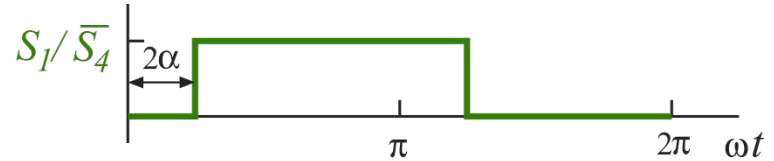
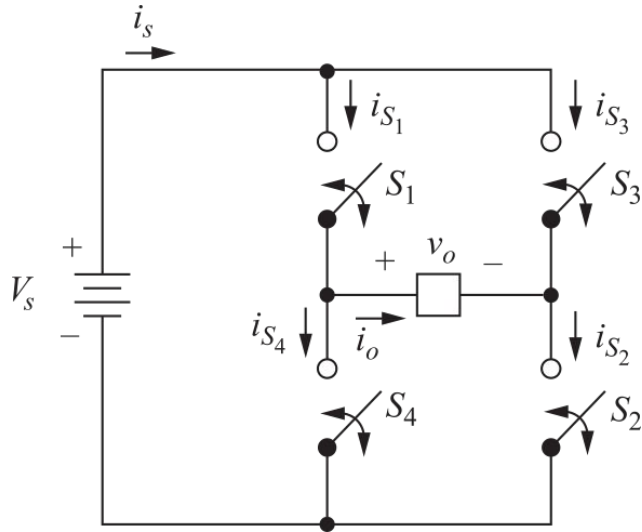
Inversor PWM Monopulso: formas de onda



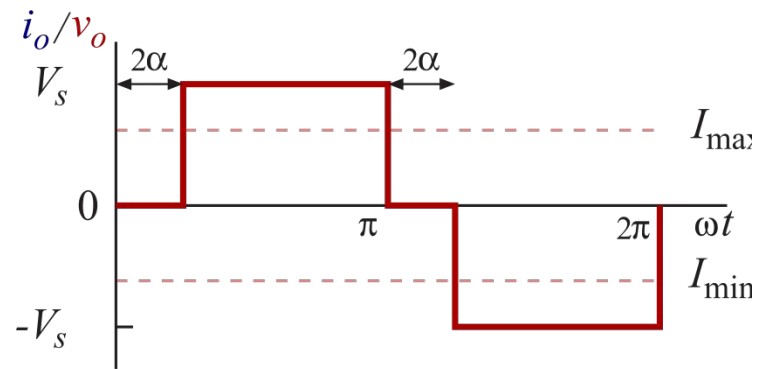
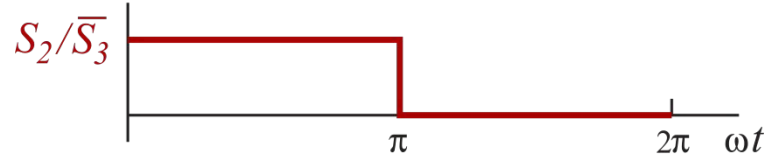
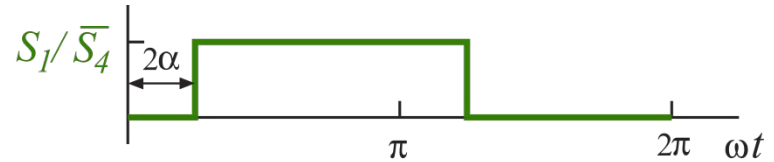
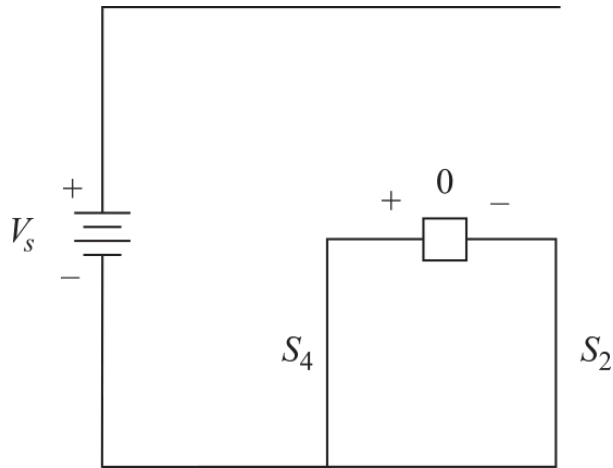
Inversor PWM Monopulso: formas de onda



Inversor PWM Monopulso: formas de onda



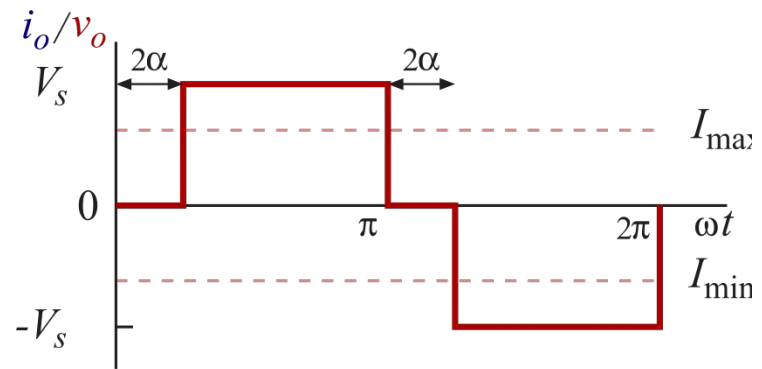
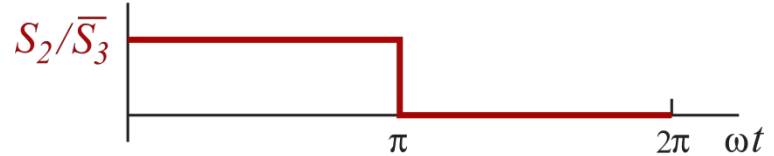
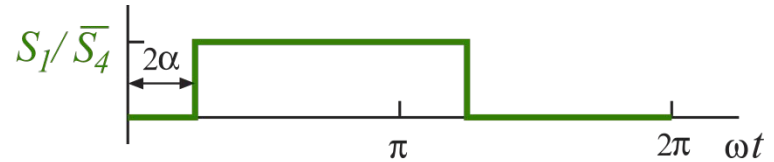
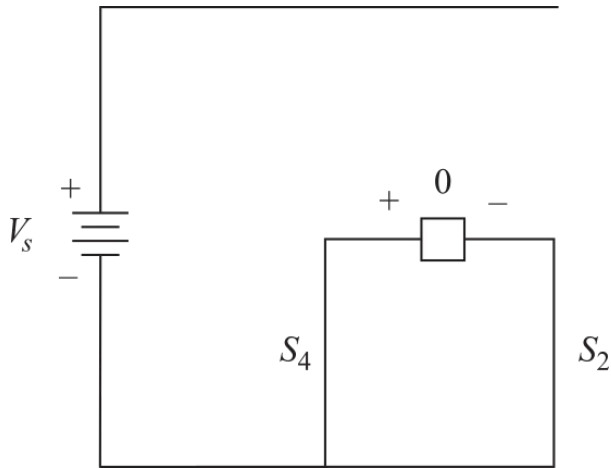
Inversor PWM Monopulso: formas de onda



Inversor PWM Monopulso: formas de onda

$$i_o(\omega t) = I_{min} e^{-\omega t / \phi}$$

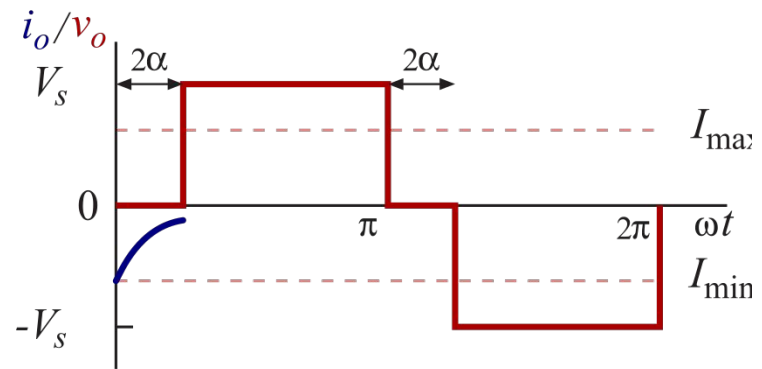
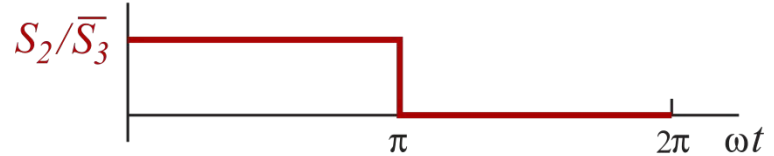
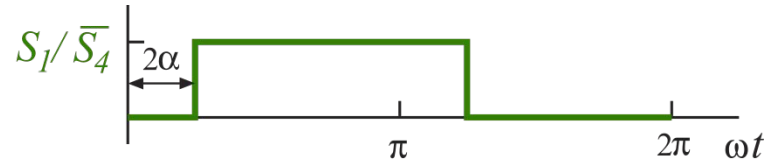
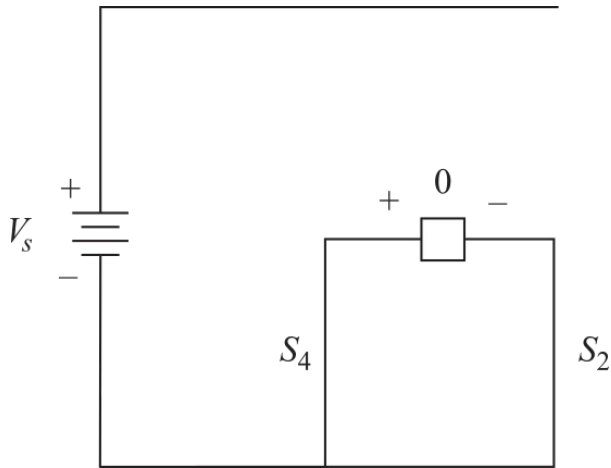
$$0 \leq \omega t \leq 2\alpha$$



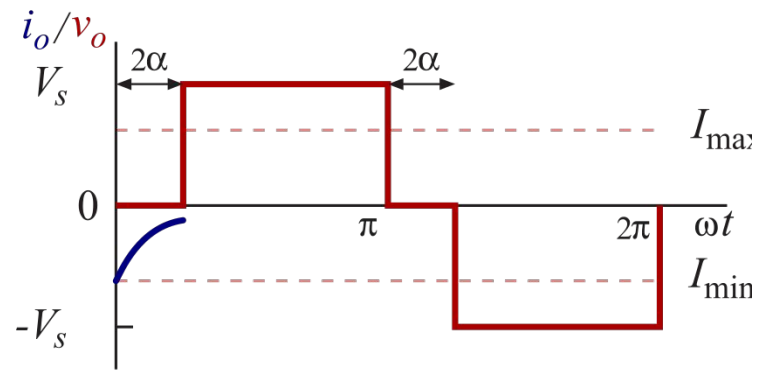
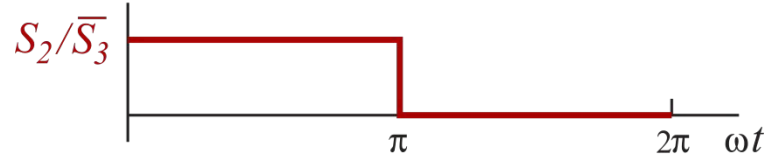
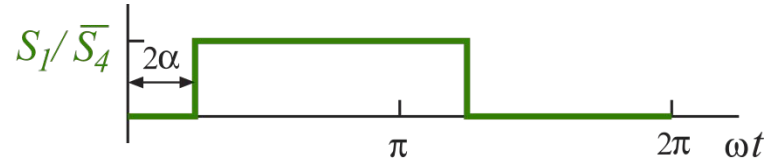
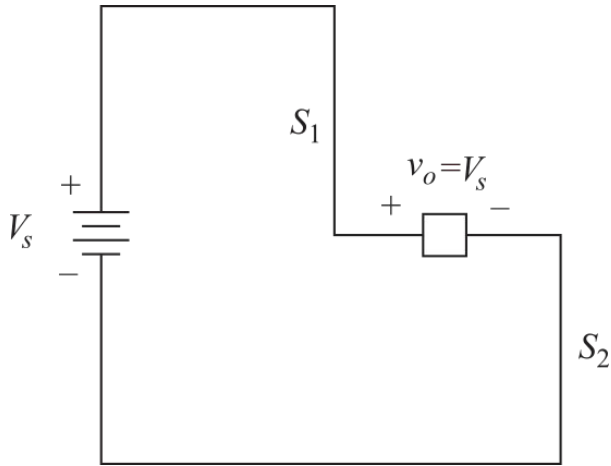
Inversor PWM Monopulso: formas de onda

$$i_o(\omega t) = I_{min} e^{-\omega t / \phi}$$

$$0 \leq \omega t \leq 2\alpha$$



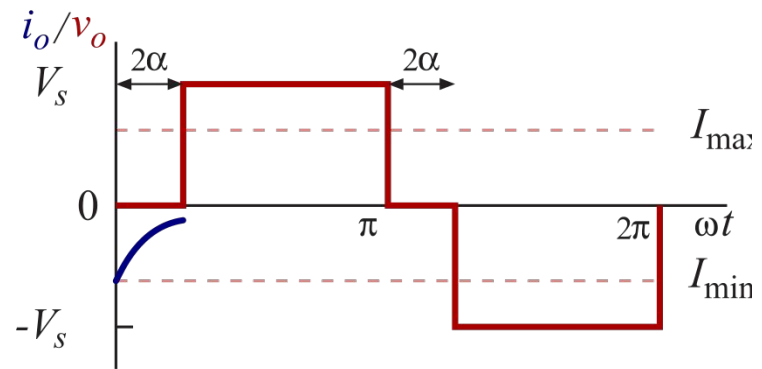
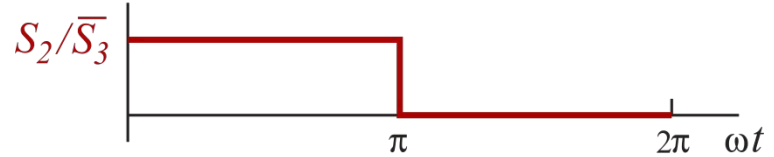
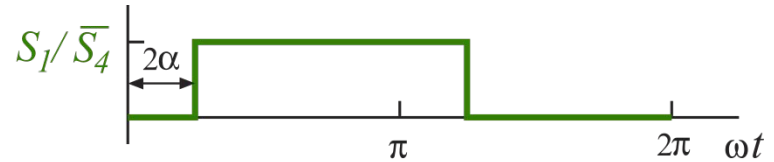
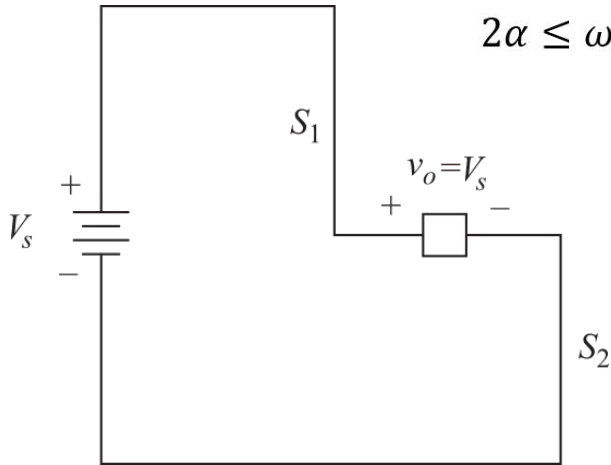
Inversor PWM Monopulso: formas de onda



Inversor PWM Monopulso: formas de onda

$$i_0(\omega t) = \frac{V_s}{R} + Ae^{-(\omega t - 2\alpha)/\phi}$$

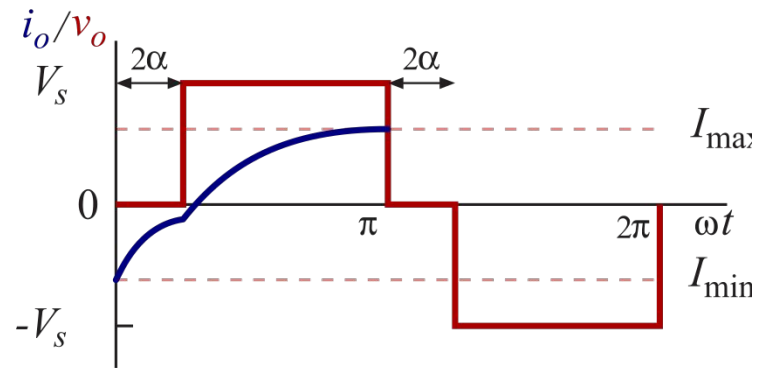
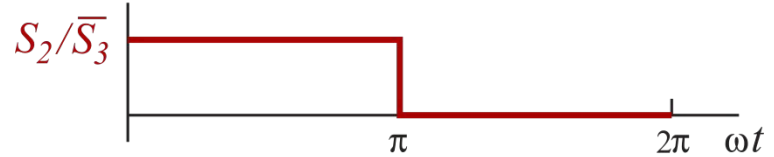
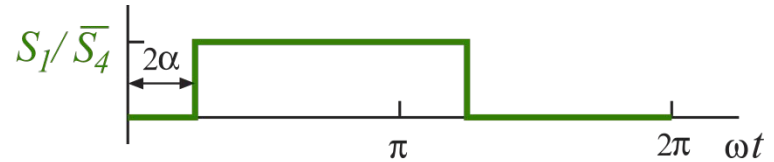
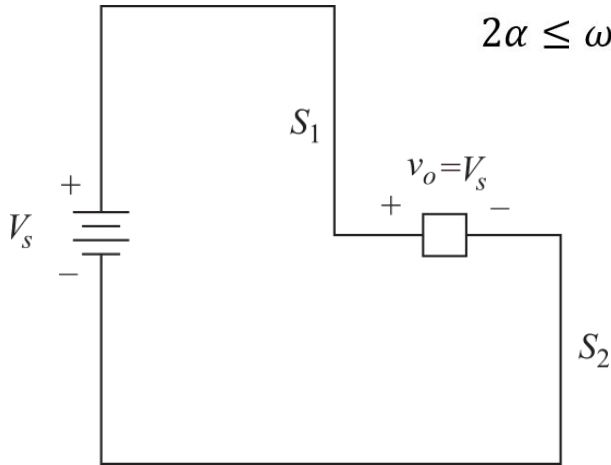
$$2\alpha \leq \omega t \leq \pi$$



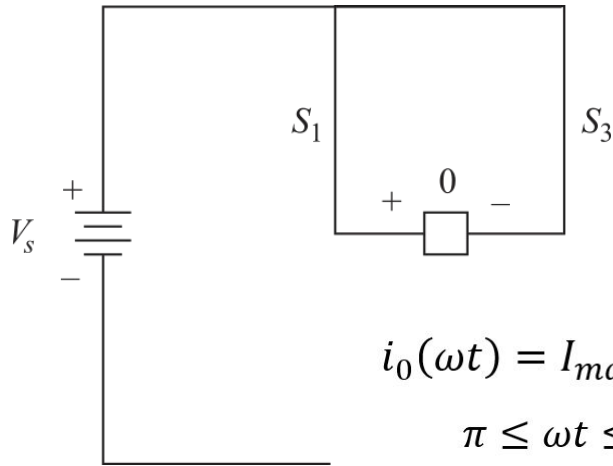
Inversor PWM Monopulso: formas de onda

$$i_o(\omega t) = \frac{V_s}{R} + Ae^{-(\omega t - 2\alpha)/\phi}$$

$$2\alpha \leq \omega t \leq \pi$$

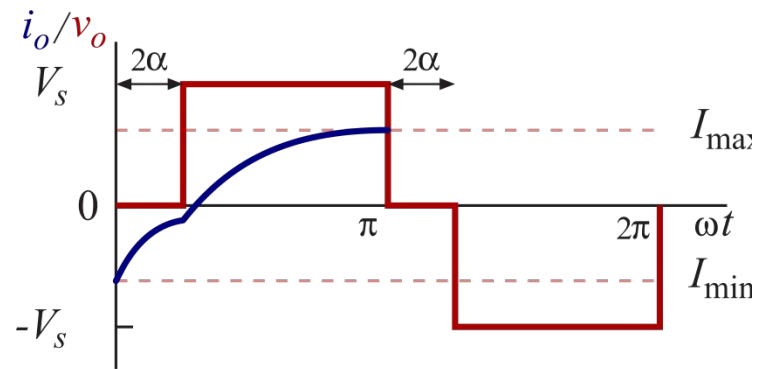
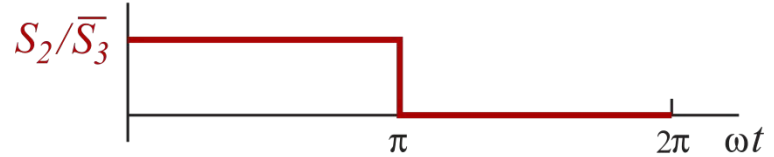
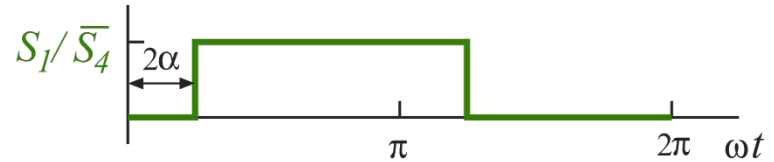


Inversor PWM Monopulso: formas de onda

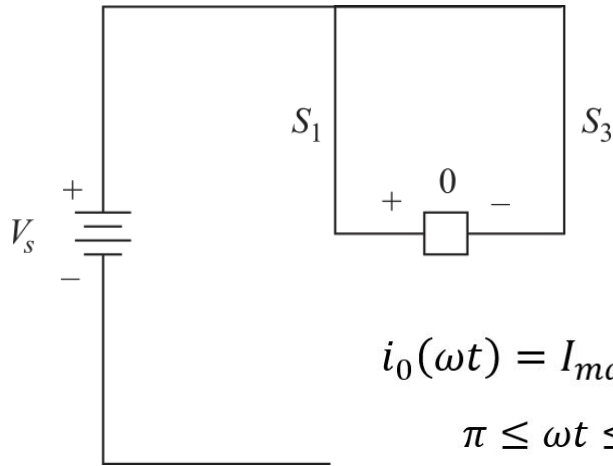


$$i_o(\omega t) = I_{max} e^{-(\omega t - \pi)/\phi}$$

$$\pi \leq \omega t \leq \pi + 2\alpha$$

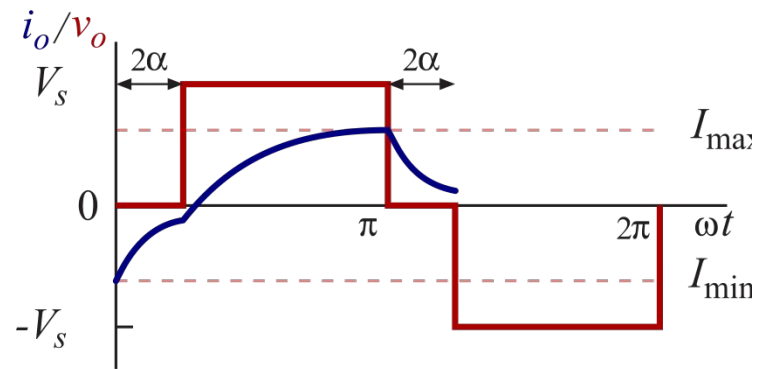
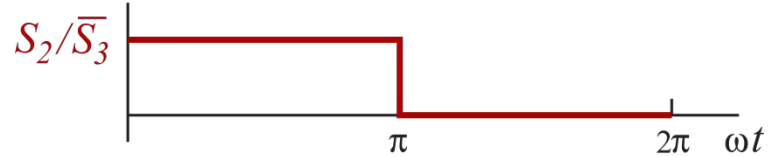
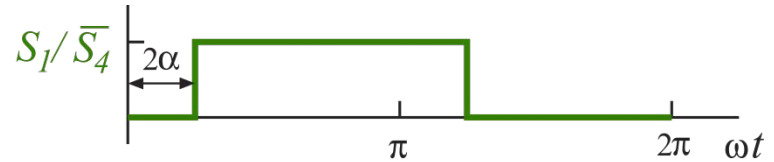


Inversor PWM Monopulso: formas de onda

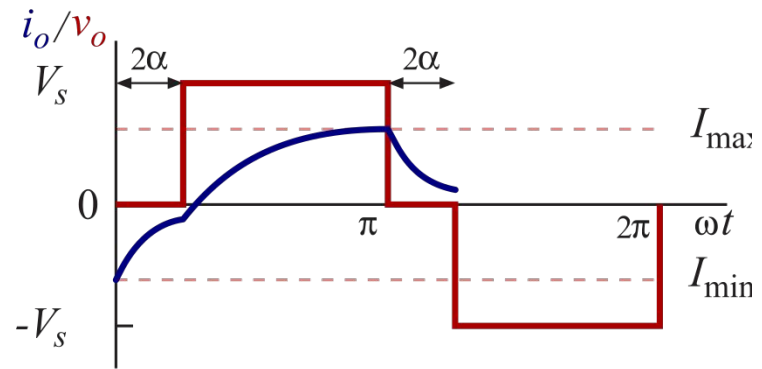
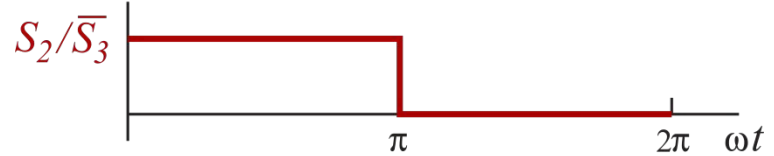
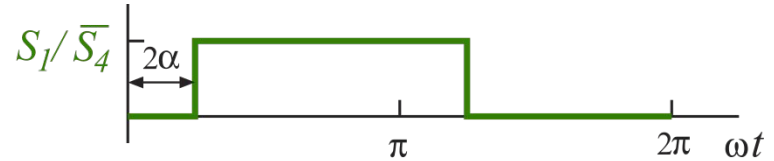
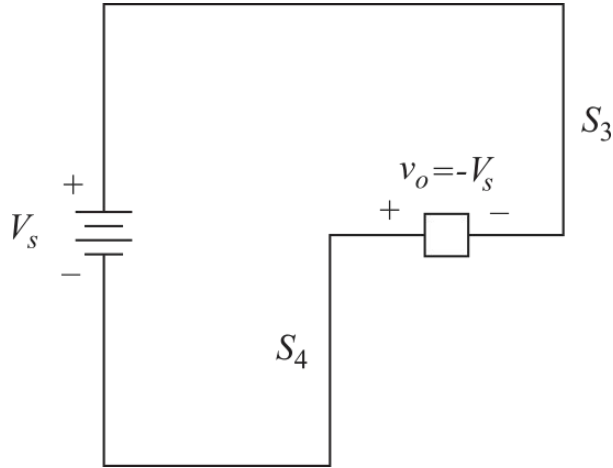


$$i_o(\omega t) = I_{max} e^{-(\omega t - \pi)/\phi}$$

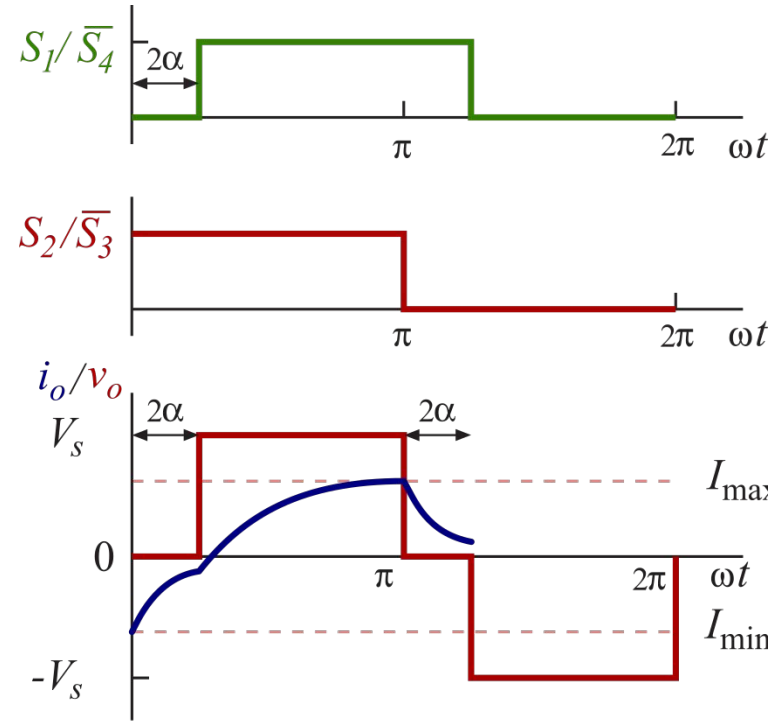
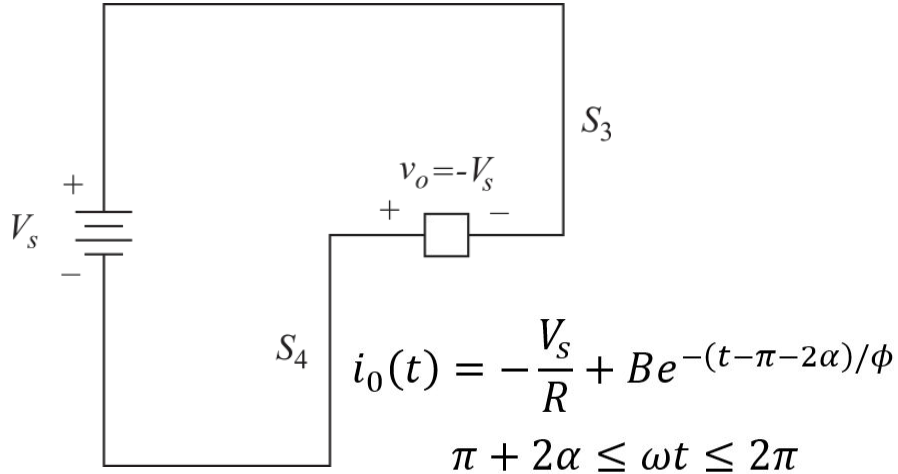
$$\pi \leq \omega t \leq \pi + 2\alpha$$



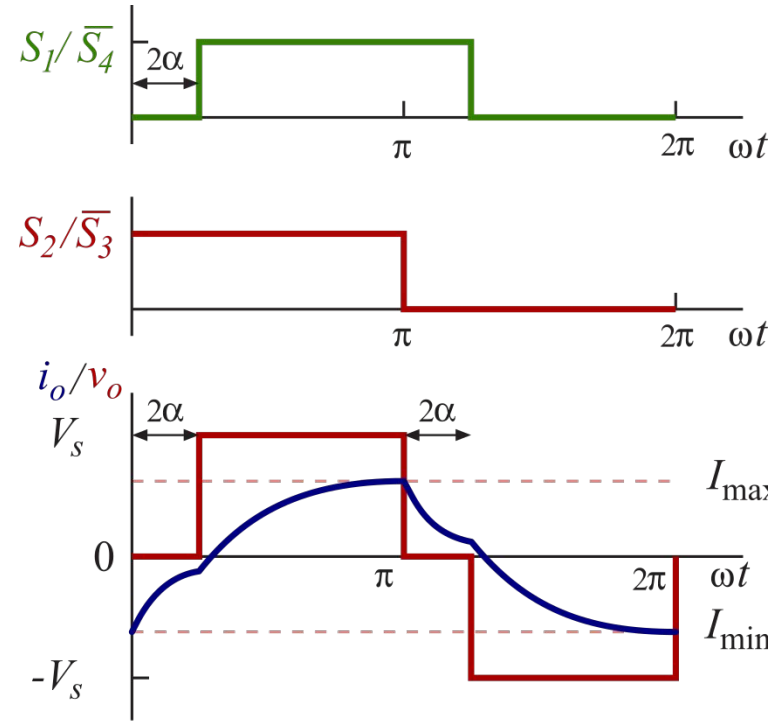
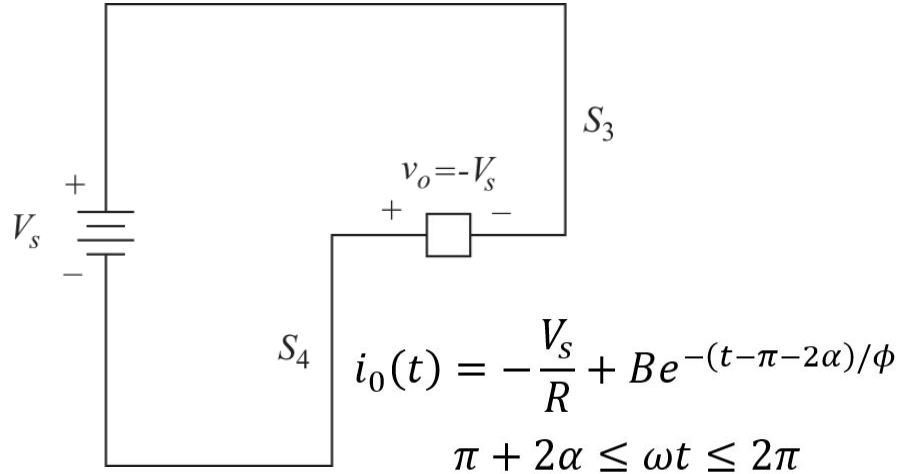
Inversor PWM Monopulso: formas de onda



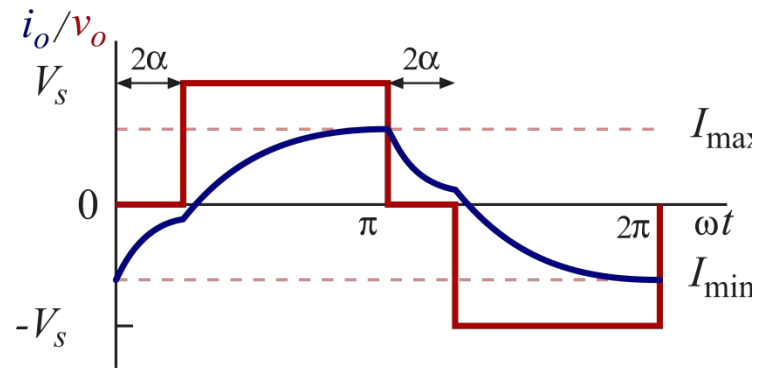
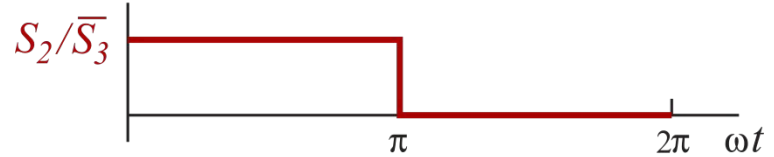
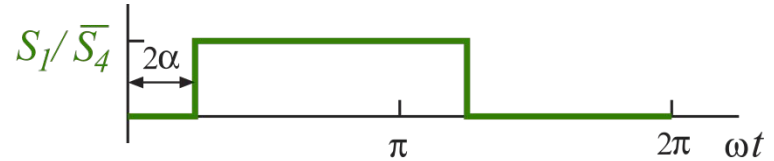
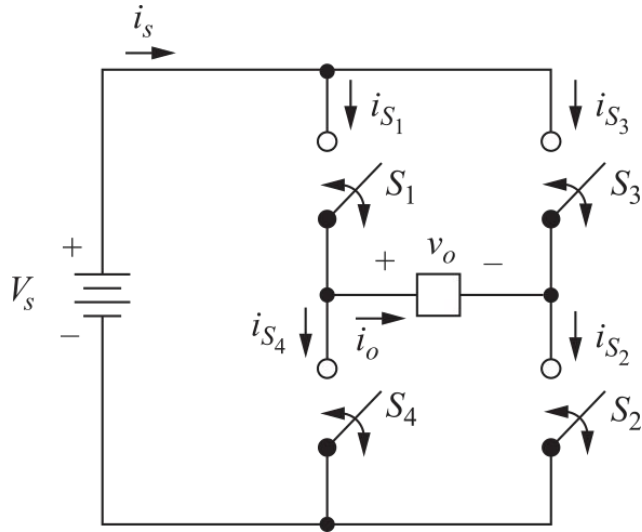
Inversor PWM Monopulso: formas de onda



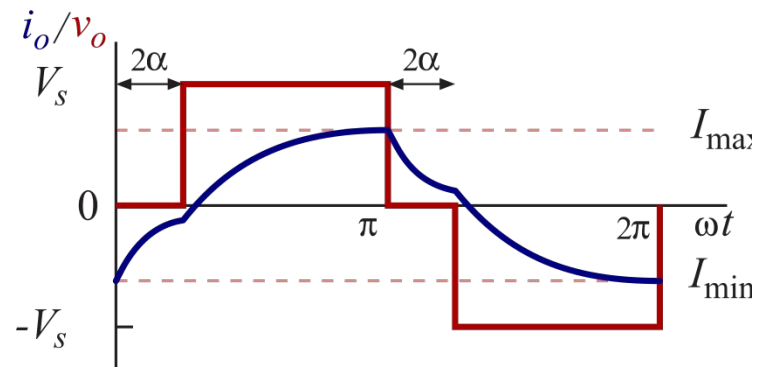
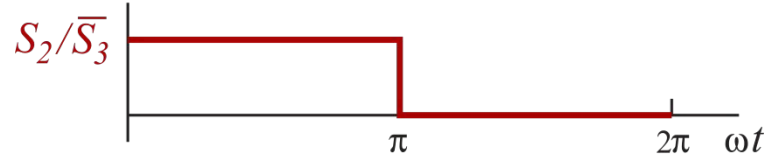
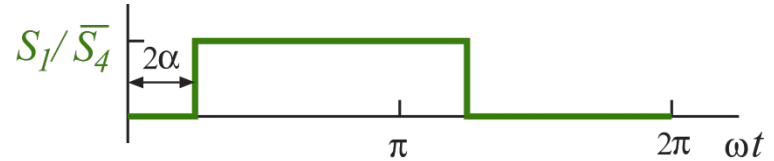
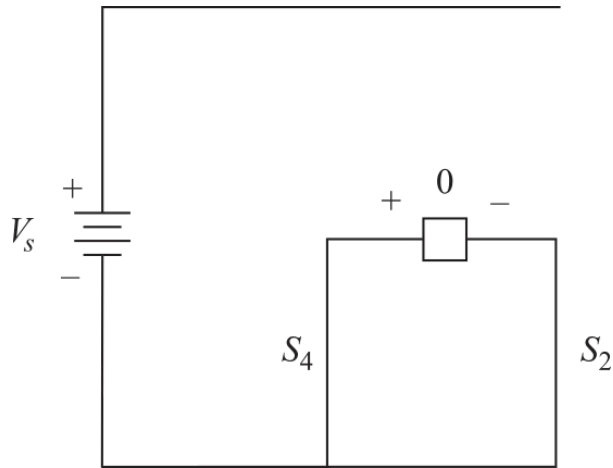
Inversor PWM Monopulso: formas de onda



Inversor PWM Monopulso: formas de onda

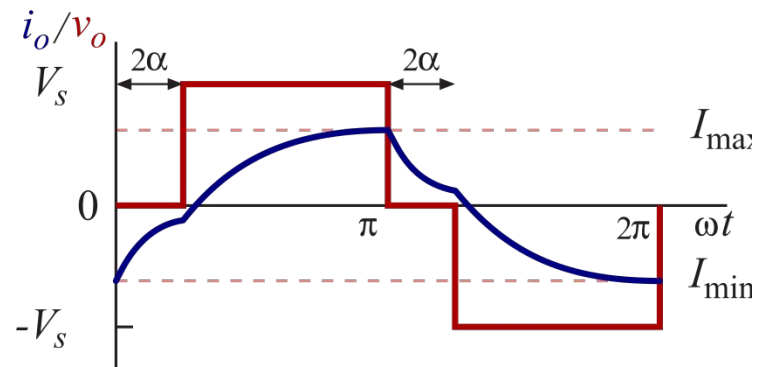
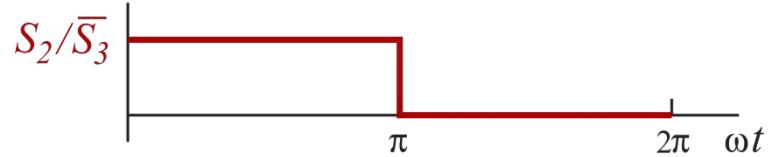
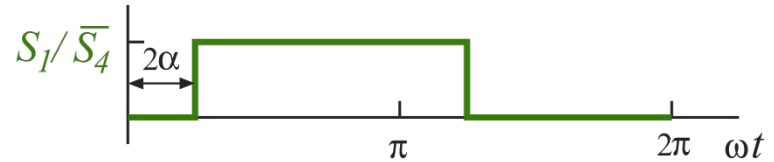
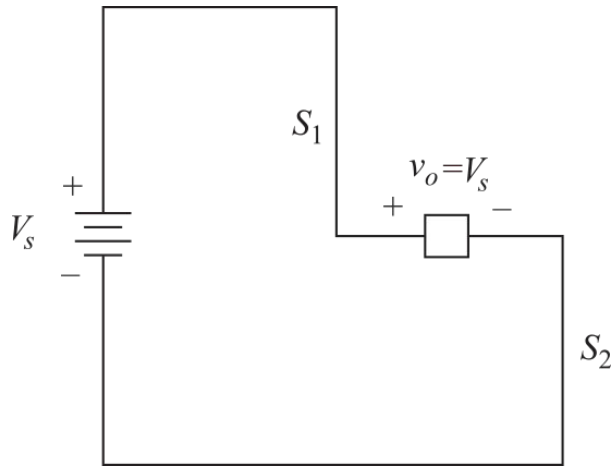


Inversor PWM Monopulso: formas de onda



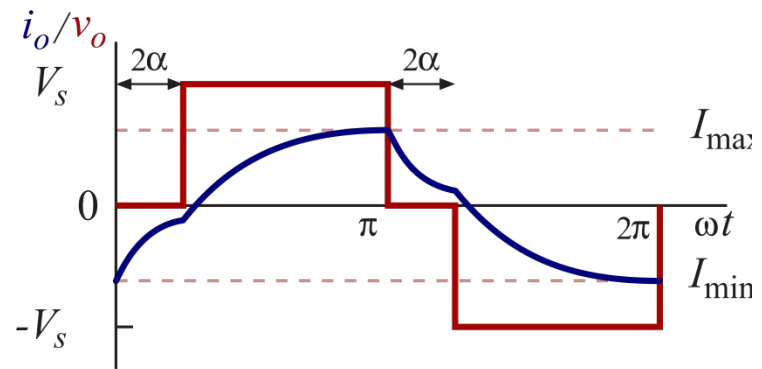
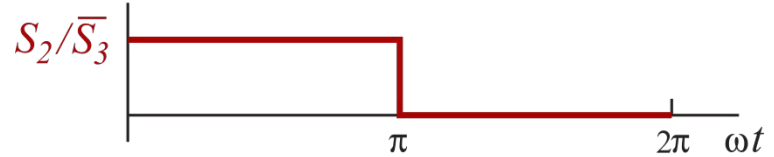
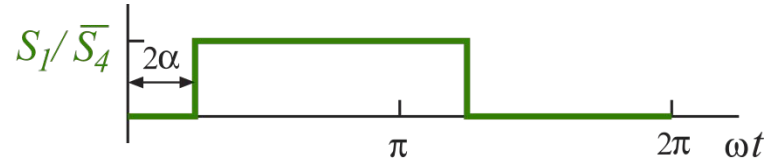
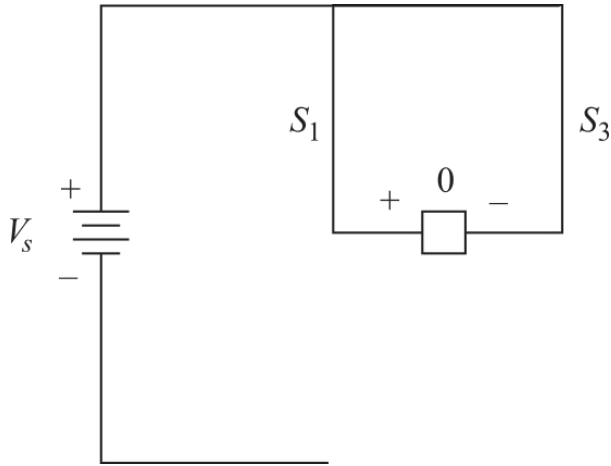
S_2
 S_4

Inversor PWM Monopulso: formas de onda



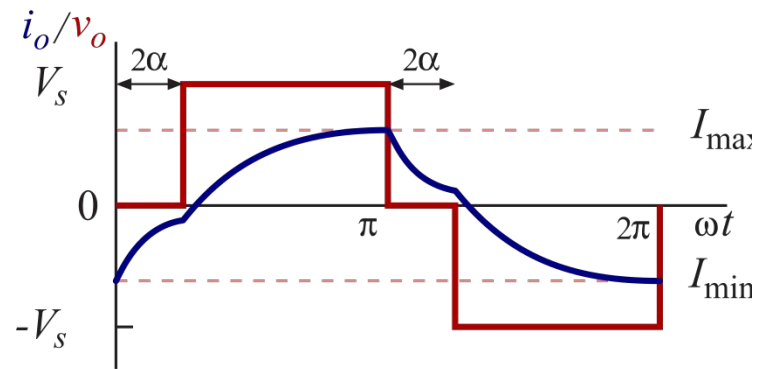
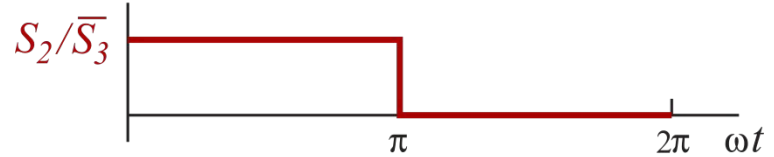
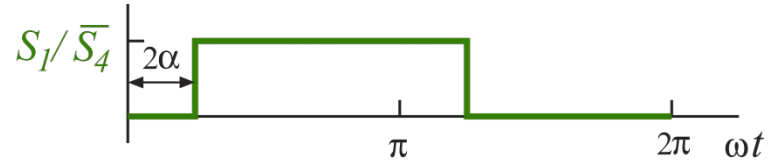
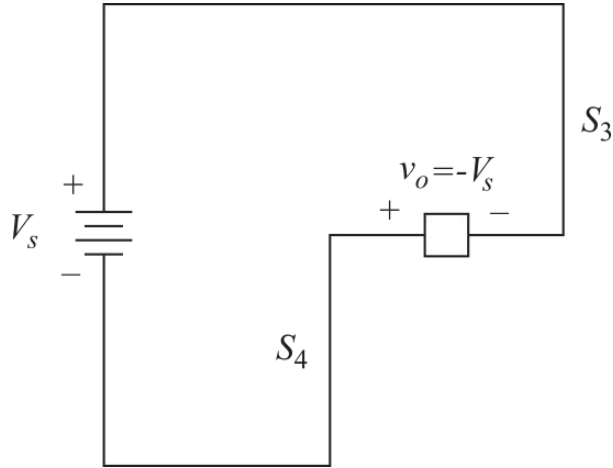
S_2	S_1
S_4	S_2

Inversor PWM Monopulso: formas de onda



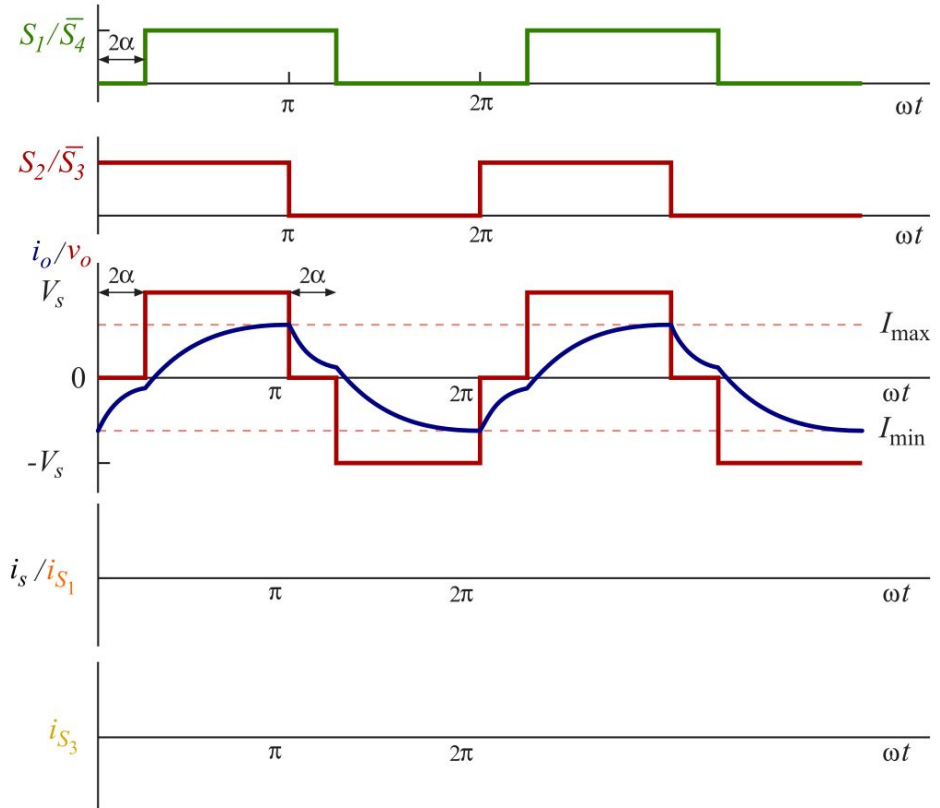
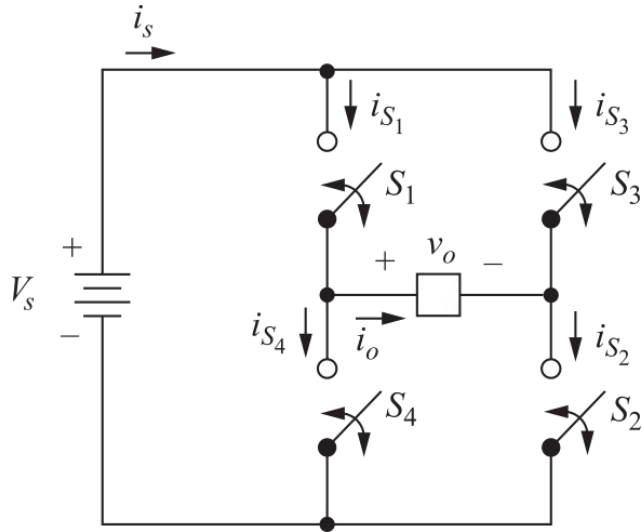
S_2	S_1	S_1
S_4	S_2	S_3

Inversor PWM Monopulso: formas de onda

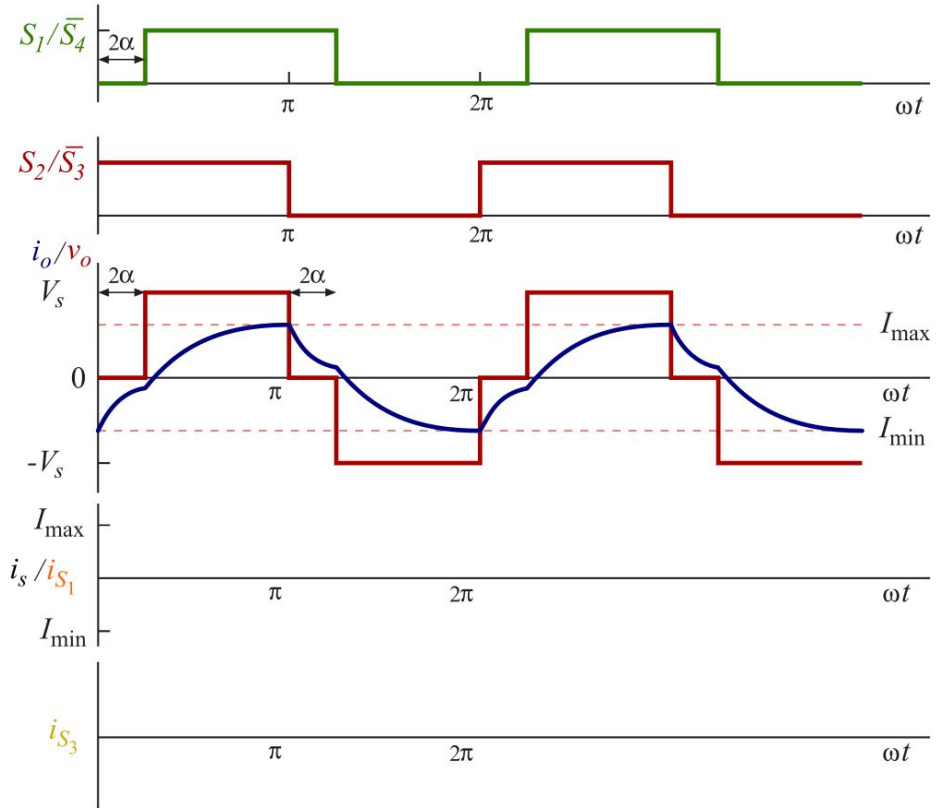
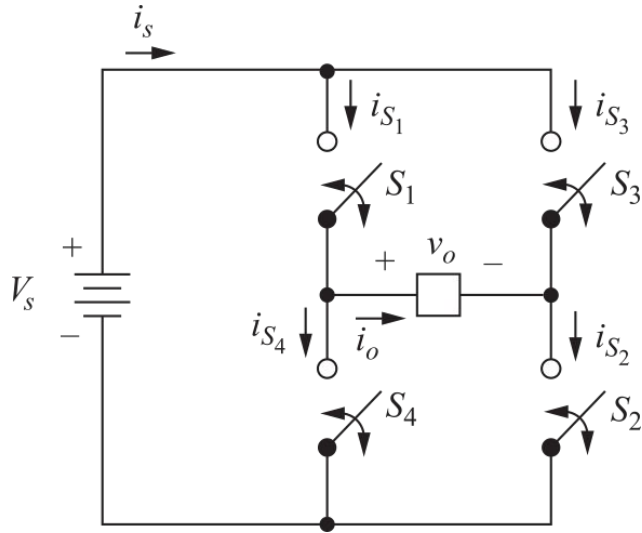


S_2	S_1	S_1	S_3
S_4	S_2	S_3	S_4

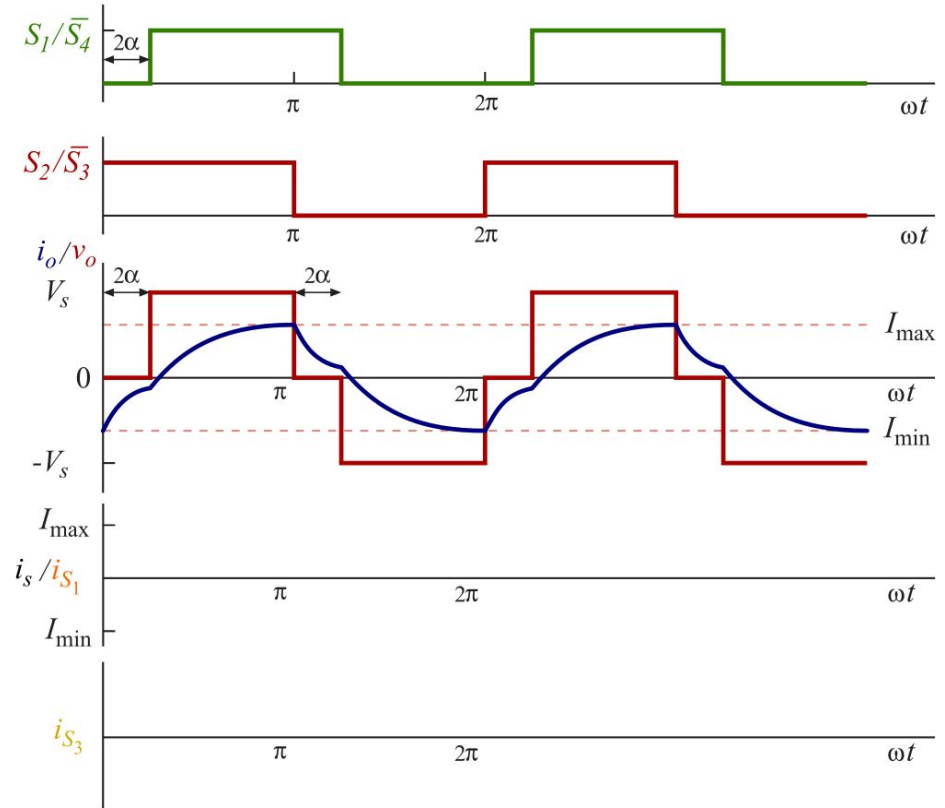
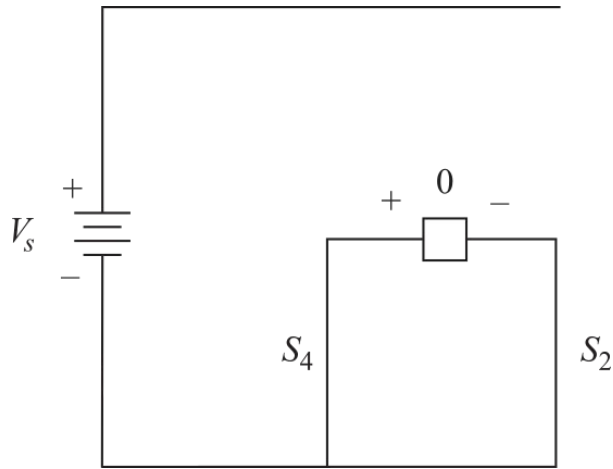
Inversor PWM Monopulso: formas de onda



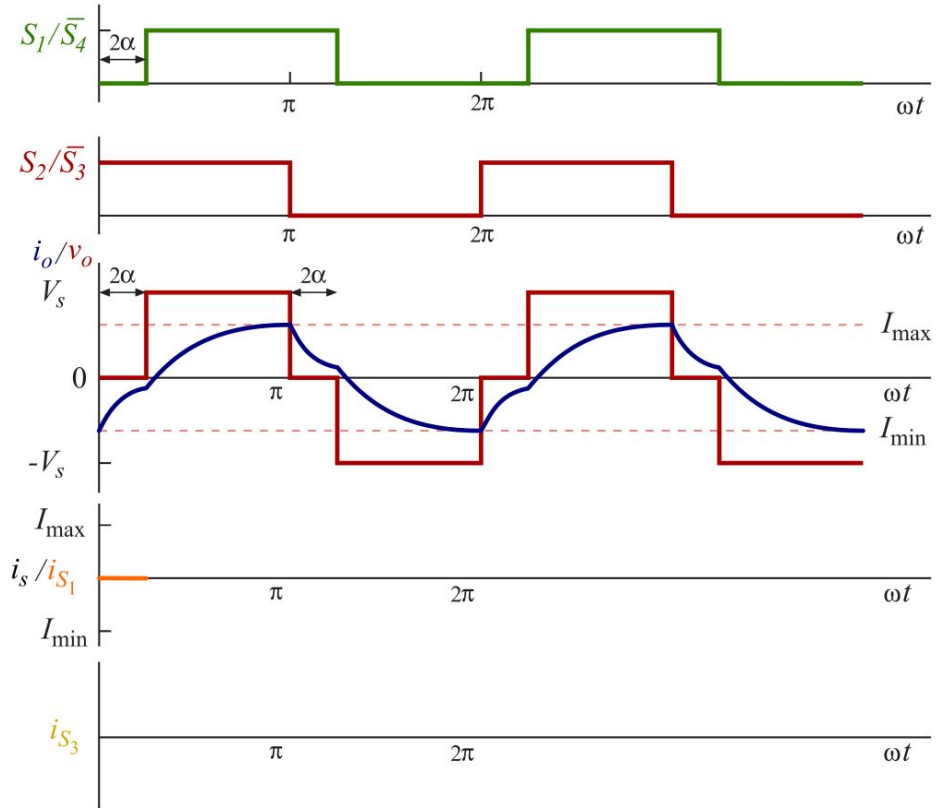
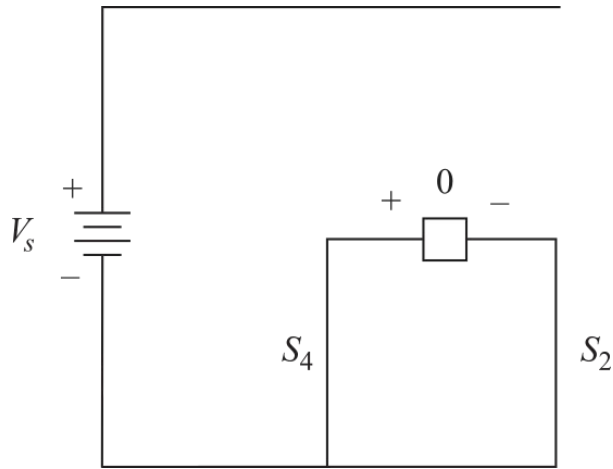
Inversor PWM Monopulso: formas de onda



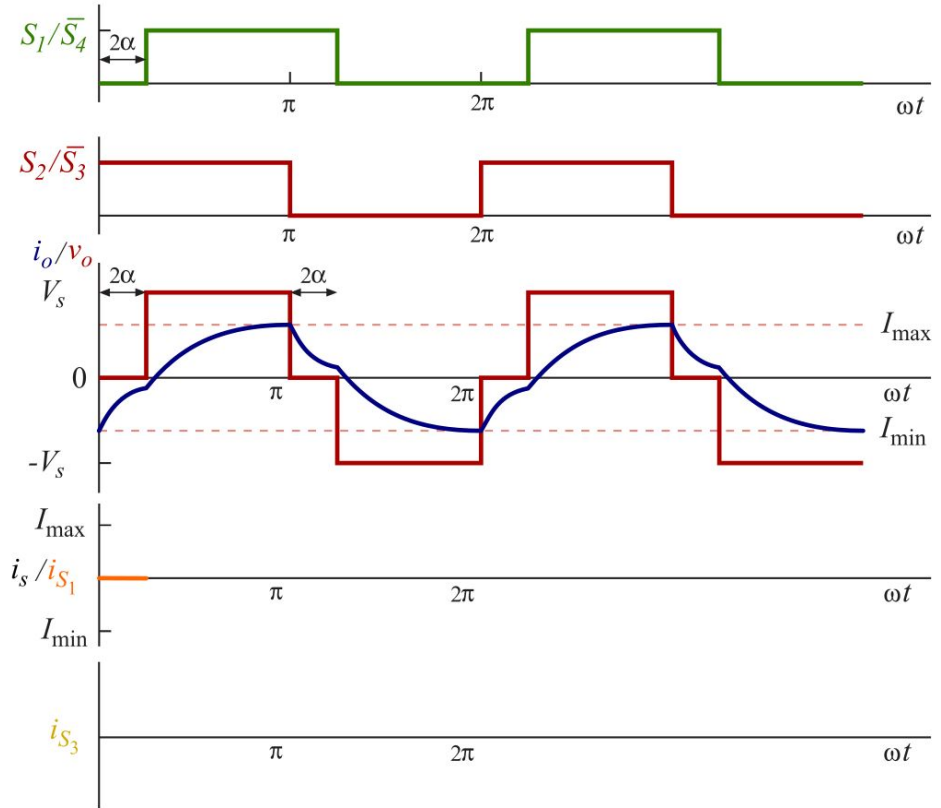
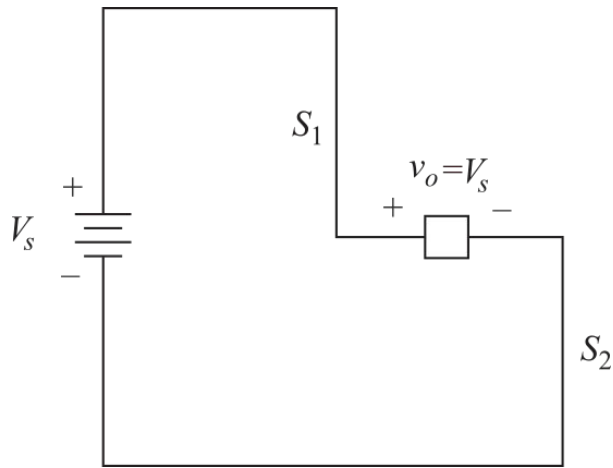
Inversor PWM Monopulso: formas de onda



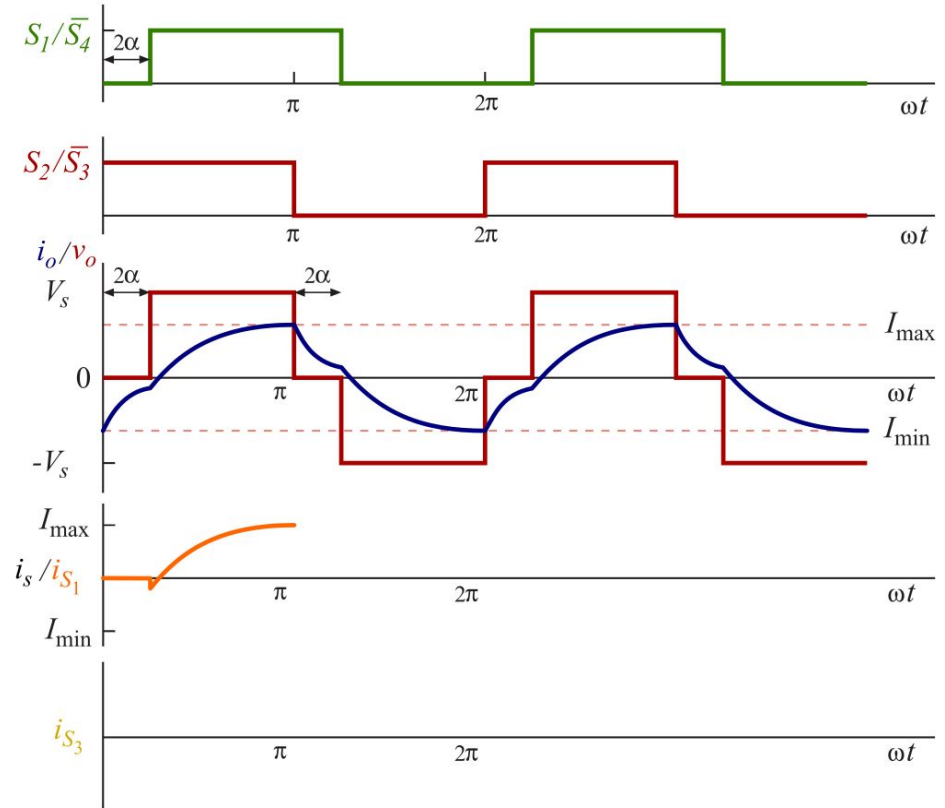
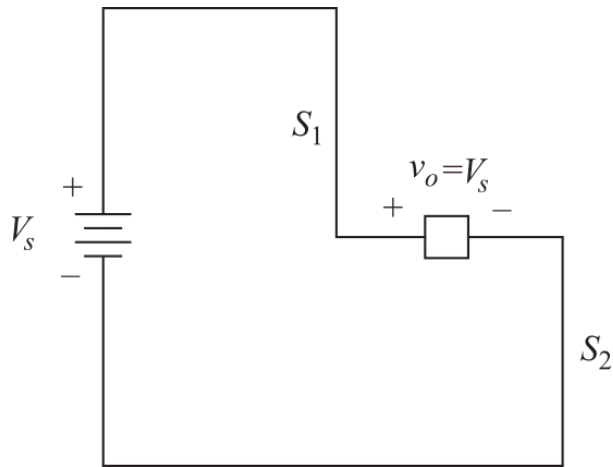
Inversor PWM Monopulso: formas de onda



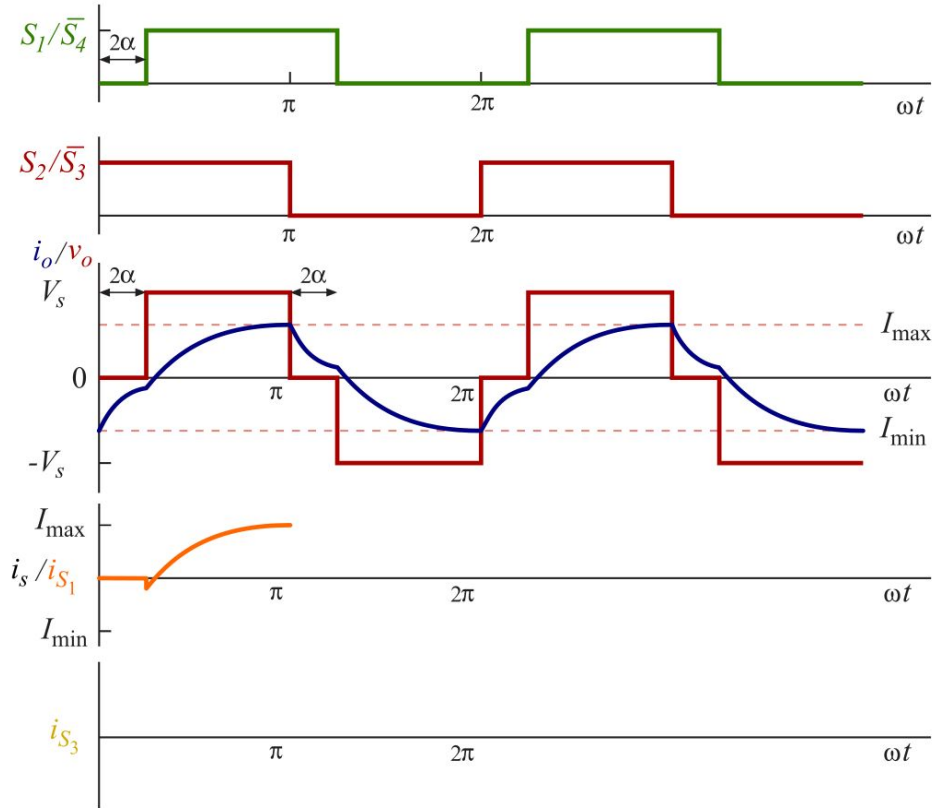
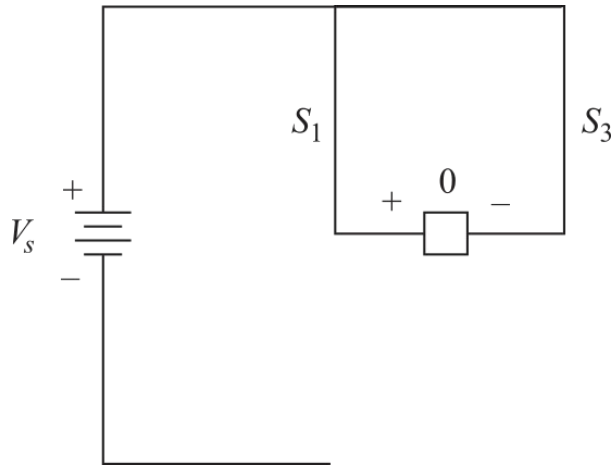
Inversor PWM Monopulso: formas de onda



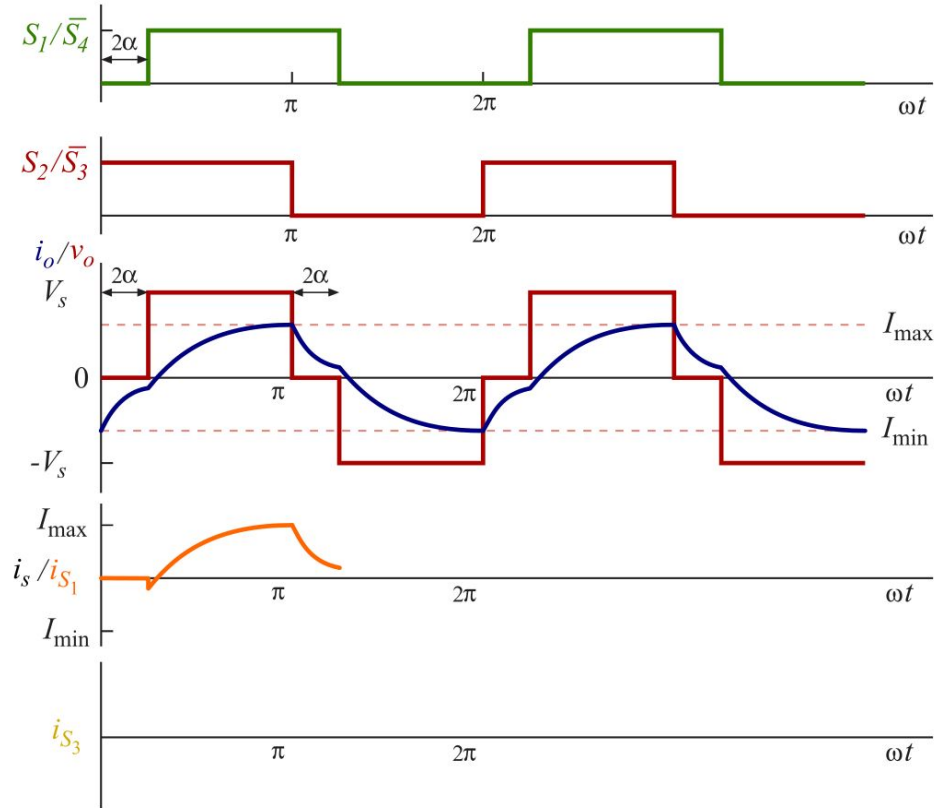
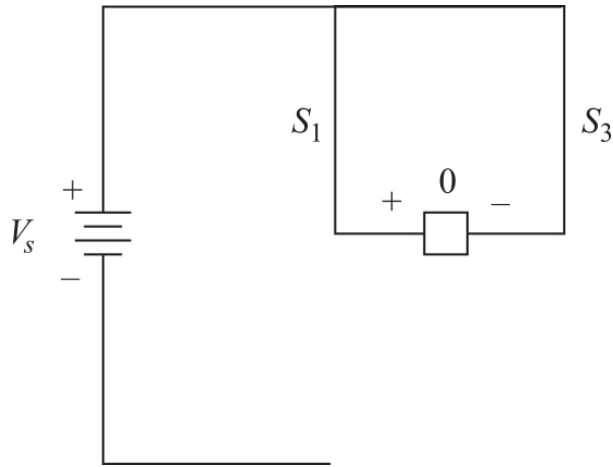
Inversor PWM Monopulso: formas de onda



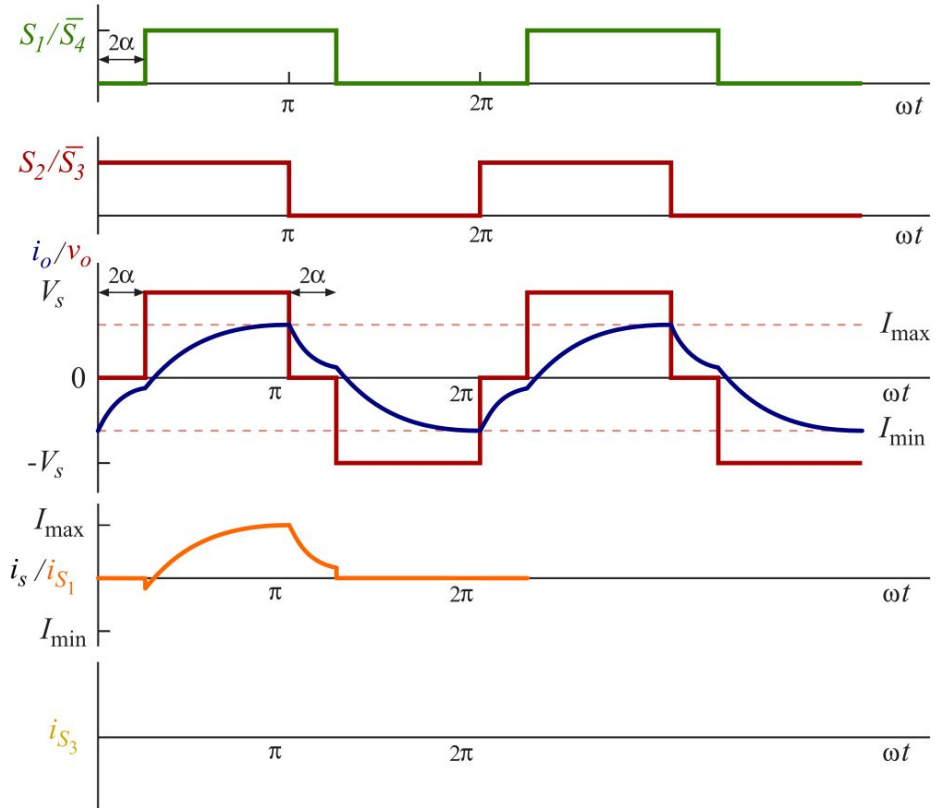
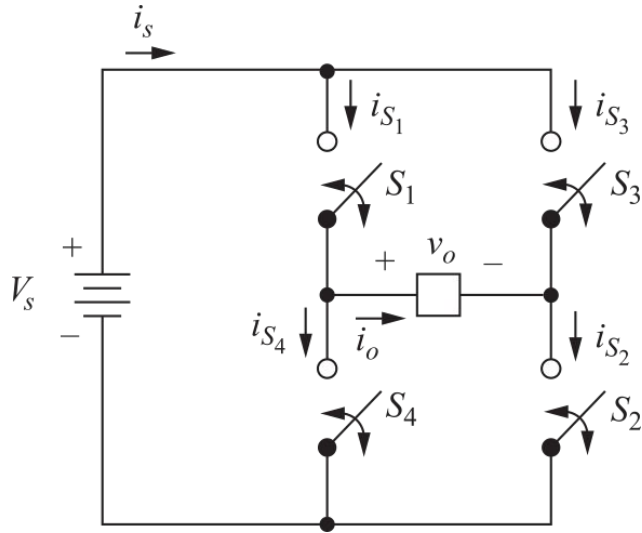
Inversor PWM Monopulso: formas de onda



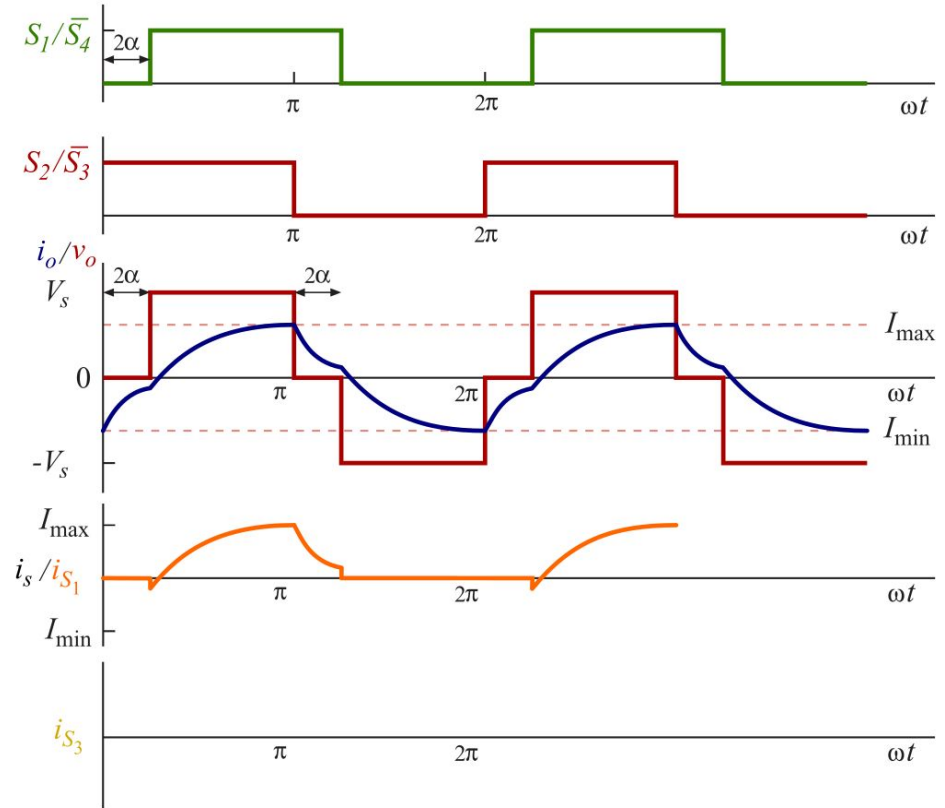
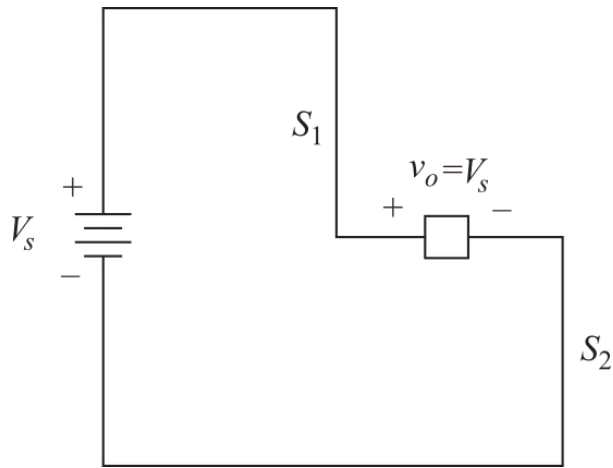
Inversor PWM Monopulso: formas de onda



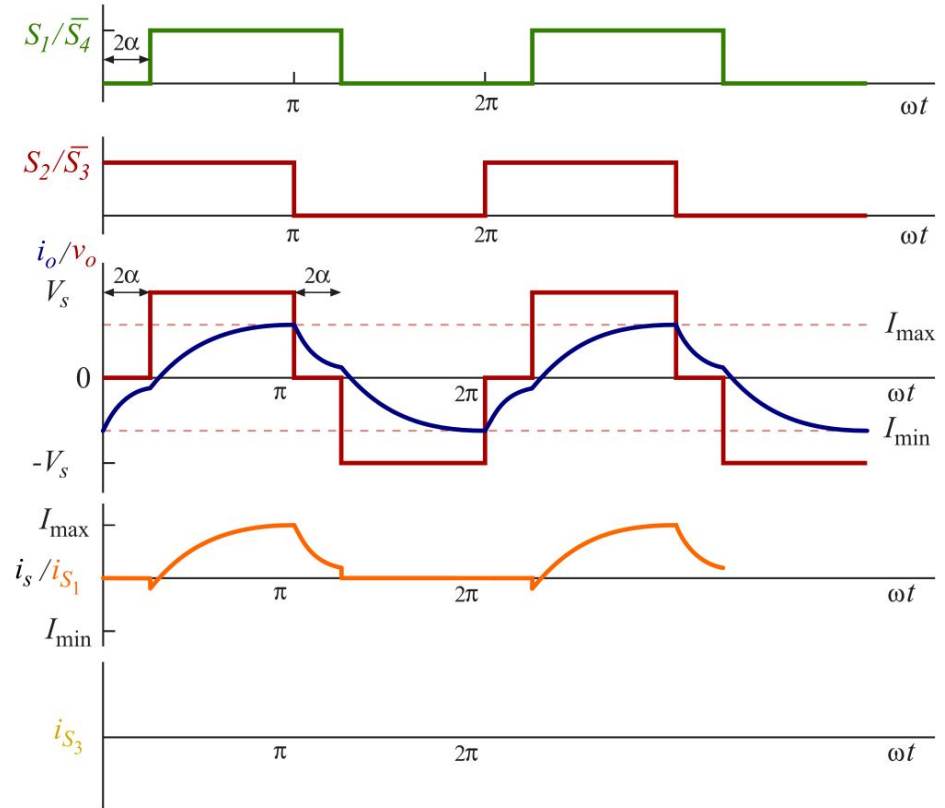
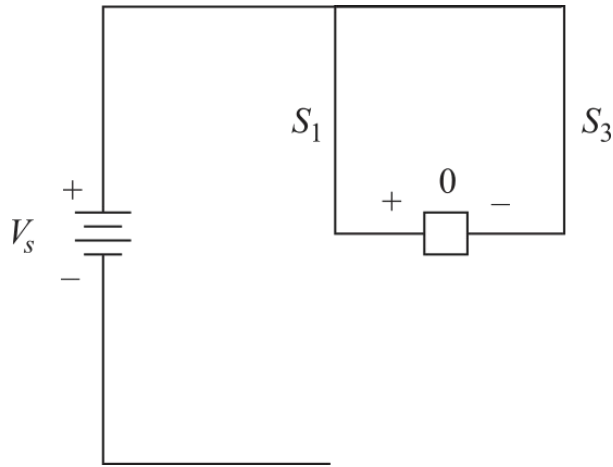
Inversor PWM Monopulso: formas de onda



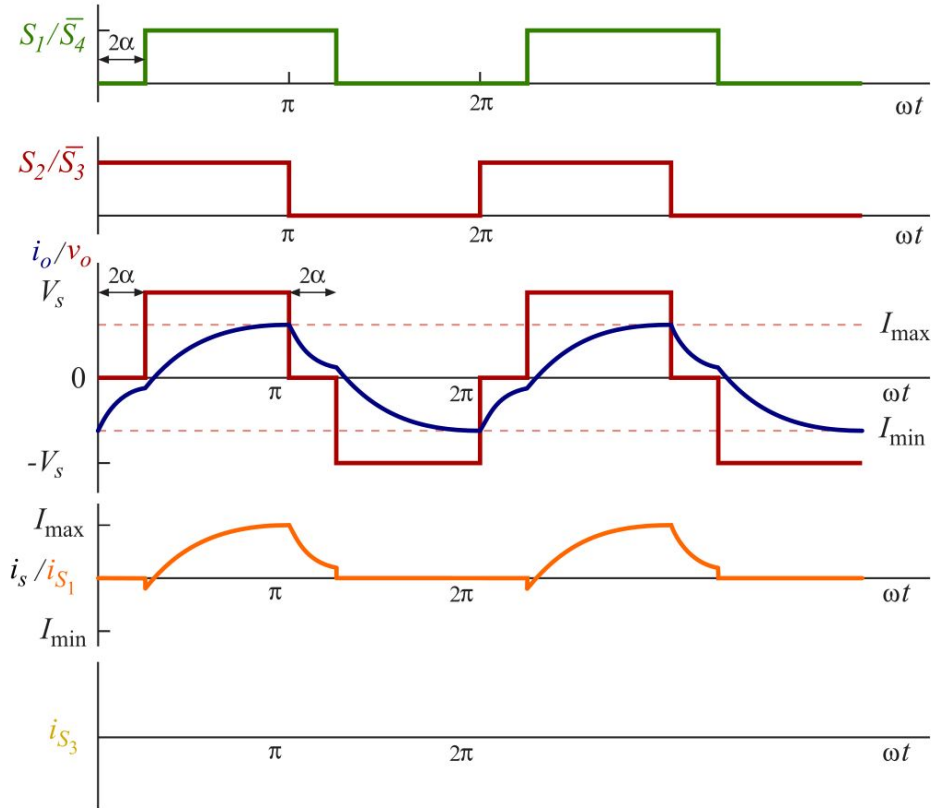
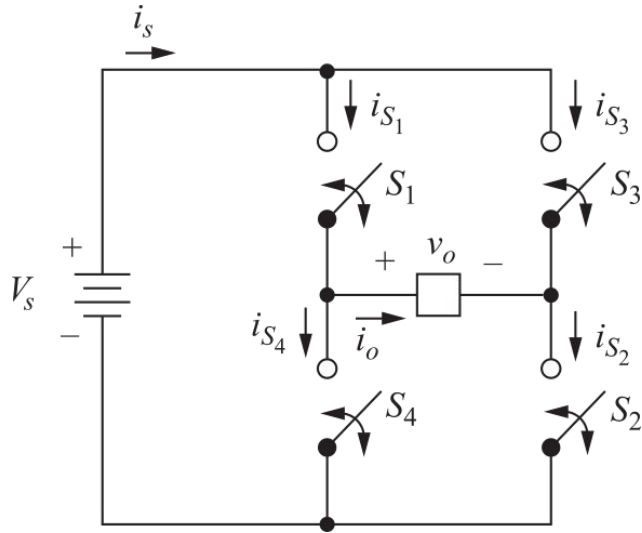
Inversor PWM Monopulso: formas de onda



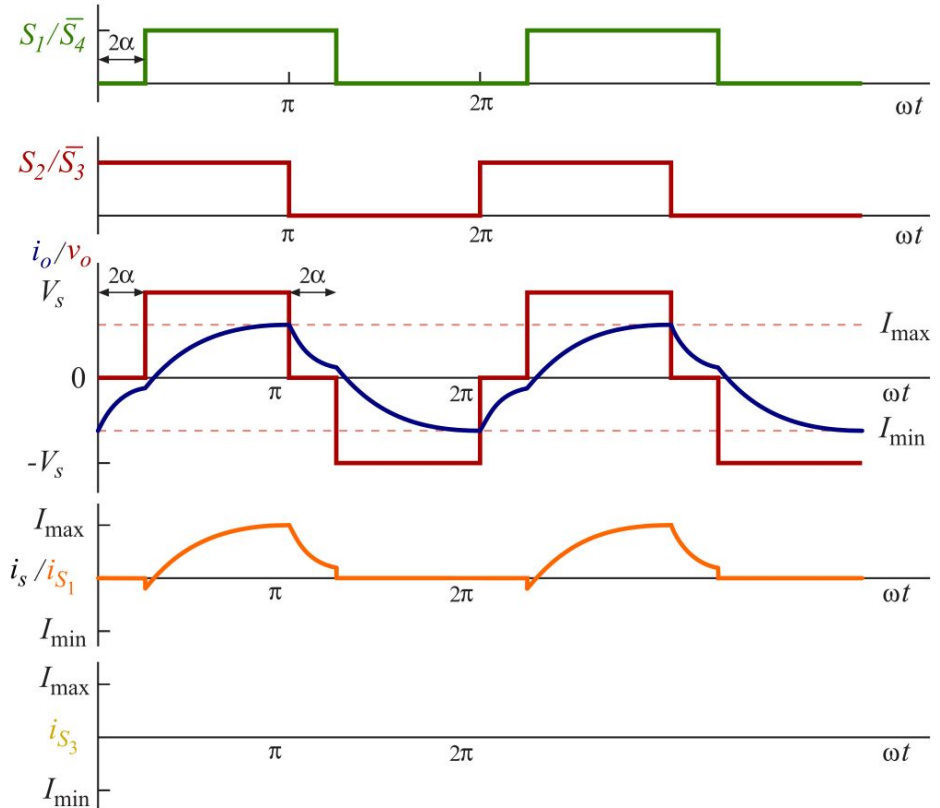
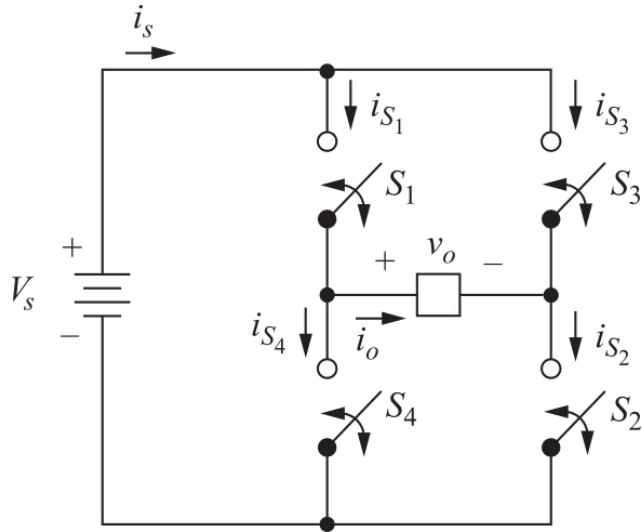
Inversor PWM Monopulso: formas de onda



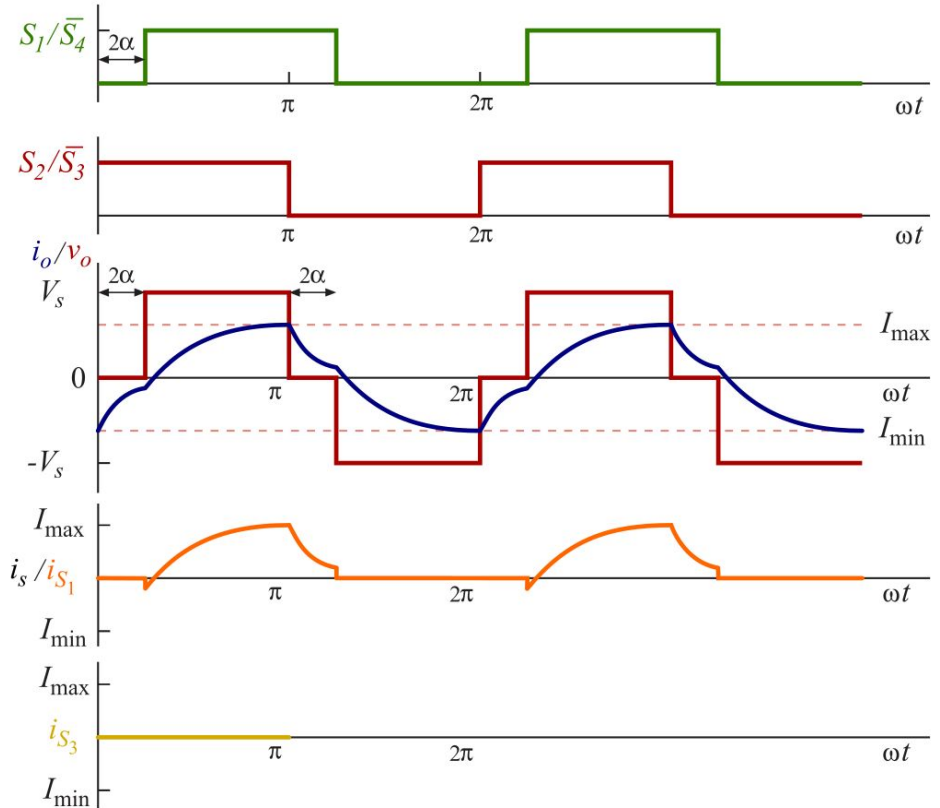
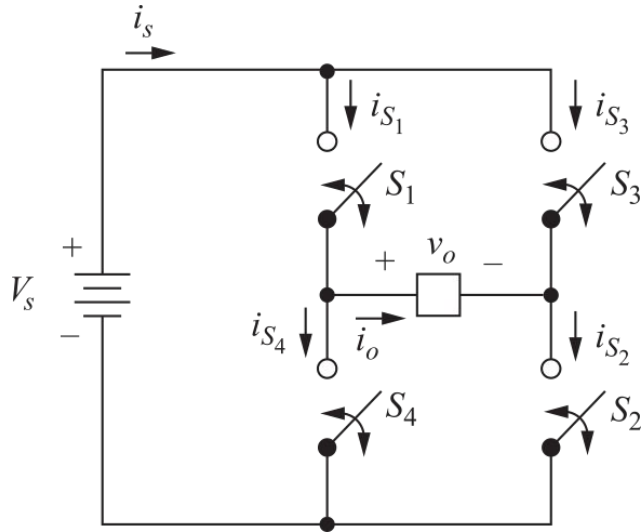
Inversor PWM Monopulso: formas de onda



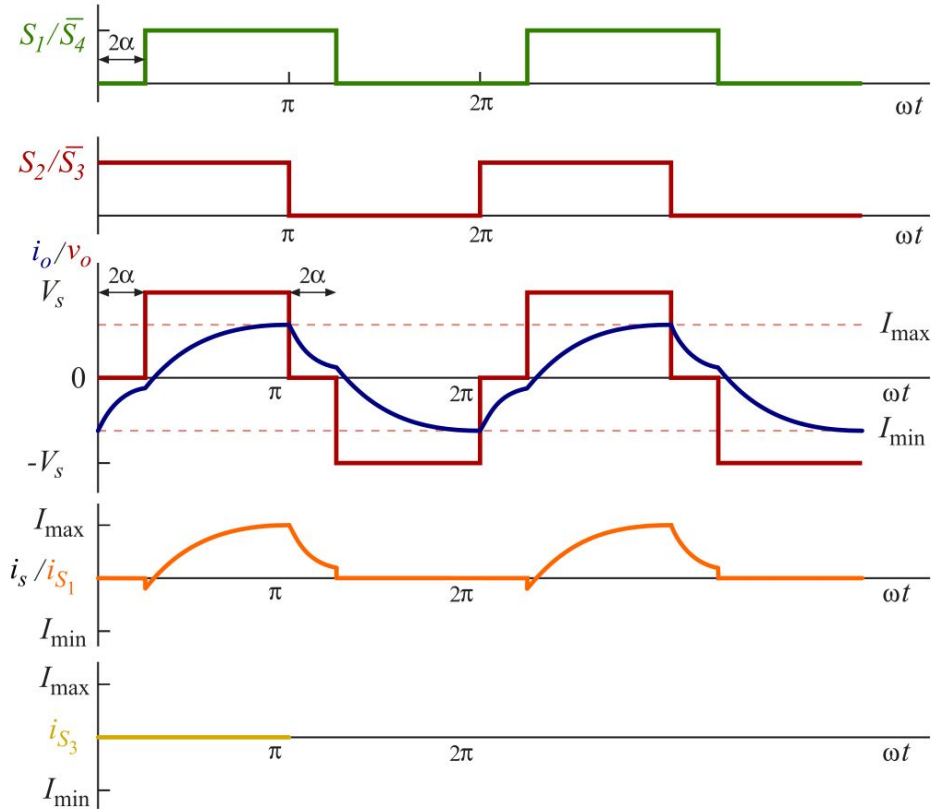
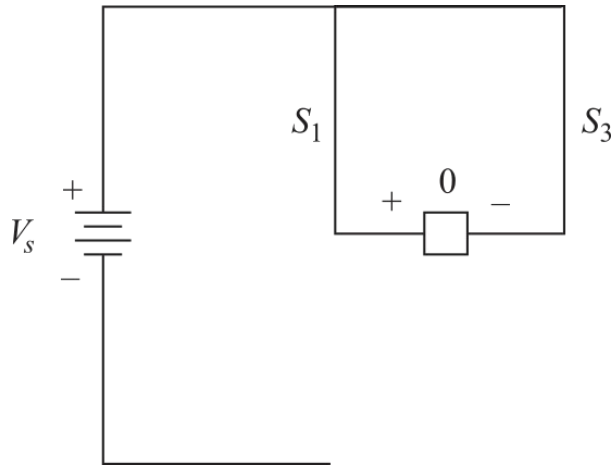
Inversor PWM Monopulso: formas de onda



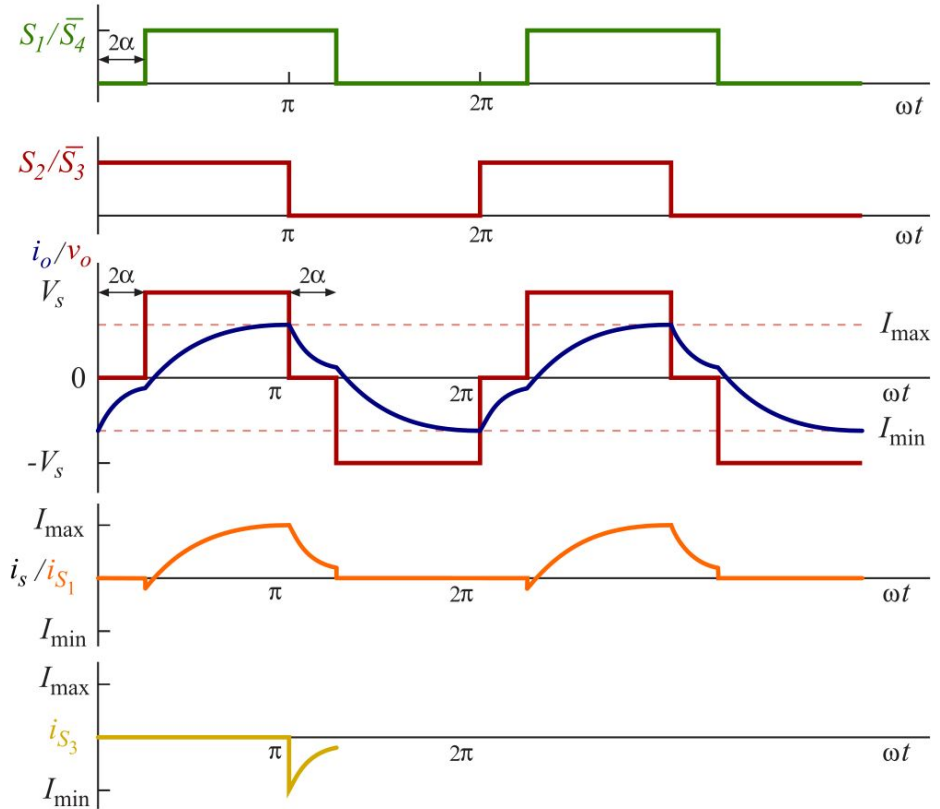
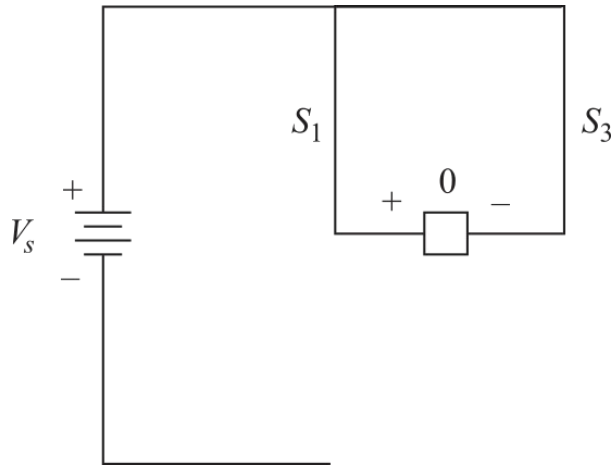
Inversor PWM Monopulso: formas de onda



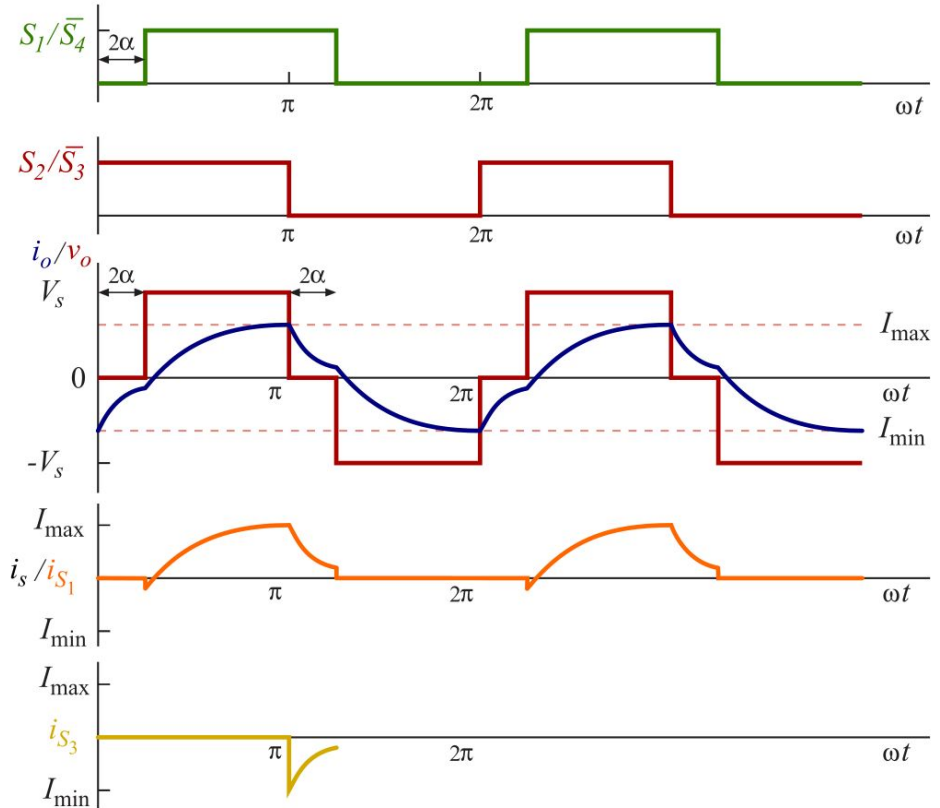
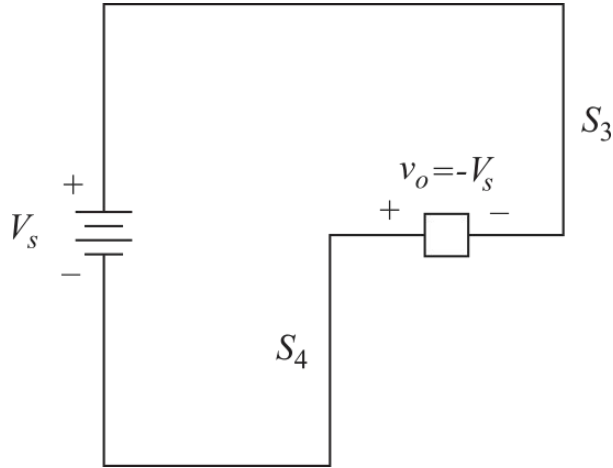
Inversor PWM Monopulso: formas de onda



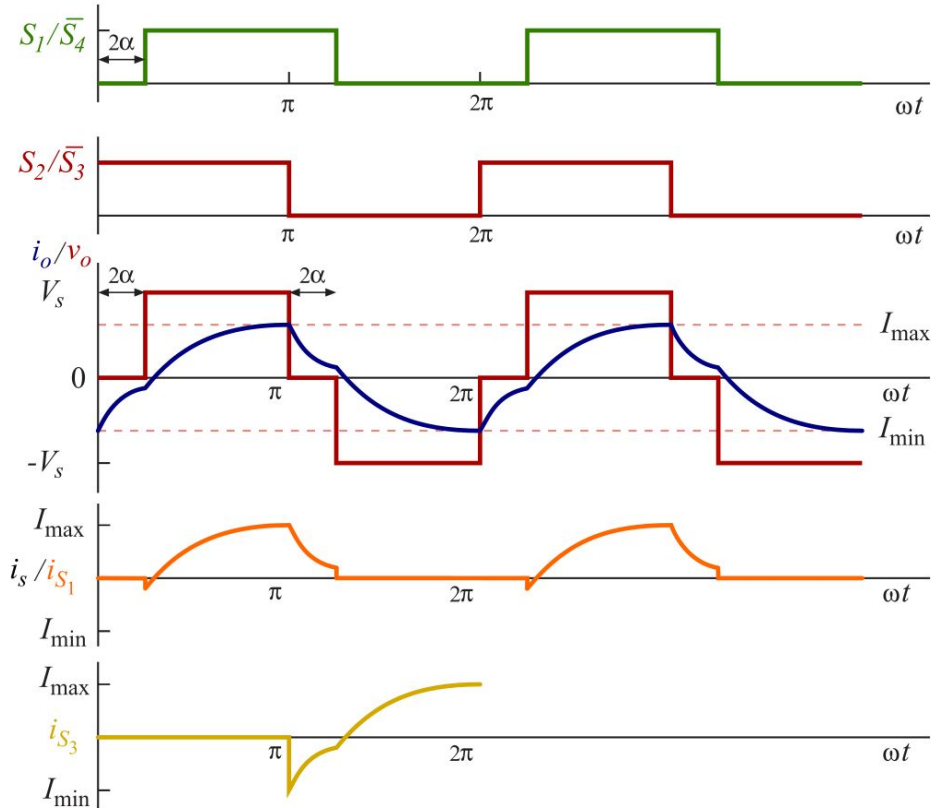
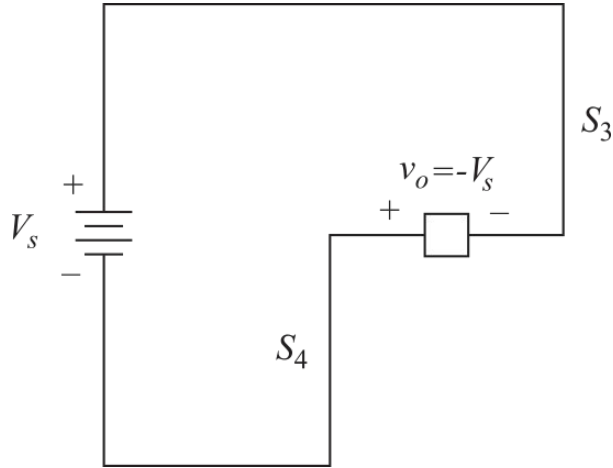
Inversor PWM Monopulso: formas de onda



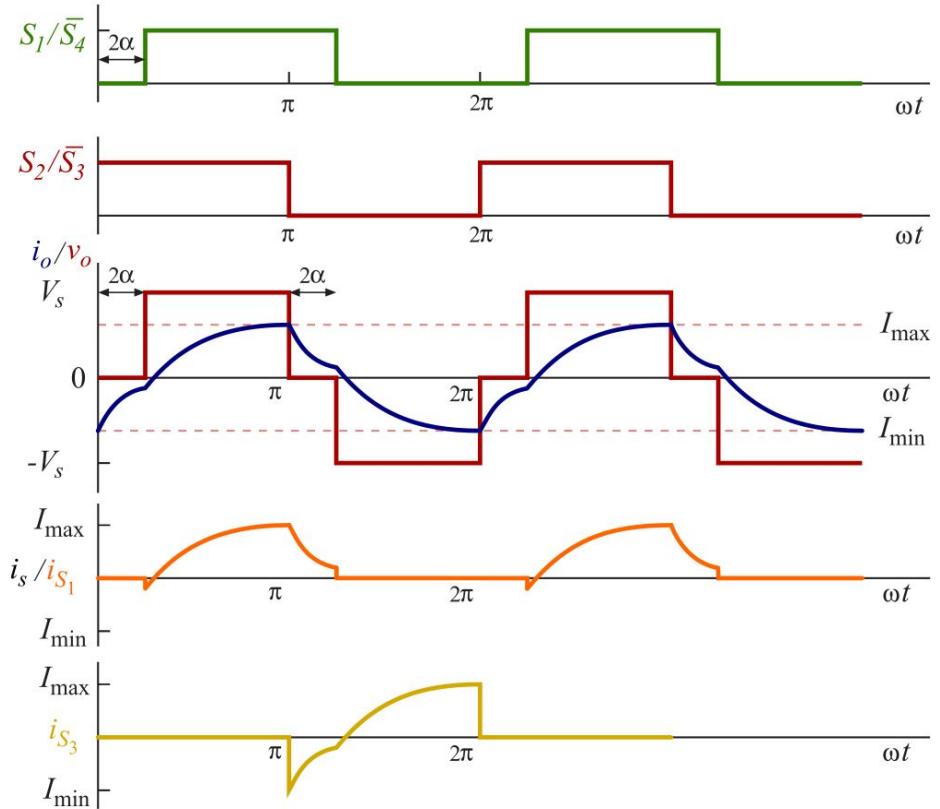
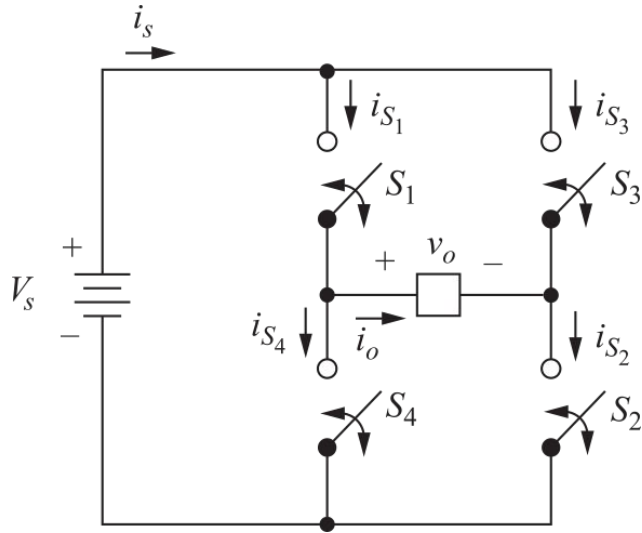
Inversor PWM Monopulso: formas de onda



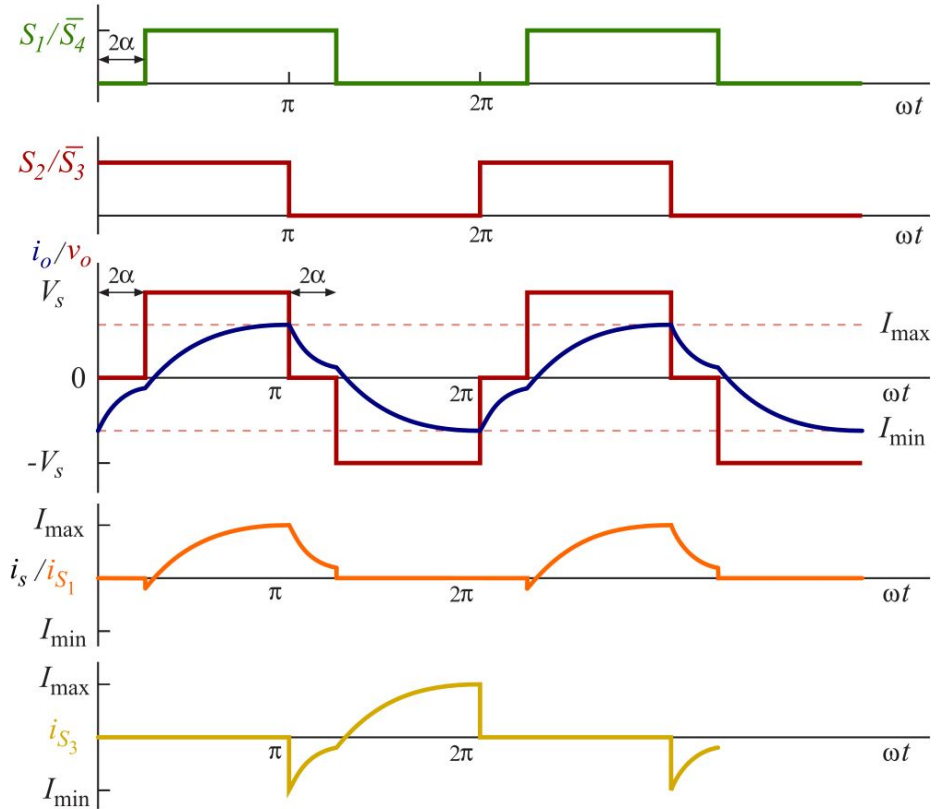
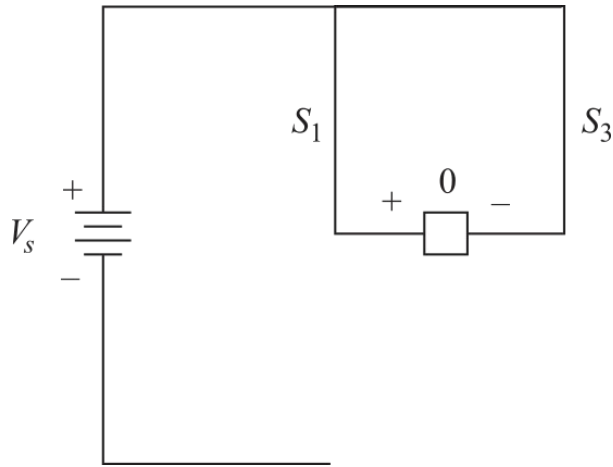
Inversor PWM Monopulso: formas de onda



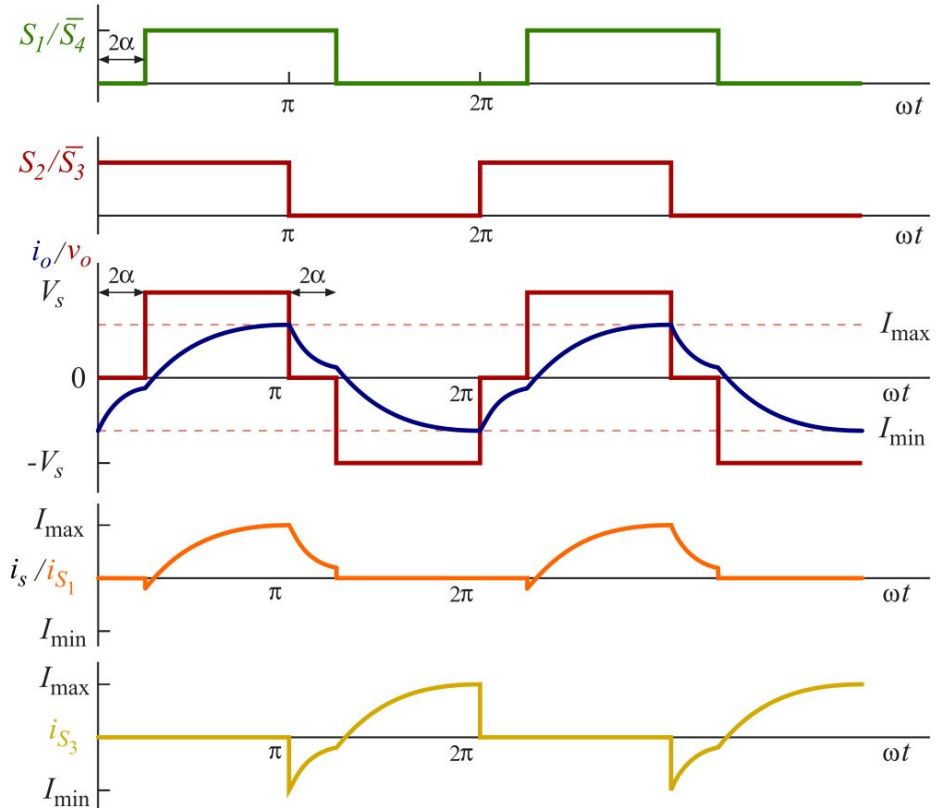
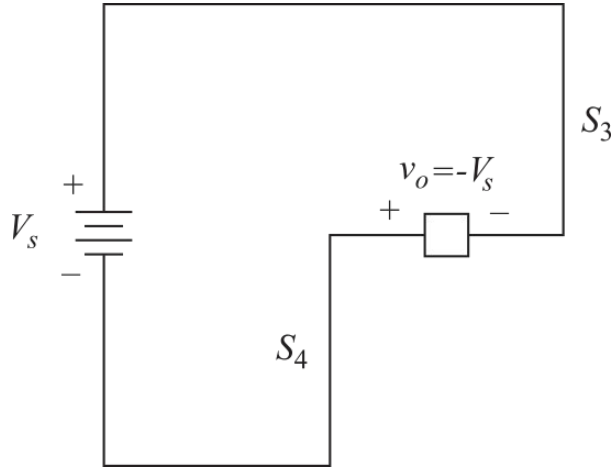
Inversor PWM Monopulso: formas de onda



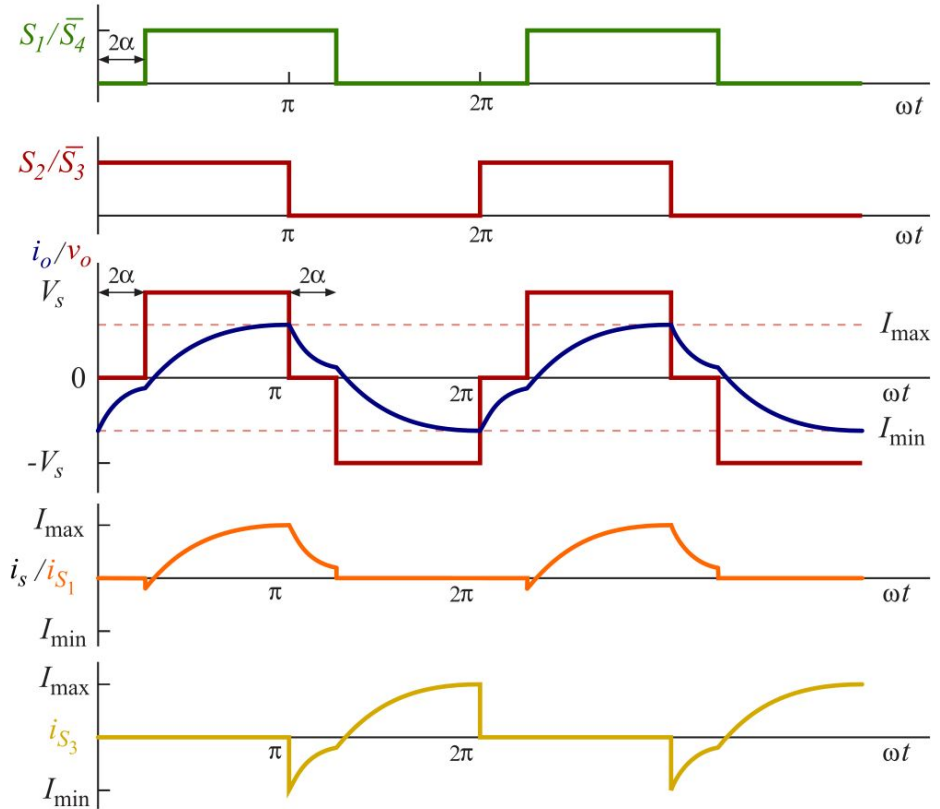
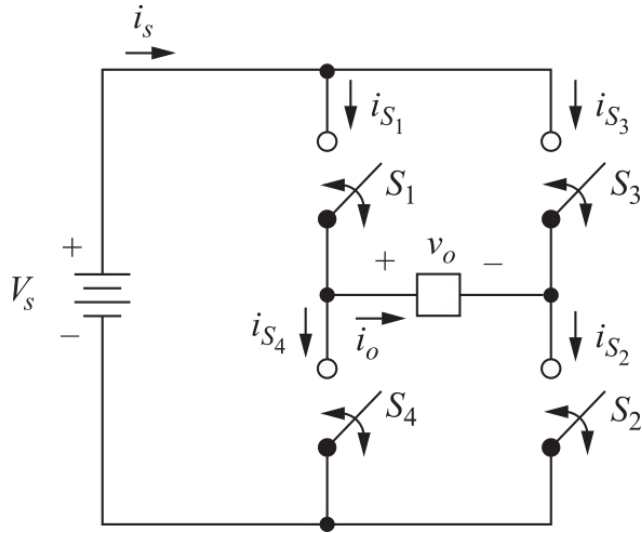
Inversor PWM Monopulso: formas de onda



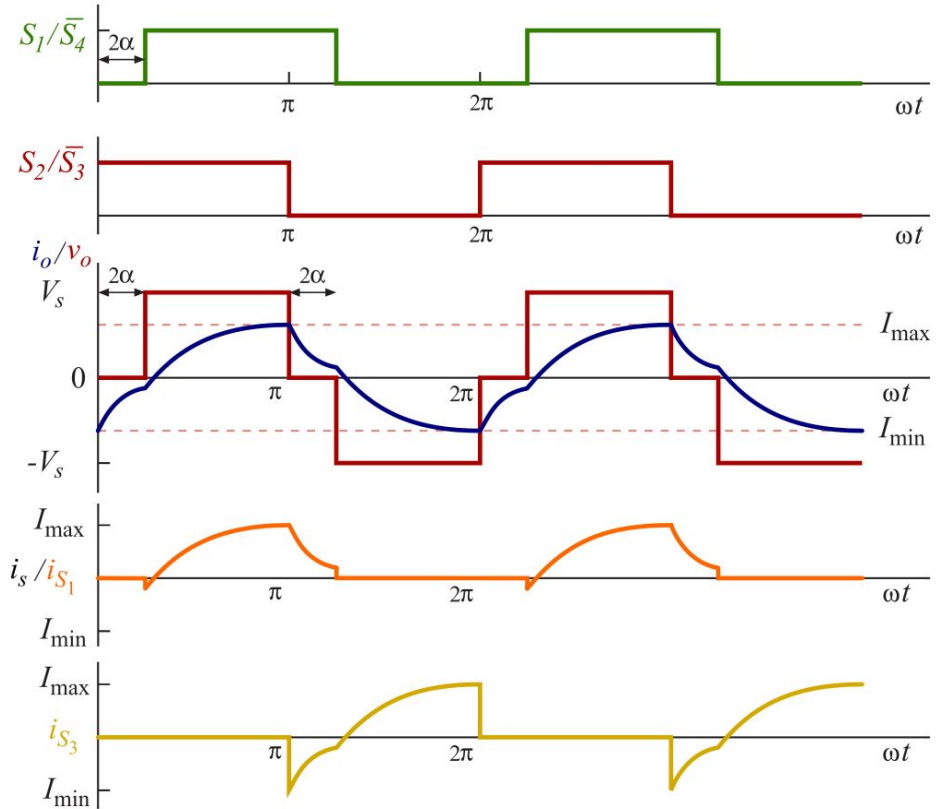
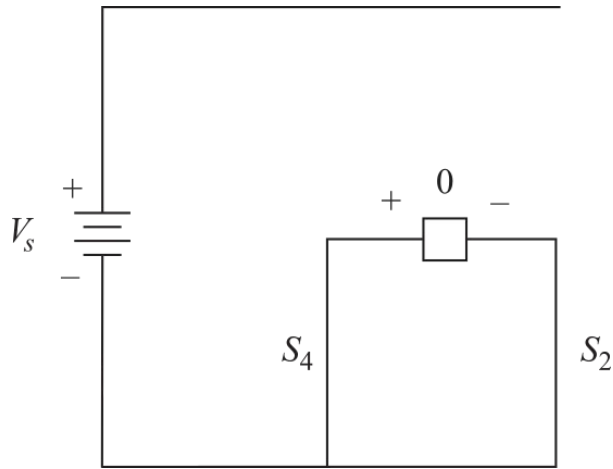
Inversor PWM Monopulso: formas de onda



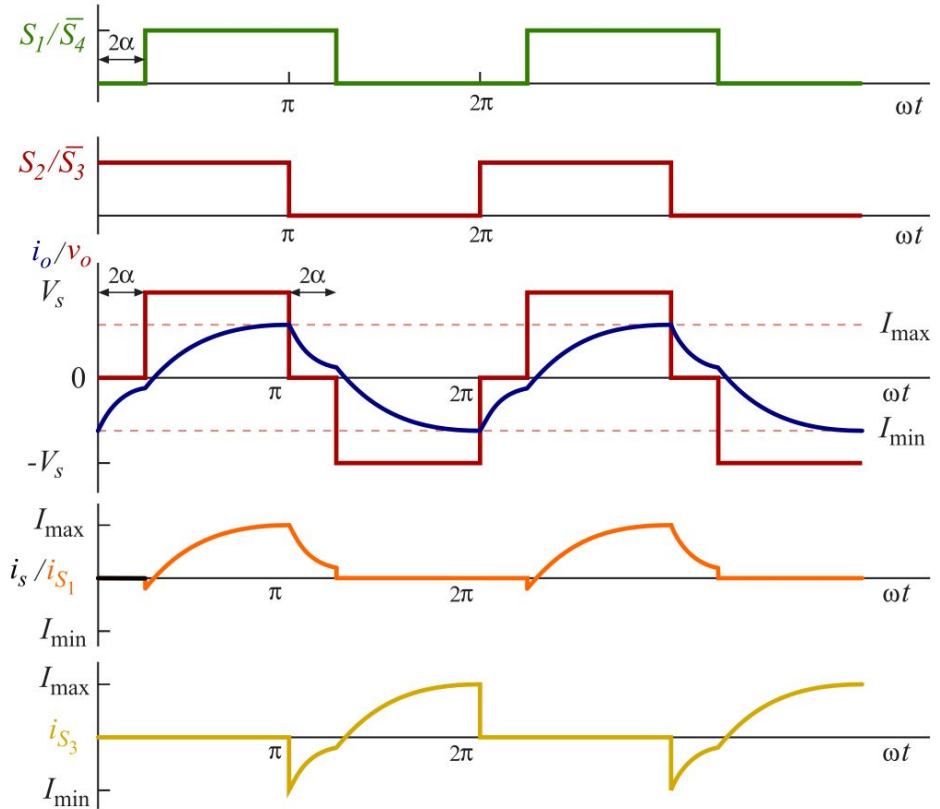
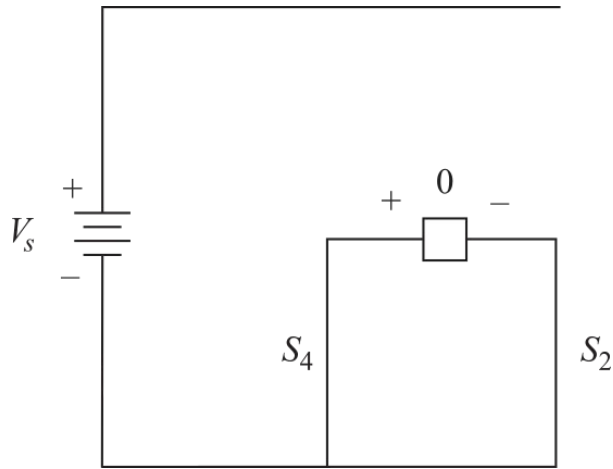
Inversor PWM Monopulso: formas de onda



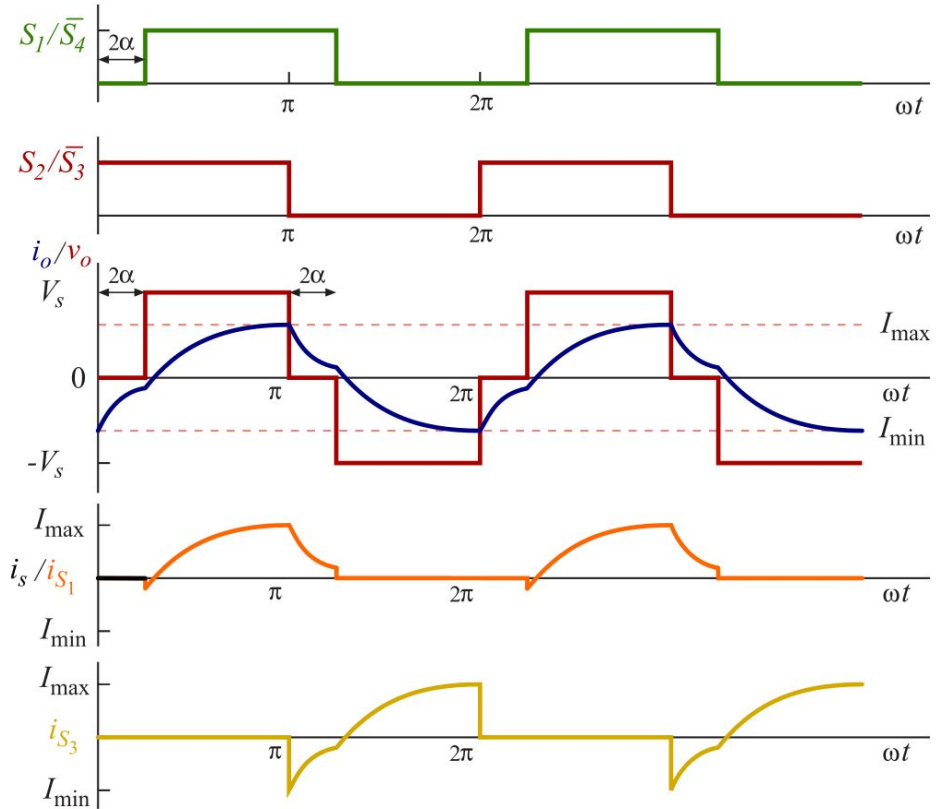
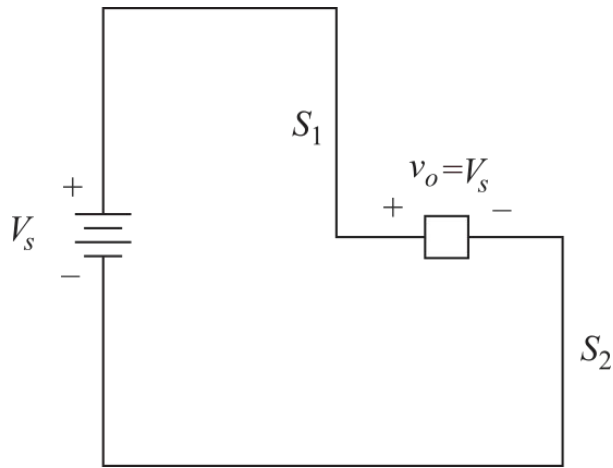
Inversor PWM Monopulso: formas de onda



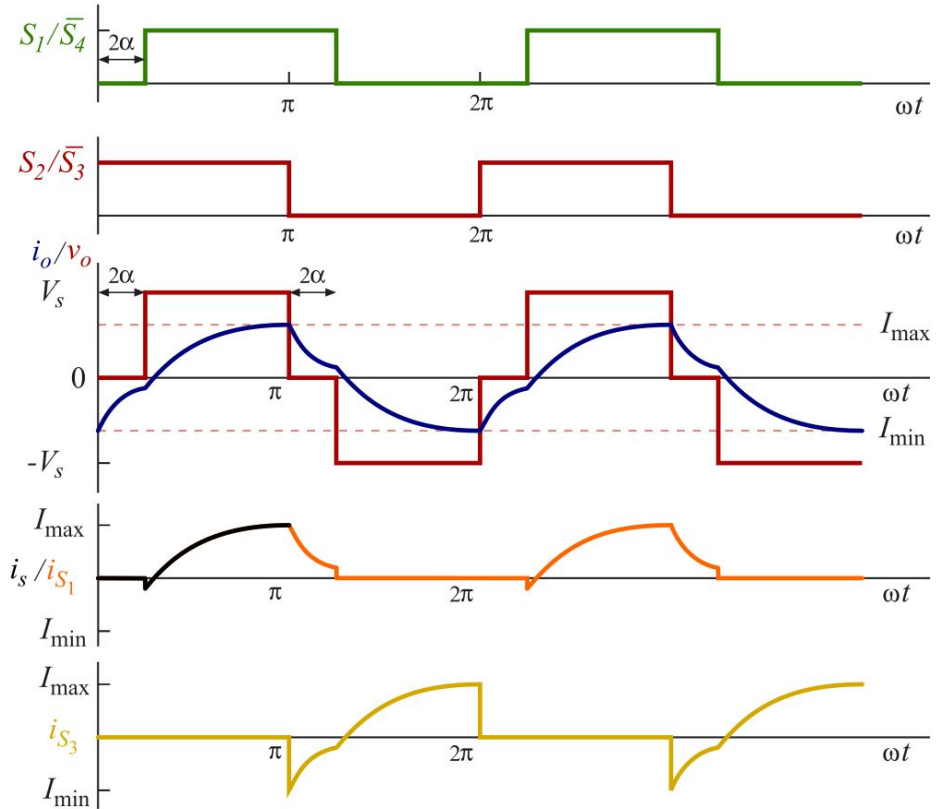
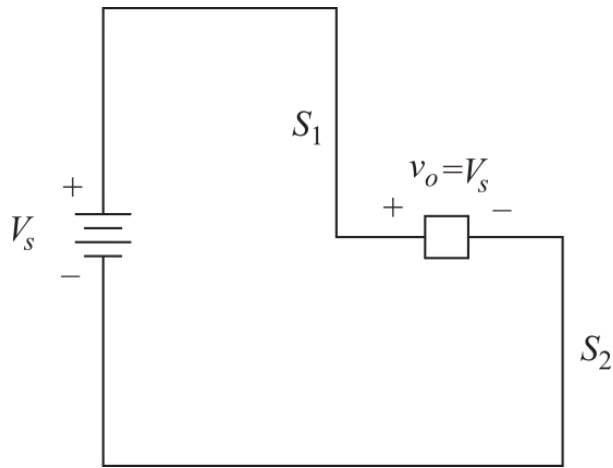
Inversor PWM Monopulso: formas de onda



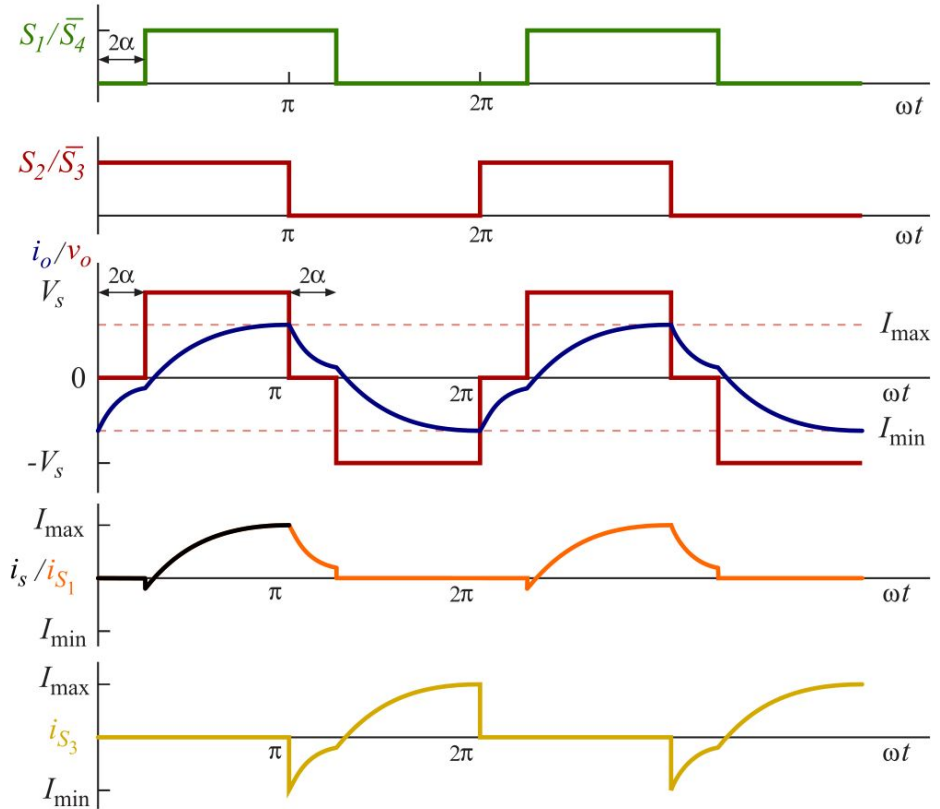
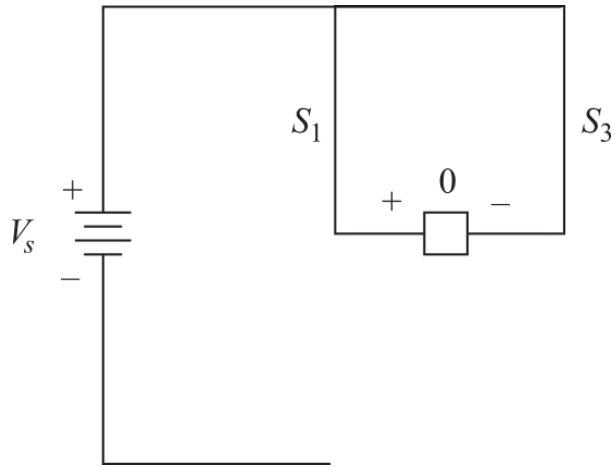
Inversor PWM Monopulso: formas de onda



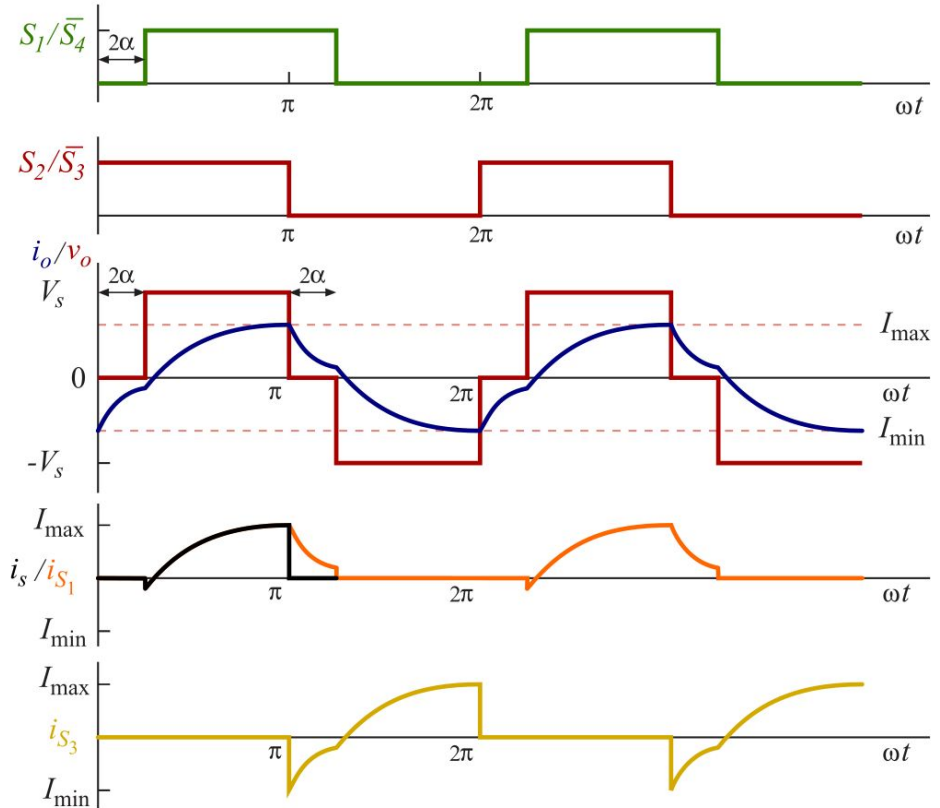
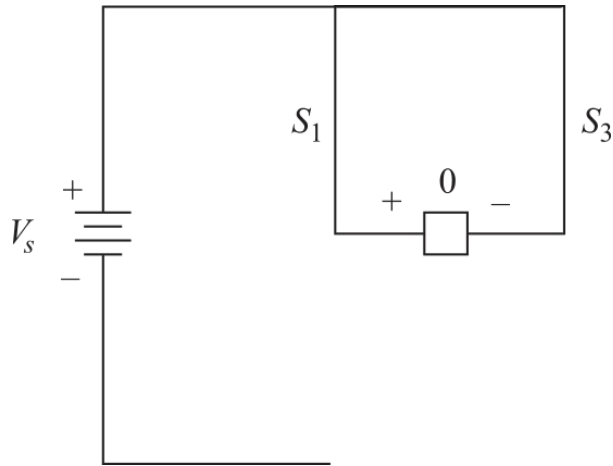
Inversor PWM Monopulso: formas de onda



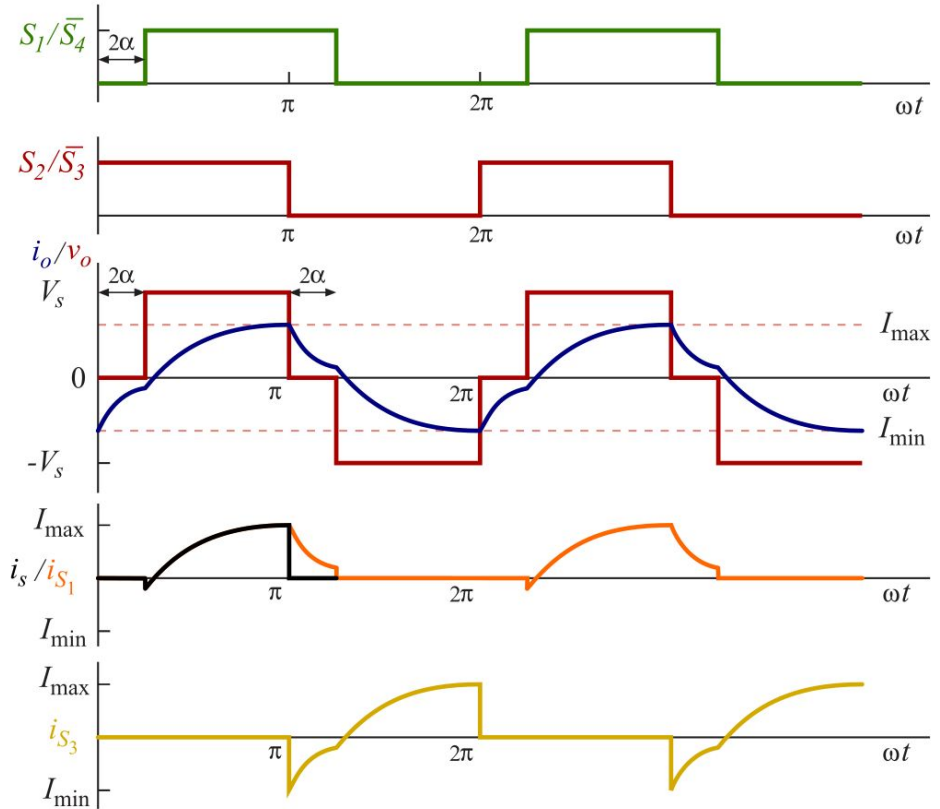
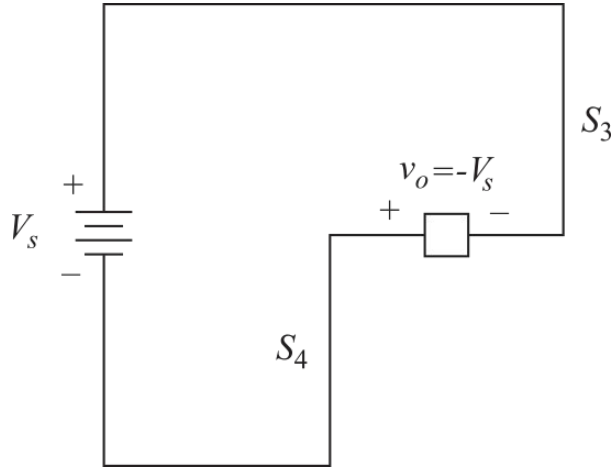
Inversor PWM Monopulso: formas de onda



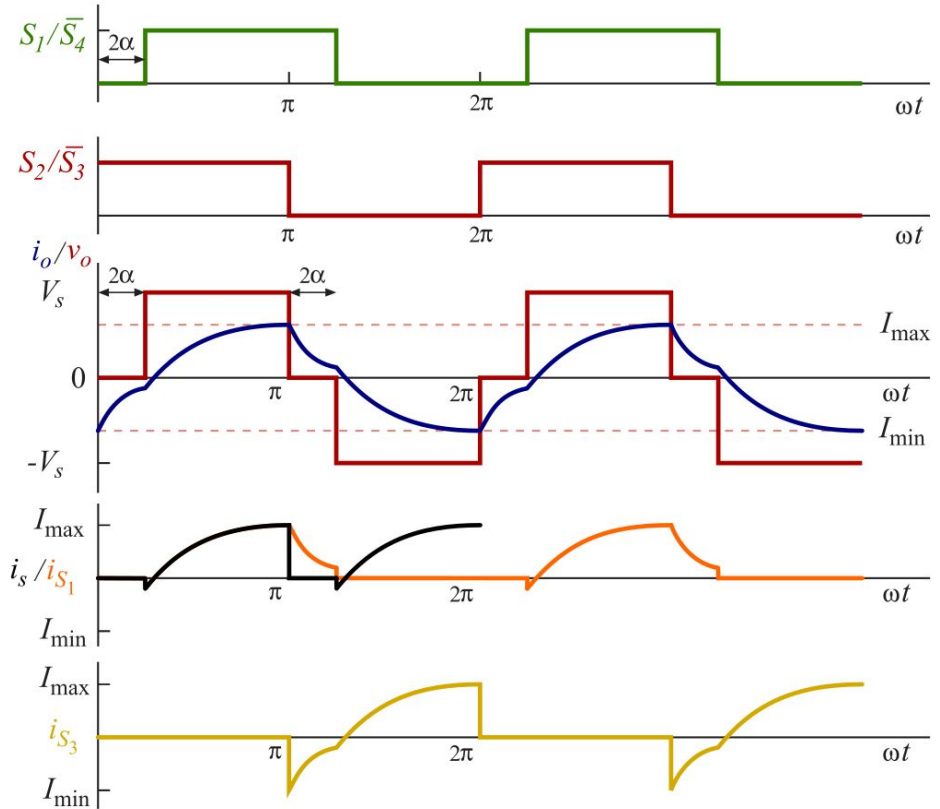
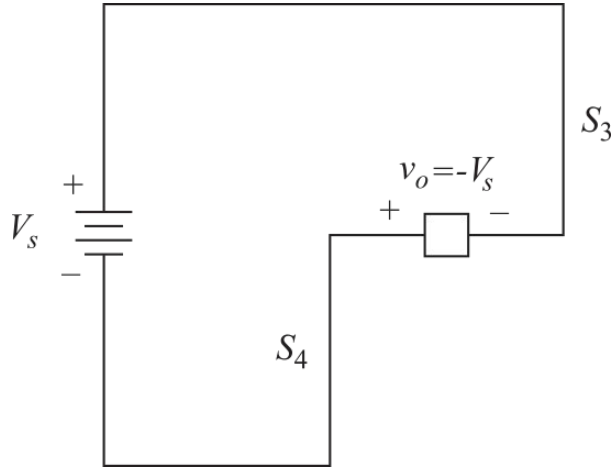
Inversor PWM Monopulso: formas de onda



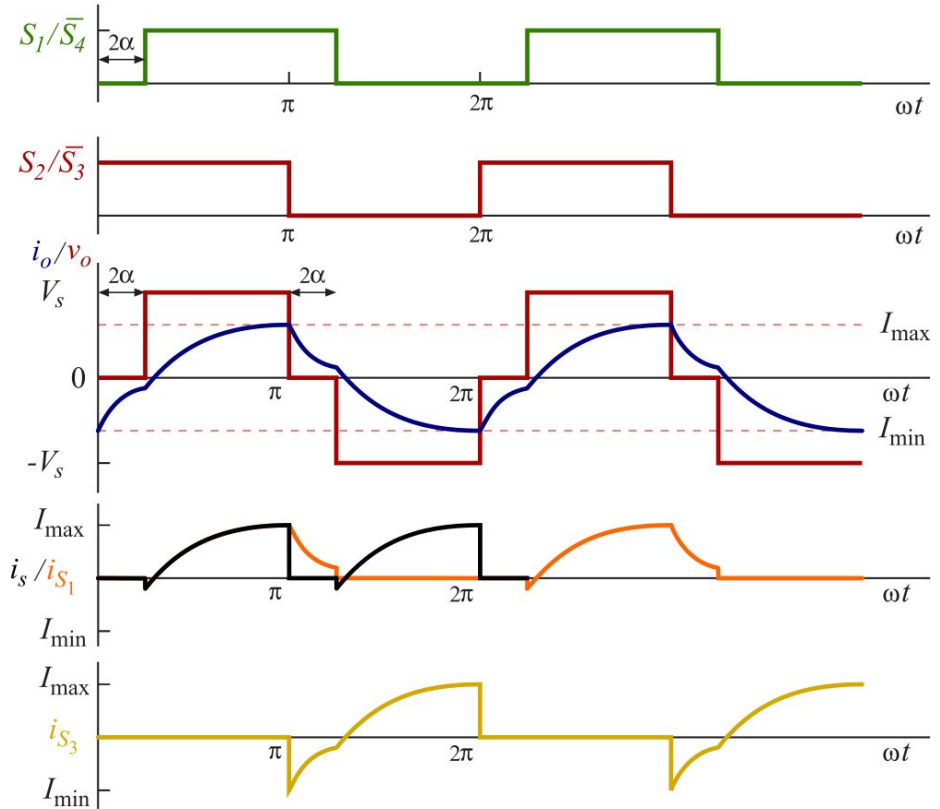
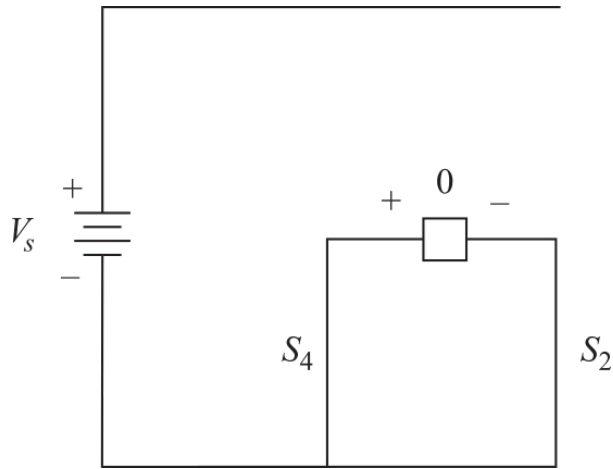
Inversor PWM Monopulso: formas de onda



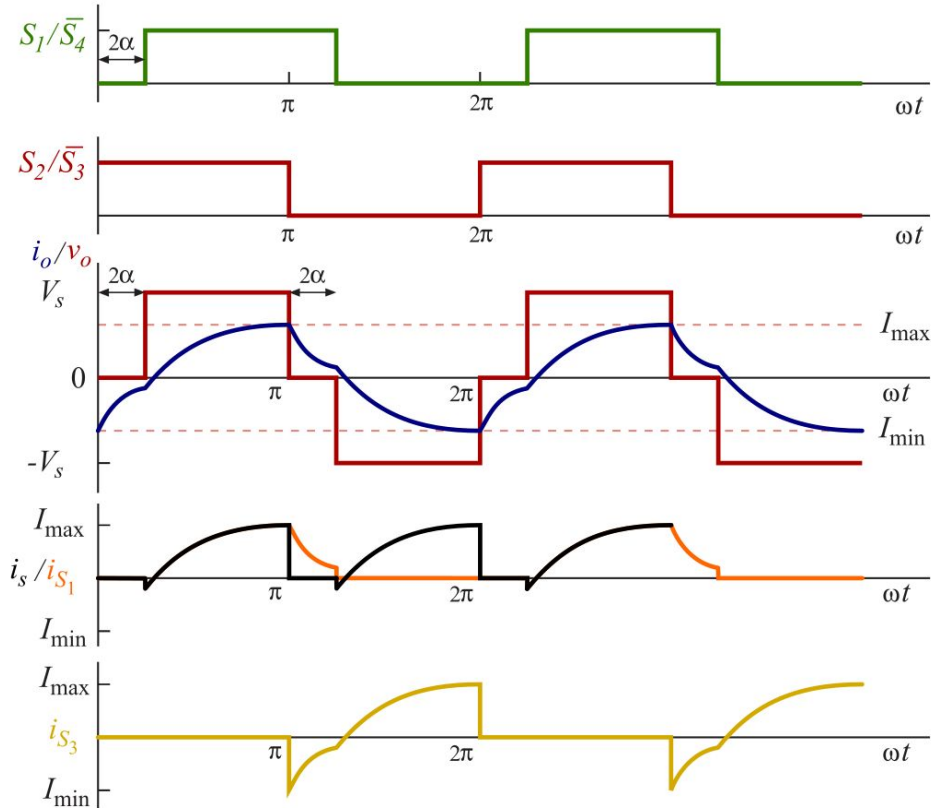
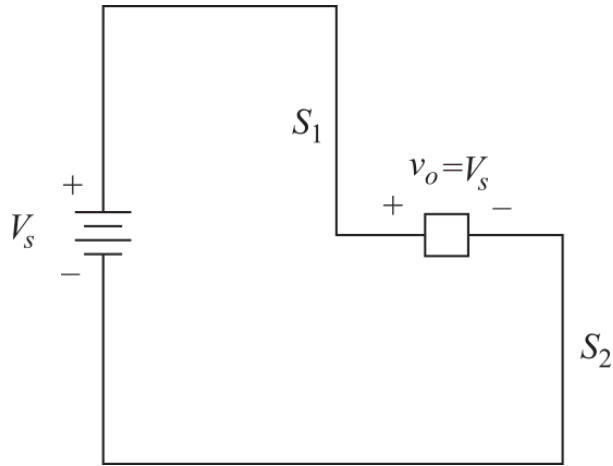
Inversor PWM Monopulso: formas de onda



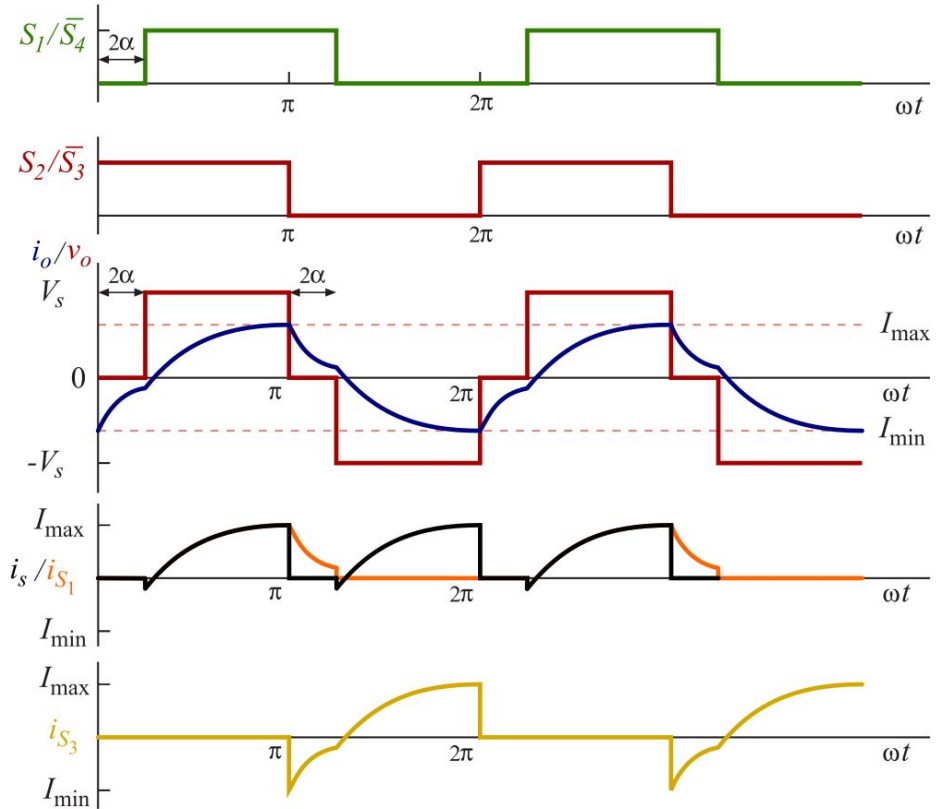
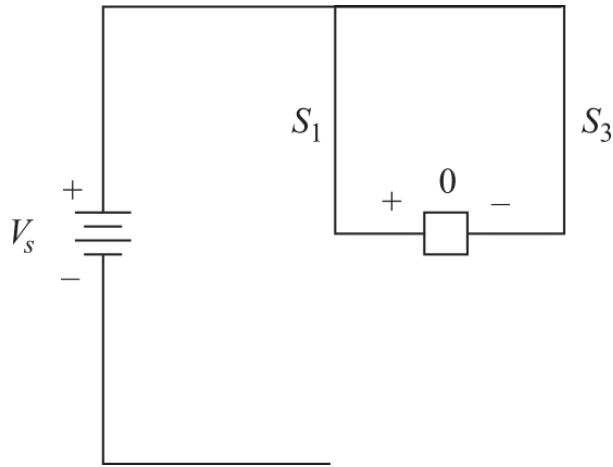
Inversor PWM Monopulso: formas de onda



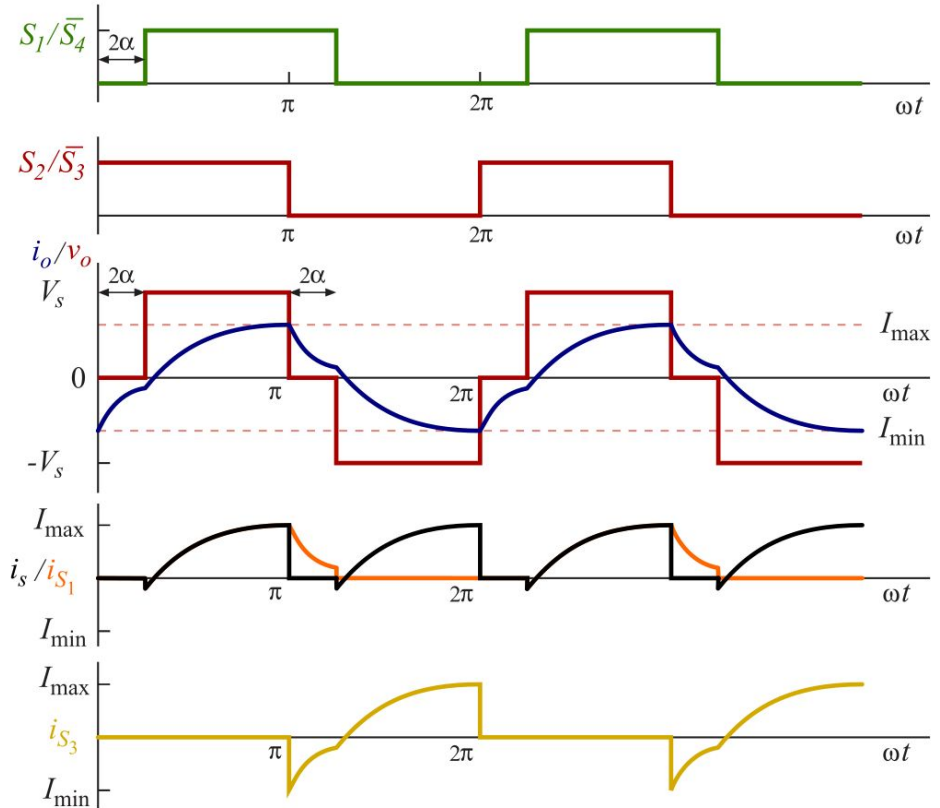
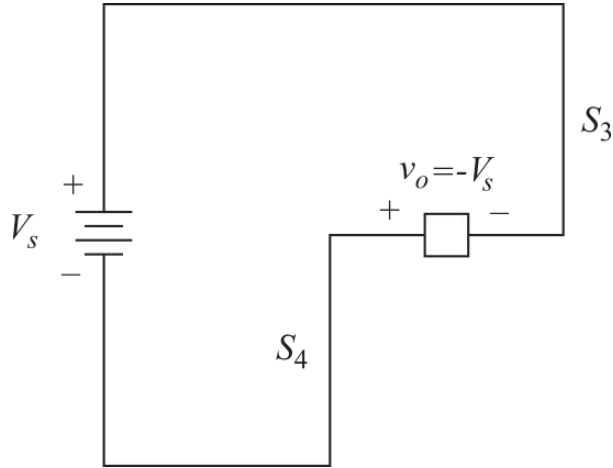
Inversor PWM Monopulso: formas de onda



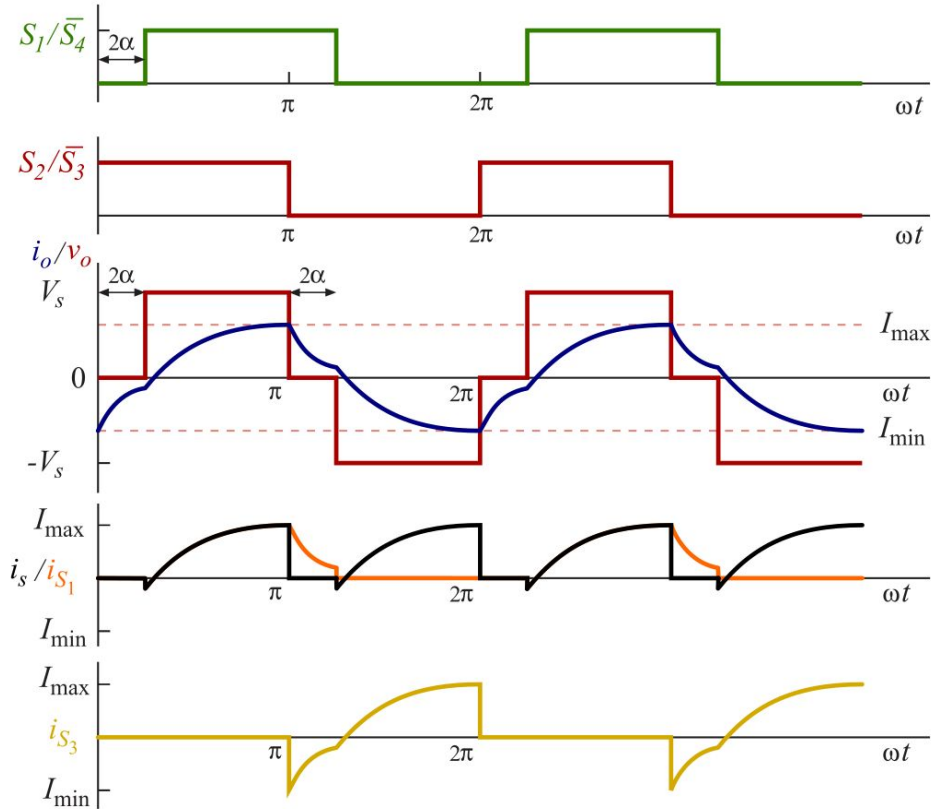
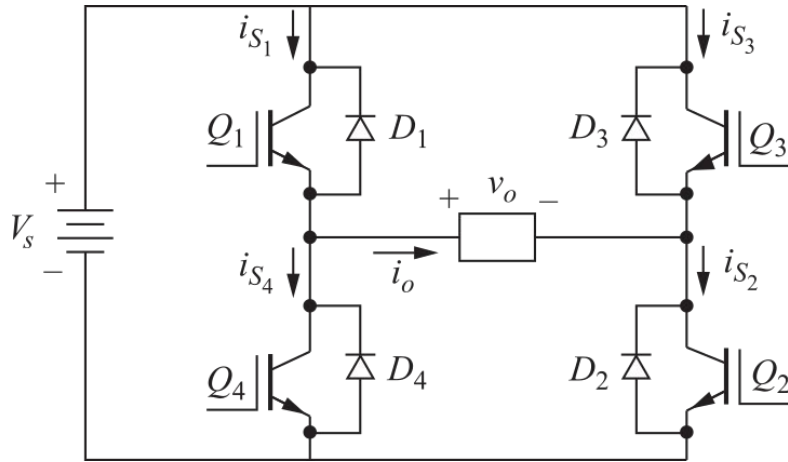
Inversor PWM Monopulso: formas de onda



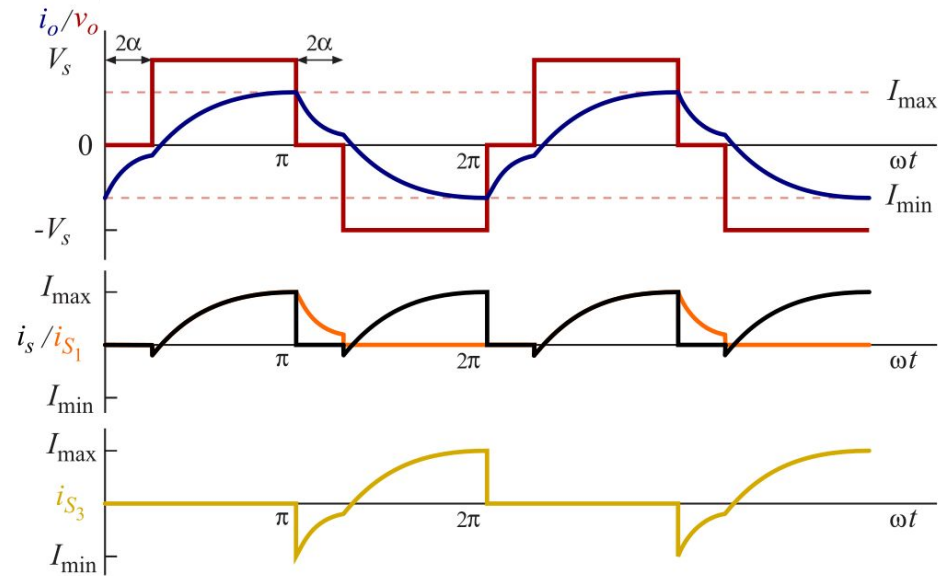
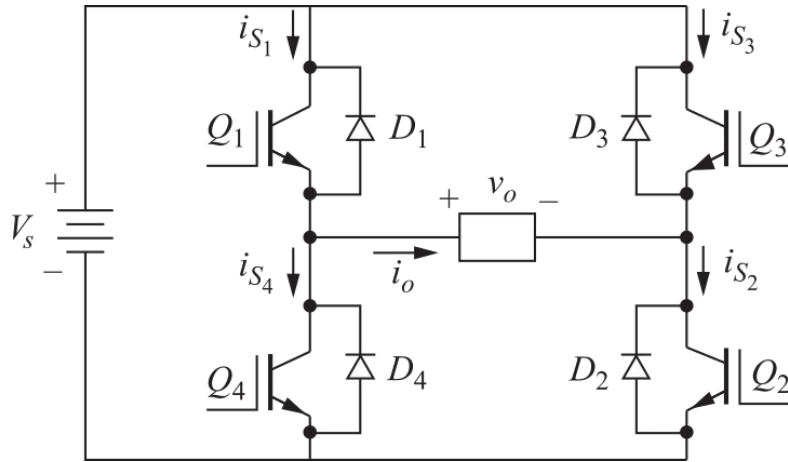
Inversor PWM Monopulso: formas de onda



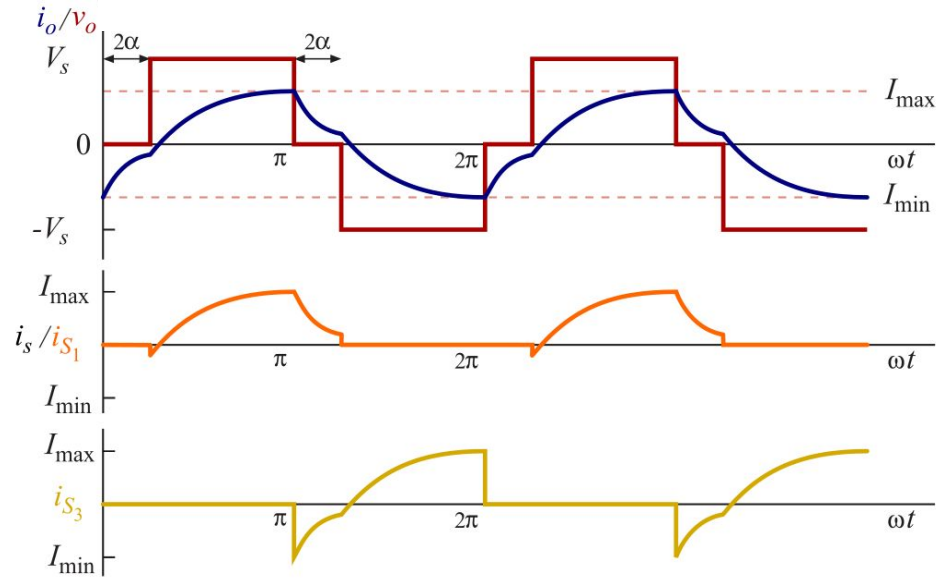
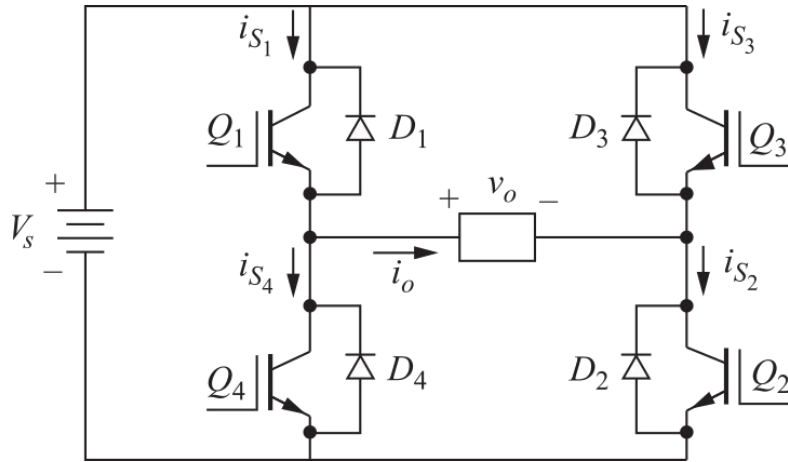
Inversor PWM Monopulso: formas de onda



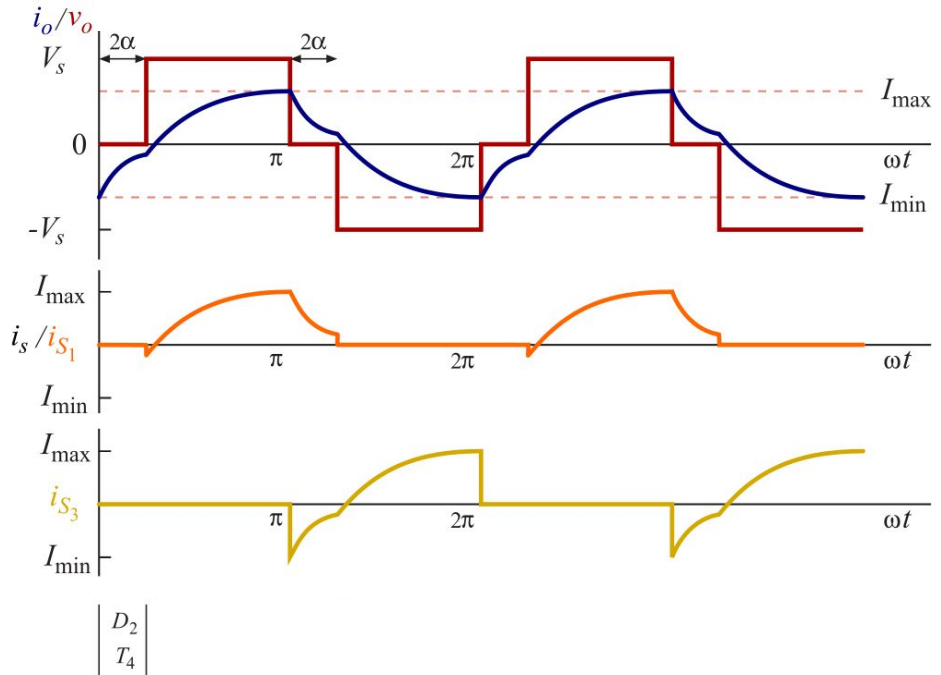
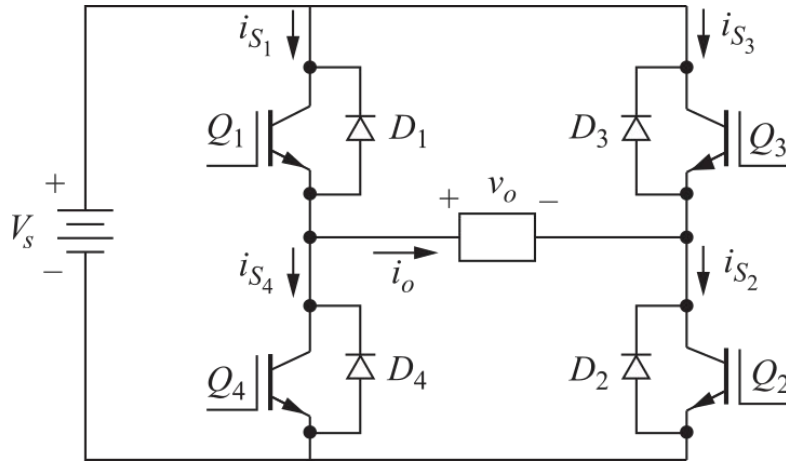
Inversor PWM Monopulso: formas de onda



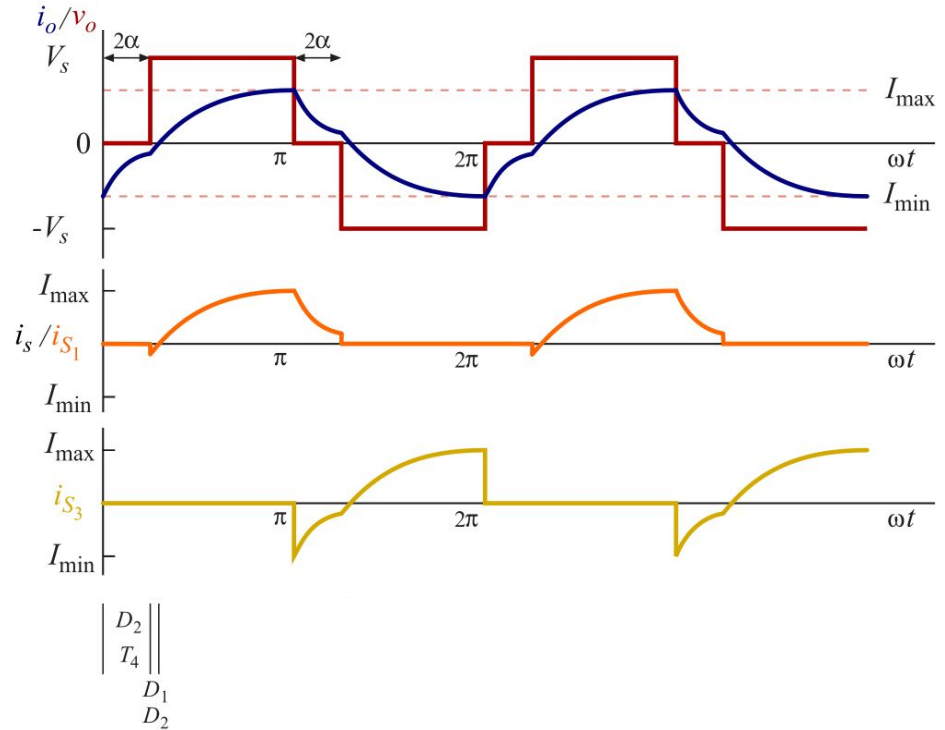
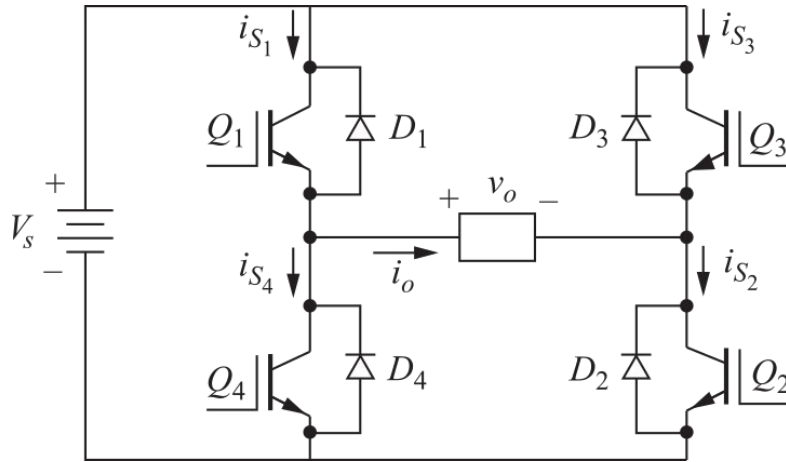
Inversor PWM Monopulso: formas de onda



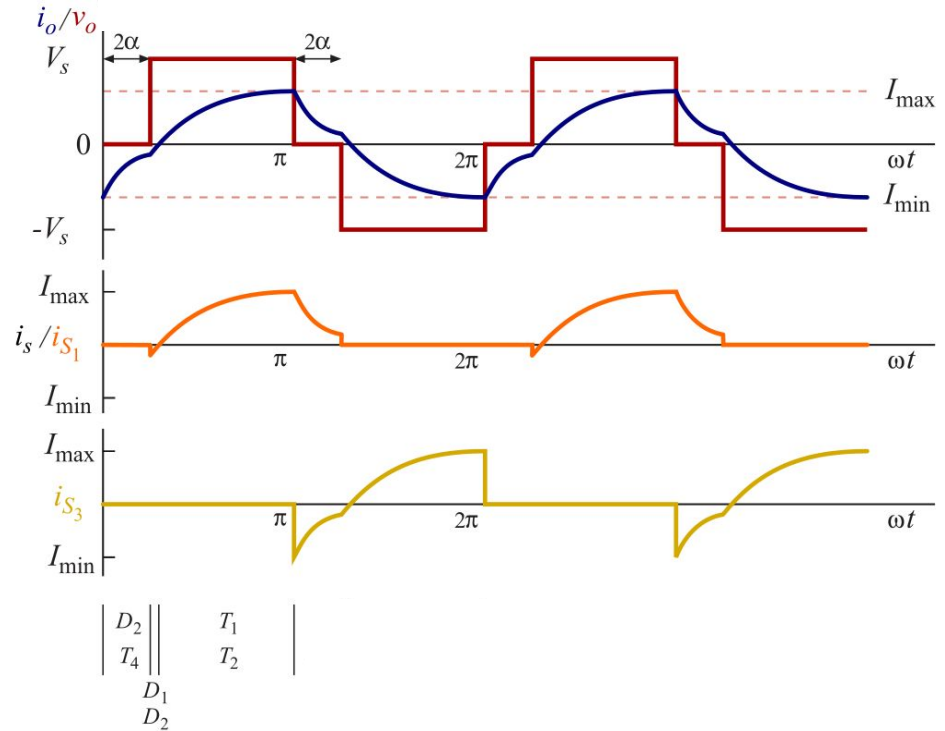
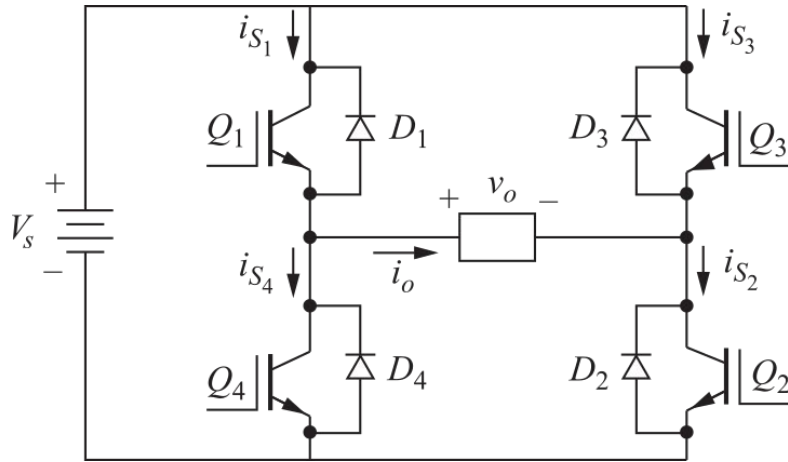
Inversor PWM Monopulso: formas de onda



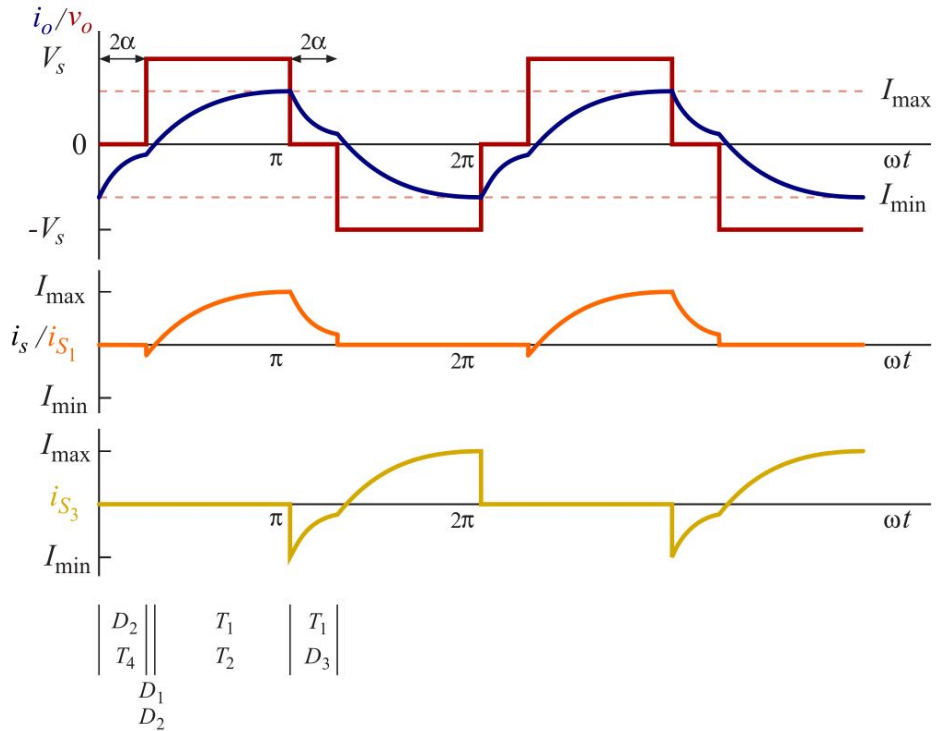
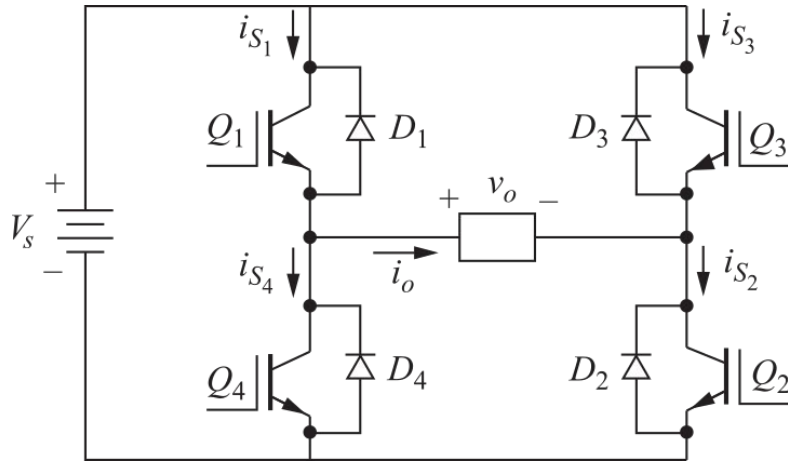
Inversor PWM Monopulso: formas de onda



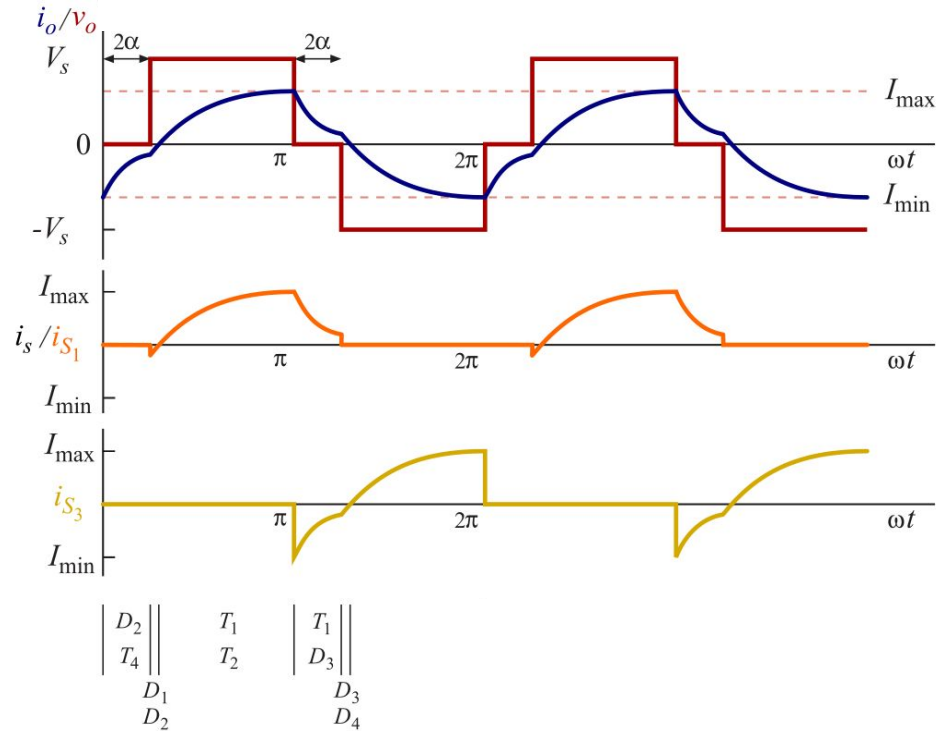
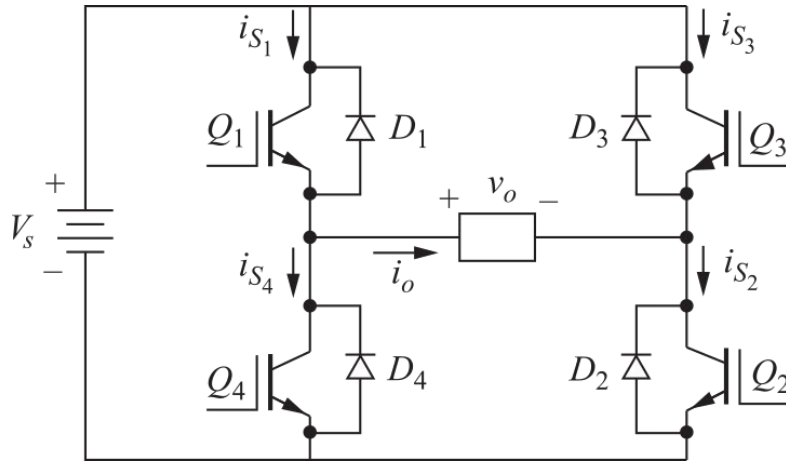
Inversor PWM Monopulso: formas de onda



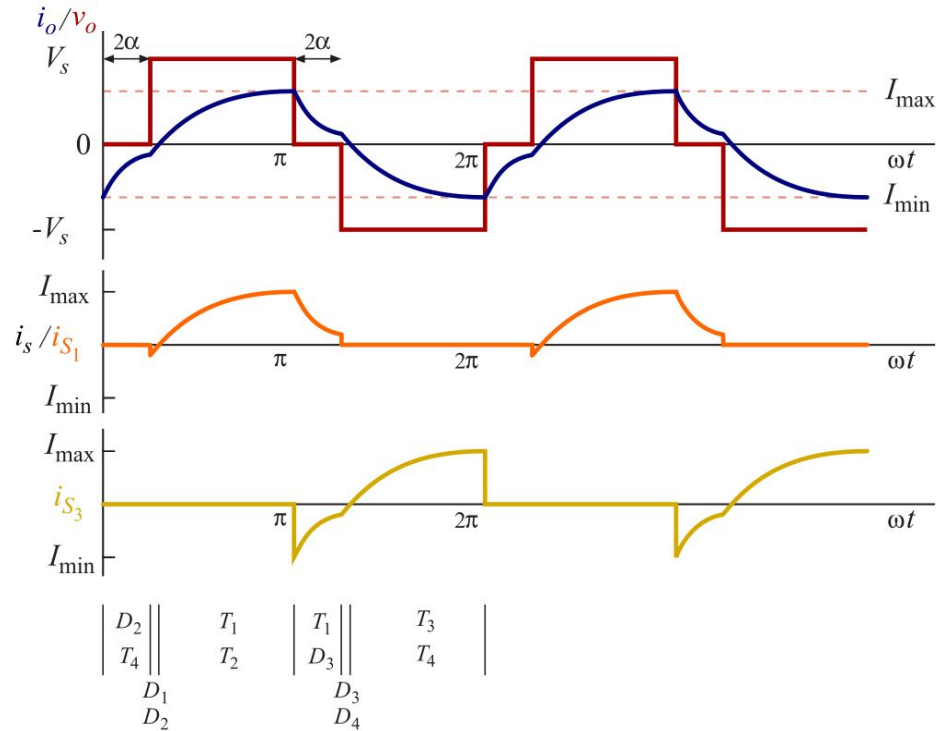
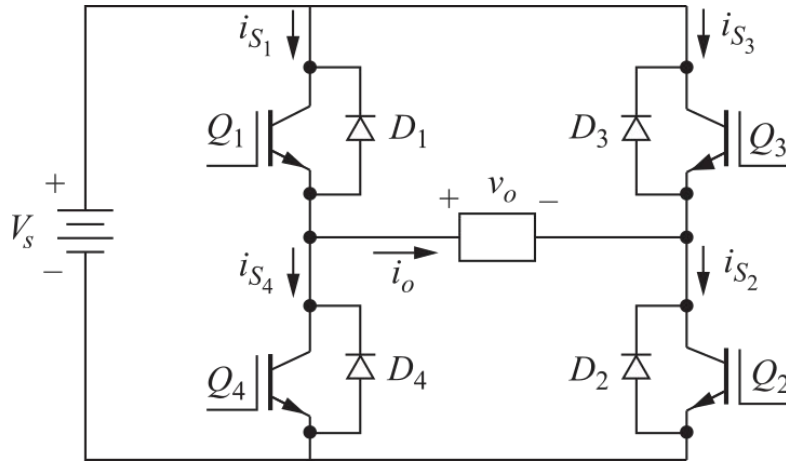
Inversor PWM Monopulso: formas de onda



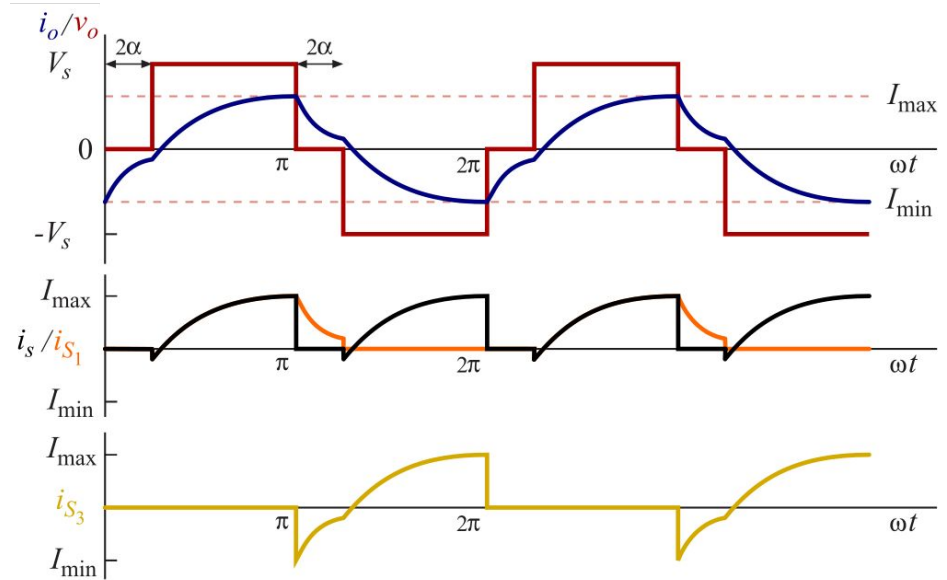
Inversor PWM Monopulso: formas de onda



Inversor PWM Monopulso: formas de onda

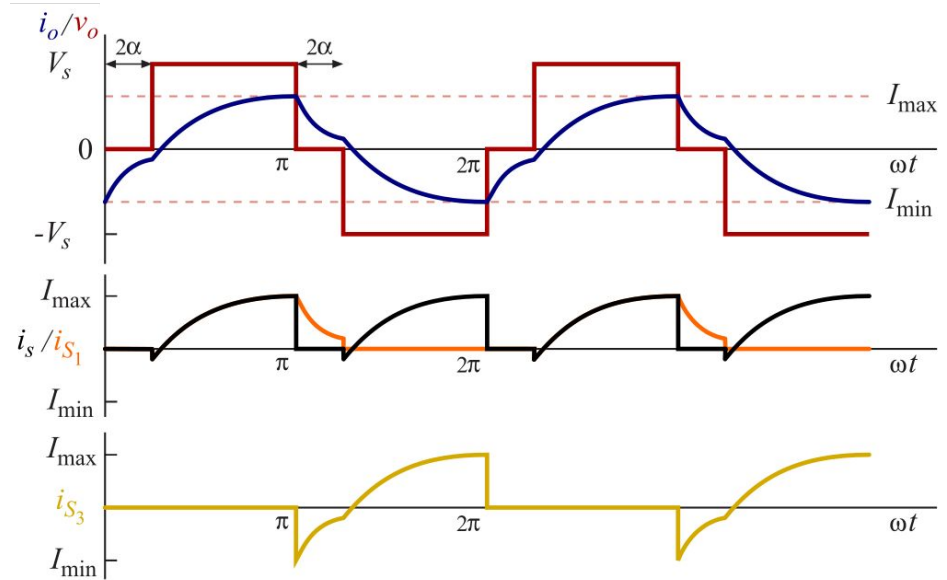


Inversor PWM Monopulso: análisis espectral



Inversor PWM Monopulso: análisis espectral

La tensión en la carga es periódica



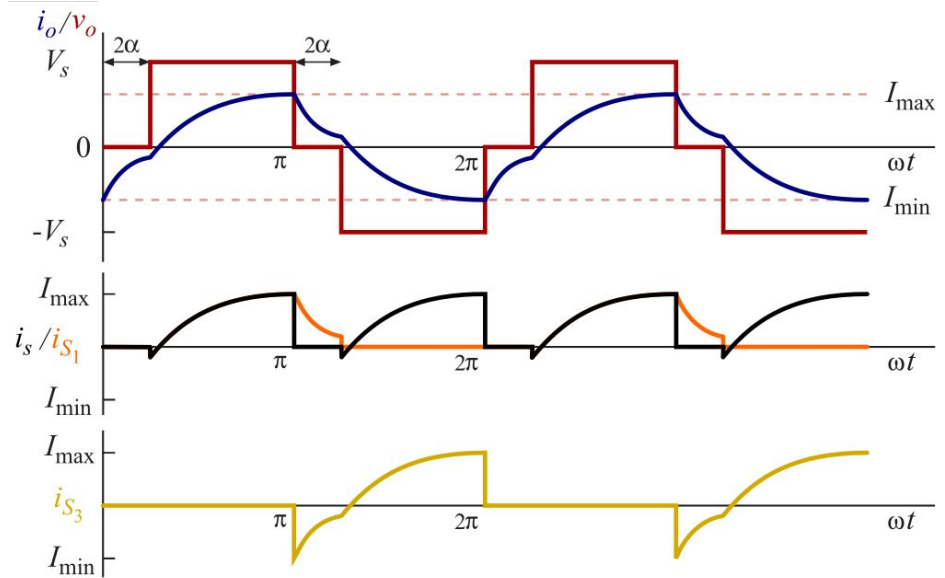
Inversor PWM Monopulso: análisis espectral

La tensión en la carga es periódica



Se puede escribir como serie de fourier:

$$v_o(t) = \sum_{n, \text{impar}} \frac{4V_s}{n\pi} \cos(n\alpha) \sin(n\omega t)$$



Inversor PWM Monopulso: análisis espectral

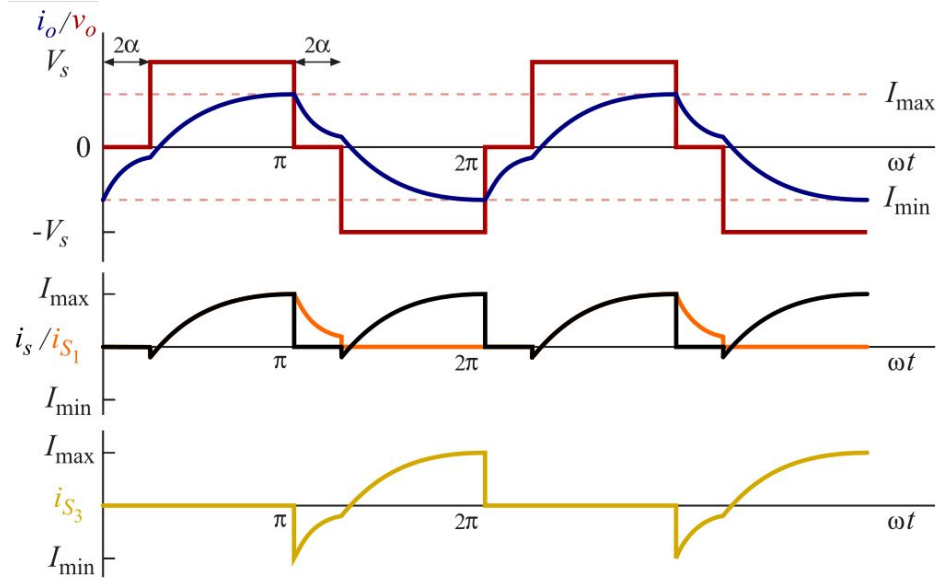
La tensión en la carga es periódica



Se puede escribir como serie de fourier:

$$v_o(t) = \sum_{n, \text{impar}} \frac{4V_s}{n\pi} \cos(n\alpha) \sin(n\omega t)$$

$$V_1 = \frac{4V_s}{\pi} \cos(\alpha)$$



Inversor PWM Monopulso: análisis espectral

La tensión en la carga es periódica



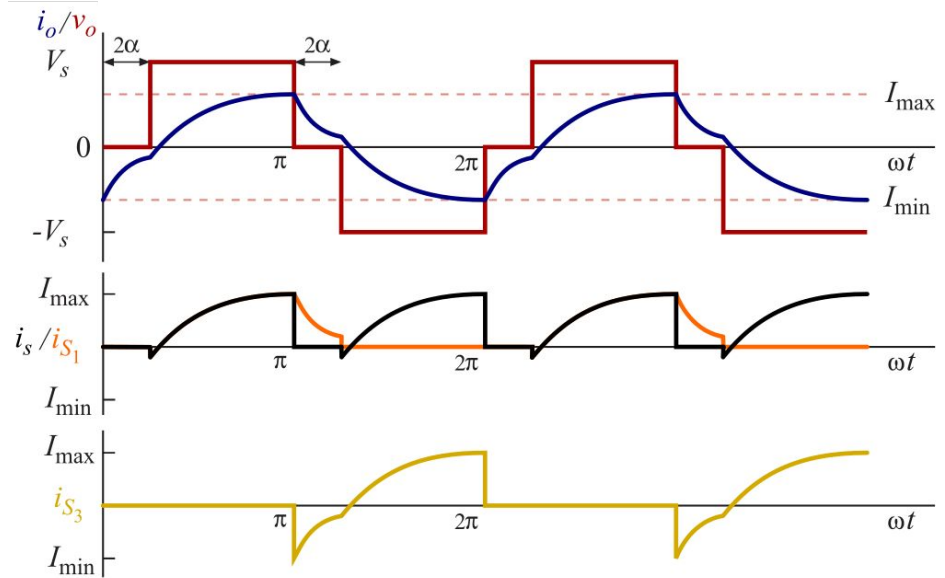
Se puede escribir como serie de fourier:

$$v_0(t) = \sum_{n, \text{impar}} \frac{4V_s}{n\pi} \cos(n\alpha) \sin(n\omega t)$$

$$V_1 = \frac{4V_s}{\pi} \cos(\alpha)$$

El armónico n se elimina si:

$$\alpha = \frac{90^\circ}{n}$$



Inversor PWM Monopulso: análisis espectral

La tensión en la carga es periódica



Se puede escribir como serie de fourier:

$$v_0(t) = \sum_{n, \text{impar}} \frac{4V_s}{n\pi} \cos(n\alpha) \sin(n\omega t)$$

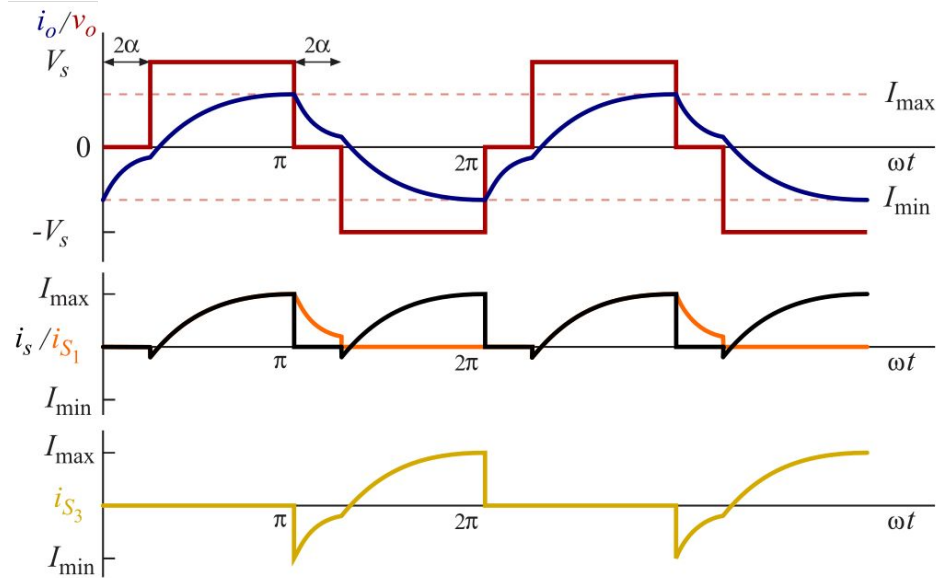
$$V_1 = \frac{4V_s}{\pi} \cos(\alpha)$$

El armónico n se elimina si:

$$\alpha = \frac{90^\circ}{n}$$



Si $\alpha = 30^\circ$



Inversor PWM Monopulso: análisis espectral

La tensión en la carga es periódica



Se puede escribir como serie de fourier:

$$v_0(t) = \sum_{n, \text{impar}} \frac{4V_s}{n\pi} \cos(n\alpha) \sin(n\omega t)$$

$$V_1 = \frac{4V_s}{\pi} \cos(\alpha)$$

El armónico n se elimina si:

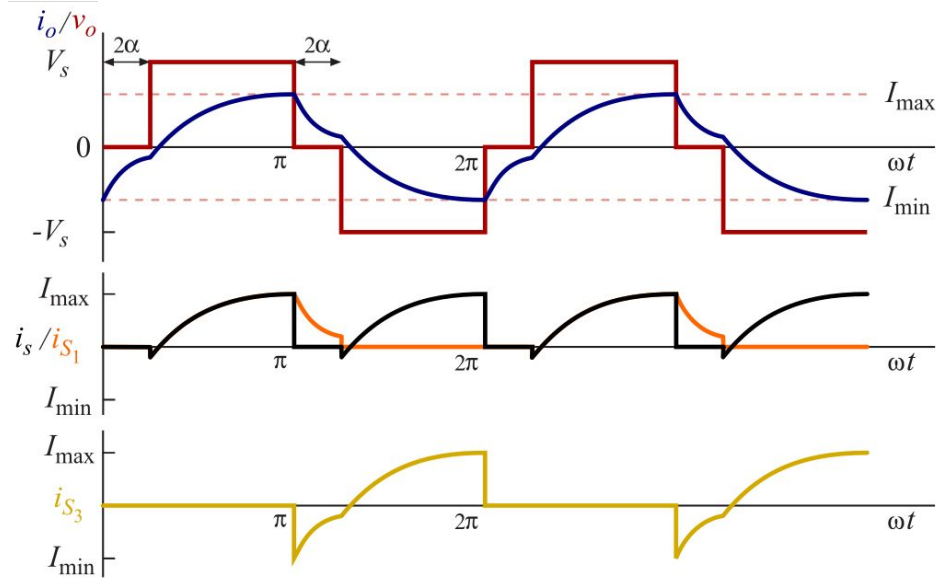
$$\alpha = \frac{90^\circ}{n}$$



Si $\alpha = 30^\circ$



$$V_3 = 0$$



Inversor PWM Monopulso: análisis espectral

La tensión en la carga es periódica



Se puede escribir como serie de fourier:

$$v_0(t) = \sum_{n, \text{impar}} \frac{4V_s}{n\pi} \cos(n\alpha) \sin(n\omega t)$$

$$V_1 = \frac{4V_s}{\pi} \cos(\alpha)$$

El armónico n se elimina si:

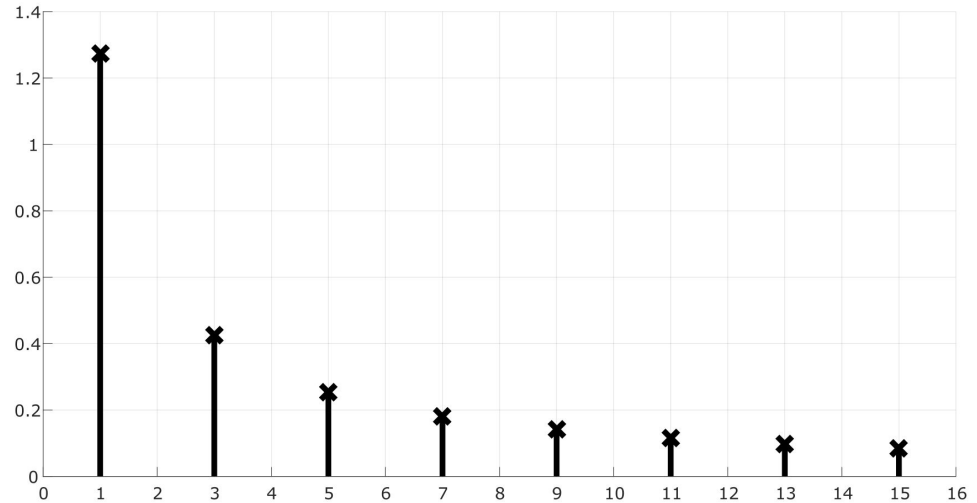
$$\alpha = \frac{90^\circ}{n}$$



Si $\alpha = 30^\circ$



$$V_3 = 0$$



Inversor PWM Monopulso: análisis espectral

La tensión en la carga es periódica



Se puede escribir como serie de fourier:

$$v_0(t) = \sum_{n, \text{impar}} \frac{4V_s}{n\pi} \cos(n\alpha) \sin(n\omega t)$$

$$V_1 = \frac{4V_s}{\pi} \cos(\alpha)$$

El armónico n se elimina si:

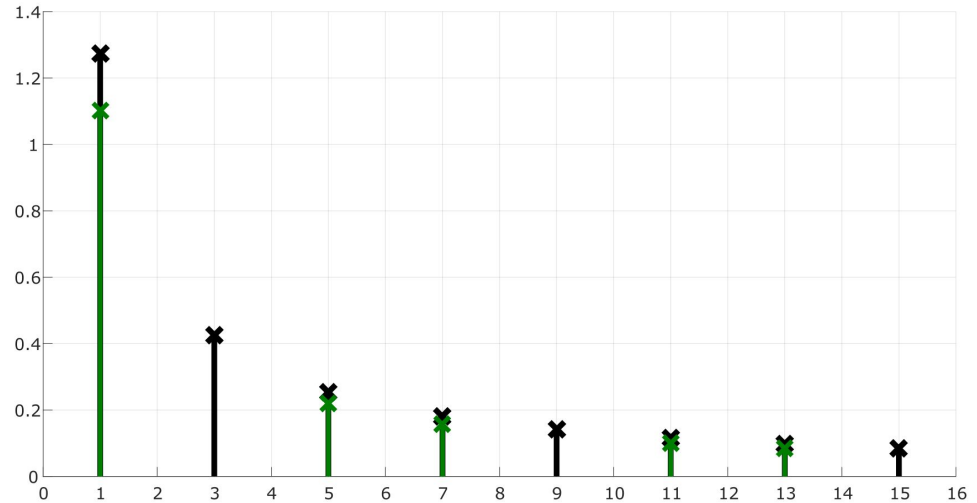
$$\alpha = \frac{90^\circ}{n}$$



Si $\alpha = 30^\circ$

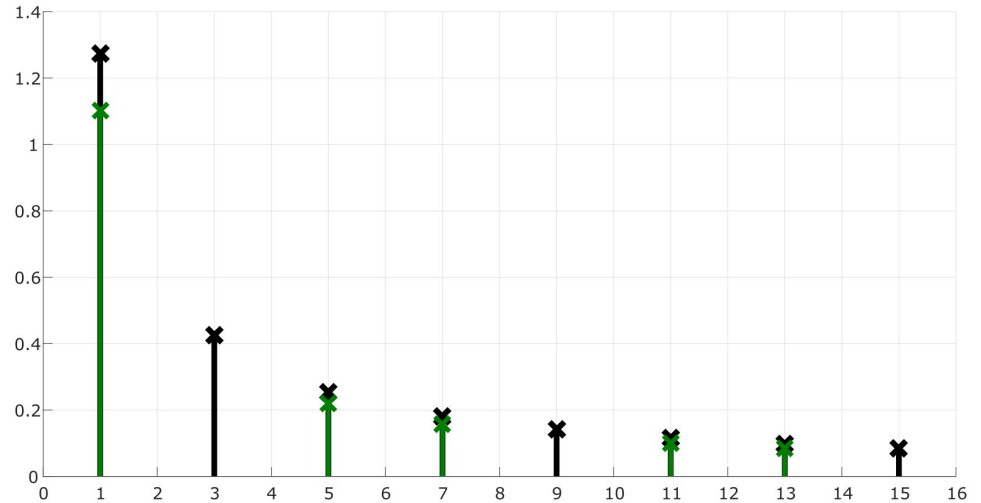


$$V_3 = 0$$



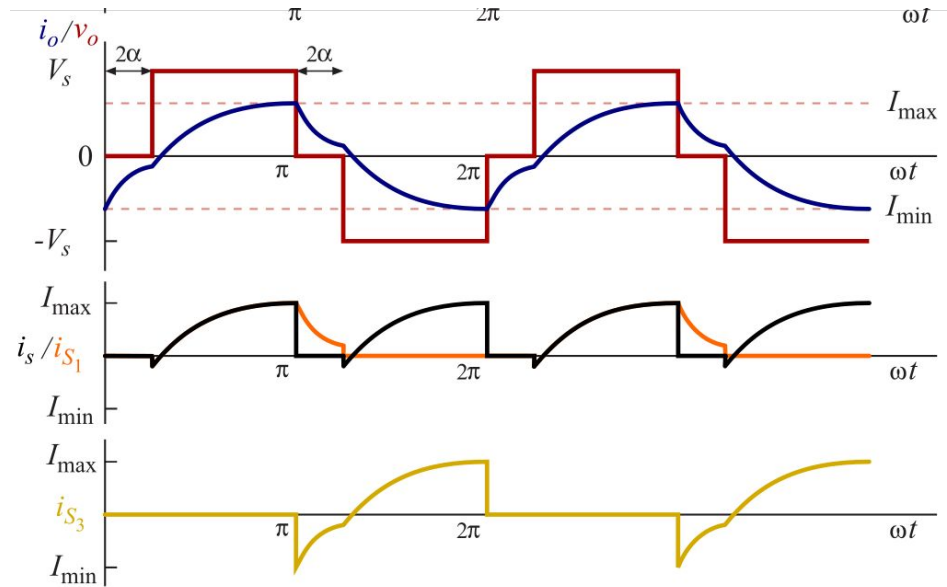
Inversor PWM Monopulso: análisis espectral

¿Cómo resulta el THD?



Inversor PWM Monopulso: análisis espectral

¿Cómo resulta el THD?

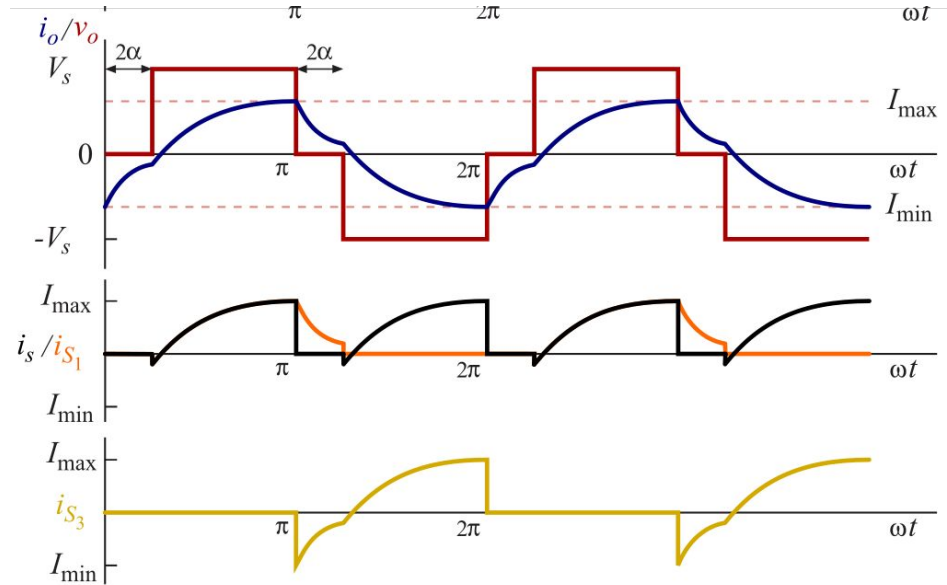


Inversor PWM Monopulso: análisis espectral

¿Cómo resulta el THD?



$$THD = \frac{\sqrt{(V_{rms})^2 - (V_{1,rms})^2}}{V_{1,rms}}$$



Inversor PWM Monopulso: análisis espectral

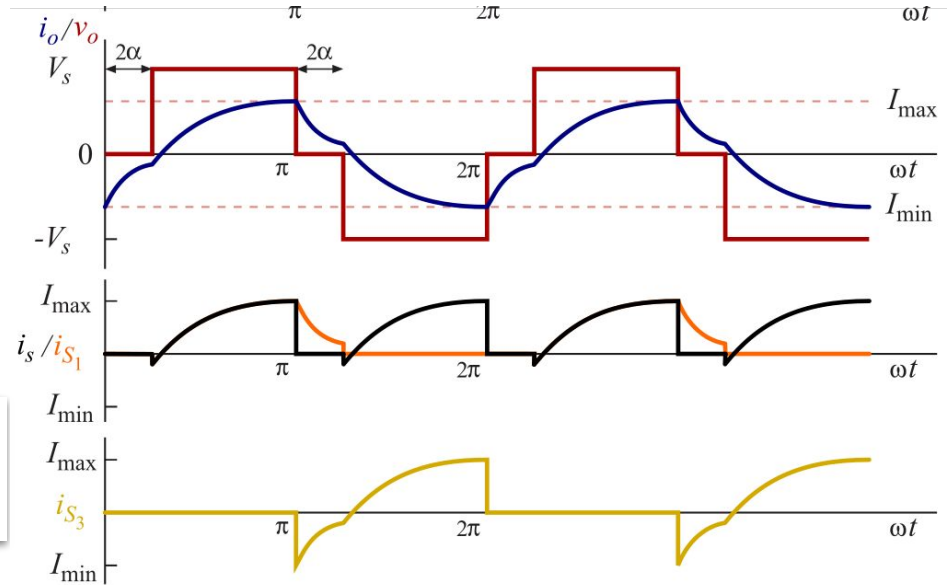
¿Cómo resulta el THD?



$$THD = \frac{\sqrt{(V_{rms})^2 - (V_{1,rms})^2}}{V_{1,rms}}$$

$$V_{rms} = V_s \sqrt{1 - \frac{2\alpha}{\pi}}$$

$$V_{1,rms} = \frac{4V_s}{\sqrt{2}\pi} \cos(\alpha)$$



Inversor PWM Monopulso: análisis espectral

¿Cómo resulta el THD?

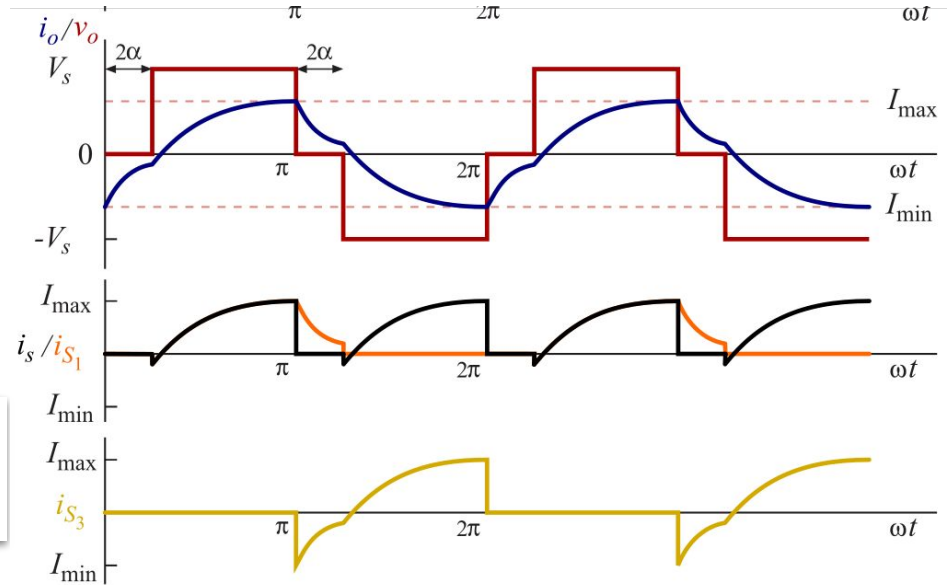


$$THD = \frac{\sqrt{(V_{rms})^2 - (V_{1,rms})^2}}{V_{1,rms}}$$

$$V_{rms} = V_s \sqrt{1 - \frac{2\alpha}{\pi}}$$

$$V_{1,rms} = \frac{4V_s}{\sqrt{2}\pi} \cos(\alpha)$$

Si $\alpha = 30^\circ \rightarrow THD\% = 31,08\%$



Inversor PWM Monopulso: análisis espectral

¿Cómo resulta el THD?

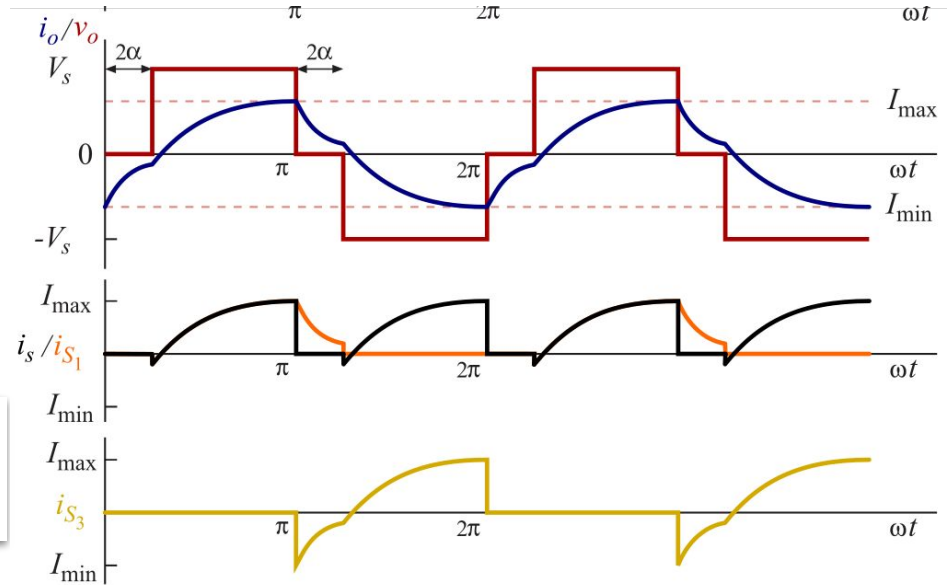


$$THD = \frac{\sqrt{(V_{rms})^2 - (V_{1,rms})^2}}{V_{1,rms}}$$

$$V_{rms} = V_s \sqrt{1 - \frac{2\alpha}{\pi}}$$

$$V_{1,rms} = \frac{4V_s}{\sqrt{2}\pi} \cos(\alpha)$$

Si $\alpha = 30^\circ \rightarrow THD\% = 31,08\%$



α	0°	10°	20°	30°	40°	50°	60°
THD%	48,34	36,15	29,44	31,08	40,98	57,19	80,3

Inversor PWM Monopulso: análisis espectral

¿Cómo resulta el THD?

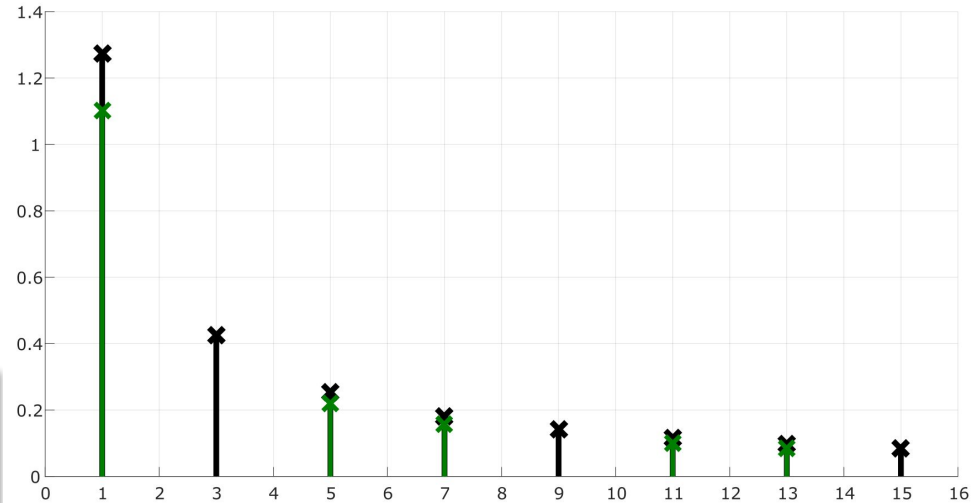


$$THD = \frac{\sqrt{(V_{rms})^2 - (V_{1,rms})^2}}{V_{1,rms}}$$

$$V_{rms} = V_s \sqrt{1 - \frac{2\alpha}{\pi}}$$

$$V_{1,rms} = \frac{4V_s}{\sqrt{2}\pi} \cos(\alpha)$$

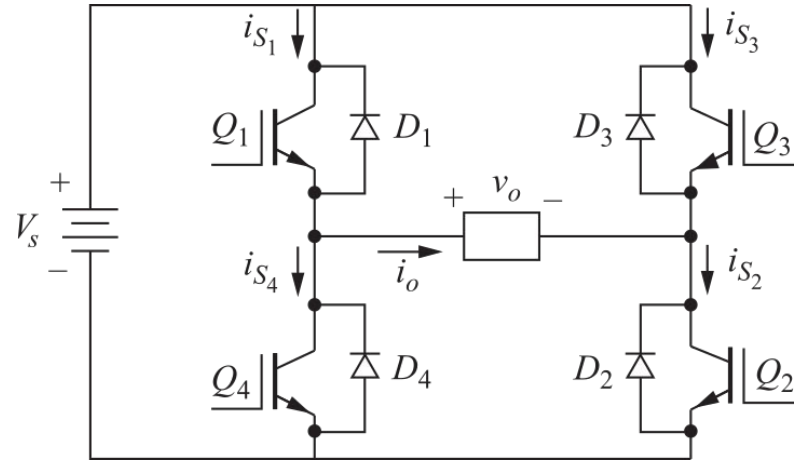
Si $\alpha = 30^\circ \rightarrow THD\% = 31,08\%$



α	0°	10°	20°	30°	40°	50°	60°
THD%	48,34	36,15	29,44	31,08	40,98	57,19	80,3

Inversor PWM Monopulso: tensión en la carga

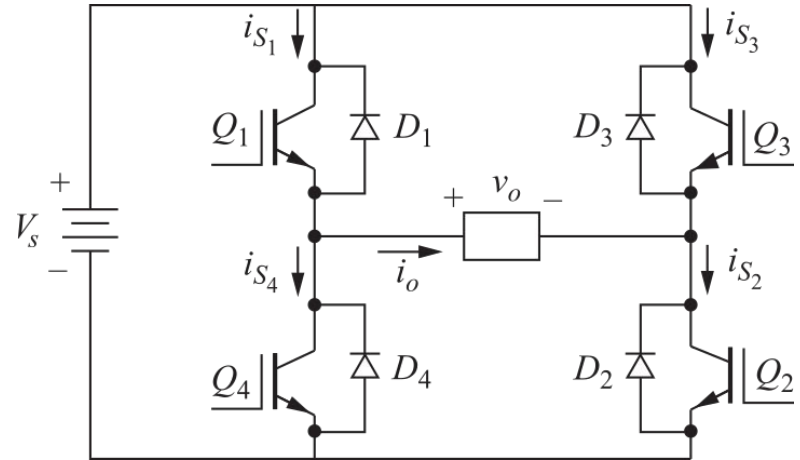
$$V_1 = \frac{4V_s}{\pi} \cos(\alpha)$$



Inversor PWM Monopulso: tensión en la carga

$$V_1 = \frac{4V_s}{\pi} \cos(\alpha)$$

¿Es posible modificar V_1 ?

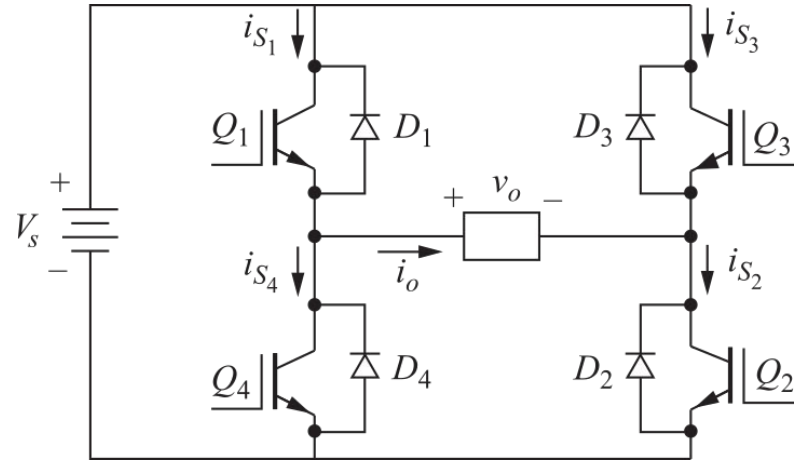


Inversor PWM Monopulso: tensión en la carga

$$V_1 = \frac{4V_s}{\pi} \cos(\alpha)$$

¿Es posible modificar V_1 ?

¡Tengo α como variable de control!



Inversor PWM Monopulso: tensión en la carga

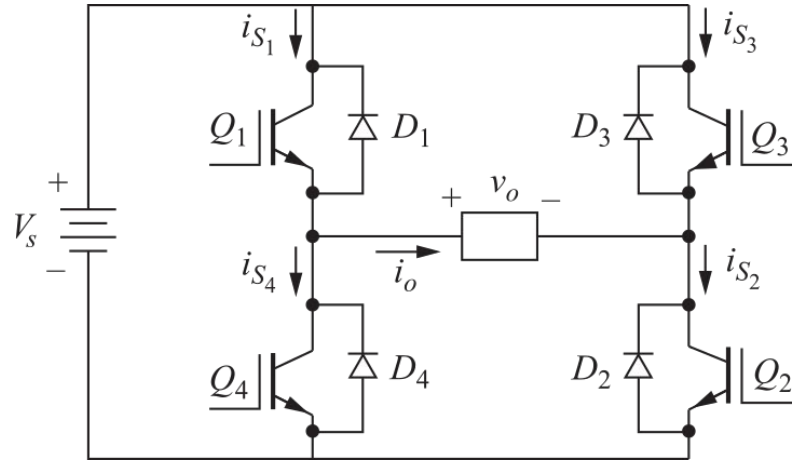
$$V_1 = \frac{4V_s}{\pi} \cos(\alpha)$$

¿Es posible modificar V_1 ?

¡Tengo α como variable de control!

Este esquema de modulación me permite
una de dos cosas:

- Controlar/modificar la amplitud de V_1
- Eliminar armónicos (sobre todo el 3^{ro})



Bibliografía

- **Apuntes de Cátedra: Inversores Estáticos**
- **Electrónica de Potencia** - Daniel W. Hart (Prentice Hall, 2001)
- **Electrónica de Potencia: Convertidores, Aplicaciones y Diseño** - N. Mohan - T.M.Undeland - W.P. Robbins (McGraw Hill, 2009)
- **Electrónica de Potencia. Circuitos, Dispositivos y Aplicaciones** - M.H. Rashid (2da Ed. Prentice Hall, 1993)
- **Power Electronics. Devices, Drivers, Applications, and Passive Components** - B. W. Williams (University of Strathclyde, Glasgow. 2006).
<http://personal.strath.ac.uk/barry.williams/book.htm>